30 Non–small-cell Lung Cancer

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EPIDEMIOLOGY AND RISK FACTORS

Lung cancer is the second most commonly diagnosed cancer worldwide and the most common cause of cancer deaths, with approximately 2.2 million new cases and 1.79 million deaths in 2020.[1] In the United States, the American Cancer Society estimates that approximately 236,000 new cases and 130,000 deaths from lung cancer will occur in 2022.[2] A study using the data from the Surveillance, Epidemiology, and End Results (SEER) program showed an initial decrease in the incidence of non–small-cell lung cancer (NSCLC) in men by 1.3% annually from 2001 to 2008 followed by a decrease by 3.1% annually from 2008 to 2016.[3] In women, the lack of changes in incidence from 2001 to 2006 was followed by a decrease of 1.5% annually from 2006 to 2016. A decrease in mortality has been observed in both men and women, particularly from 2013 to 2016.

Cigarette smoking is by far the leading cause of lung cancer, accounting for approximately 80% to 90% of cases. Combustion of tobacco produces multiple known carcinogens, with the most important including polycyclic aromatic hydrocarbons, tobacco-specific nitrosamines, and nitrates.[4] The additive component of tobacco is nicotine. Smokers in the United States have an approximately 20-fold increase in lung cancer risk compared to never-smokers.[5] Furthermore, despite the decreased risk for developing lung cancer among individuals who quit smoking, the risks remain elevated compared to never-smokers even after prolonged abstinence. Although cigar smoking is also associated with the development of lung cancer, the risk is lower than for cigarette smoking due to differences in smoking frequency and depth of inhalation.

Other risk factors include environmental tobacco smoke (ETS), also known as second-hand smoking; environmental exposures such as asbestos; high doses of ionizing radiation; radon and air pollution including combustion of fossil fuels; and products from heating and cooking. ETS consists of both exhaled smoke and sidestream smoke from the burning end of cigarettes.[6] The risk for lung cancer is increased in nonsmokers who live with smokers, and it has been estimated that approximately 17% of lung cancers among never-smokers may be attributed to high levels of ETS during childhood and adolescence. Several occupational carcinogens have been implicated in the development of lung cancer, of which asbestos is the most common. Asbestos is a naturally occurring mineral, consisting of two types (serpentine and amphibole), that has been used for construction and insulating materials. The risk for lung cancer is essentially restricted to occupational exposure, dose dependent, and higher for amphibole fibers. The combination of asbestos exposure and cigarette smoking has a synergistic effect on the risk for lung cancer. Radon is a radioactive gas produced by the uranium decay on soil.[7] Residential radon is the second most common risk factor for lung cancer, accounting for approximately 10% of cases. Indoor air pollution from the use of unprocessed solid fuels, predominantly coal, and wood for heating and cooking has been associated with increased risk for lung cancer, mostly in the developing world.[4]

The presence of familial clustering and the development of lung cancer only in a minority of smokers suggests a genetic component in addition to the environmental factors. Genome-wide association studies have reported several lung cancer susceptibility loci with varying strength of evidence.[8] The strongest associations have been described on loci 15q24.25, 6p21, and 5p15. The locus 5p15 has been associated only with adenocarcinoma, whereas 6p21 has a stronger association with squamous cell lung cancer.

CLINICAL MANIFESTATIONS, DIAGNOSIS, AND STAGING

A systematic approach to the initial evaluation of patients suspected of having lung cancer should include an assessment of clinical symptoms and signs, a thoughtful strategy to efficiently establish a diagnosis, and careful determination of the extent of disease.

Clinical Presentation

Currently, most patients with lung cancer present with advanced disease, and the evaluation is prompted by the onset of symptoms or signs. As lung cancer screening becomes broadly implemented, it is expected that more patients will be identified with early-stage disease, when they are likely to be asymptomatic. The evaluation of every patient with suspected or known lung cancer starts with a thorough history and physical examination. Symptoms and signs will reflect the consequences of the primary tumor, extension into adjacent structures, distant metastases, or paraneoplastic syndromes (Table 30.1).

Symptoms and Signs Related to Endobronchial Disease

Endobronchial tumor involvement typically presents with persistent cough, dyspnea, or hemoptysis. Cough and dyspnea may result from direct airway involvement, bronchorrhea, or postobstructive pneumonitis. Dyspnea may also reflect disease spread or complications such as pulmonary thromboembolism. Cough and dyspnea are nonspecific. Their attribution to comorbidities common in lung cancer patients such as chronic obstructive pulmonary disease may delay diagnosis. Hemoptysis typically is low grade and rarely massive but less likely to be ascribed to a benign process.

TABLE 30.1

Symptoms and Signs of Lung Cancer

Suggestive of endobronchial involvement	
	Cough
	Dyspnea
	Hemoptysis
	Localized wheezing
	Recurrent or nonresolving pneumonia or pulmonary infiltrate
Suggestive of intrathoracic spread	
	Cough
	Dyspnea
	Localized or pleuritic chest pain
	Hoarseness due to recurrent laryngeal nerve palsy
	Unilateral hemidiaphragm elevation due to phrenic nerve palsy
	Pancoast syndrome with ptosis, meiosis, anhidrosis, shoulder and arm pain, paresthesias, or weakness
	Superior vena cava syndrome with facial and neck swelling, dilated venous pattern on chest
	Pericardial effusion
Suggestive of systemic spread	
	Weight loss, anorexia
	Weakness, fatigue
	Chest pain
	Palpable supraclavicular lymphadenopathy
	Unexplained focal bony pain
	Unexplained focal neurologic findings, seizures, confusion
	Unexplained laboratory abnormalities (anemia, abnormal liver function tests or alkaline phosphatase, hypercalcemia)
Paraneoplastic syndromes	
	Hypercalcemia
	Syndrome of inappropriate antidiuretic hormone (SIADH)
	Cushing syndrome
	Lambert-Eaton myasthenic syndrome
	Paraneoplastic encephalitis
	Unexplained hematologic abnormalities (anemia, leukocytosis, thrombocytosis)
	Hypercoagulability with venous thromboembolic disease
	Hypertrophic pulmonary osteoarthropathy
	Dermatomyositis and polymyositis

Symptoms and Signs Related to Intrathoracic Spread of Disease

Intrathoracic spread of cancer with direct extension to adjacent thoracic structures such as the chest wall, pleura, and mediastinum may present as dyspnea, chest discomfort, or pain. Hoarseness should prompt concern for vocal cord dysfunction because the left recurrent laryngeal nerve circuitously courses under the aortic arch and is susceptible to compression or invasion by left-sided cancers or adenopathy. Lung cancer is the most common cause of superior vena cava syndrome, with facial and neck swelling or fullness, headache, dysphagia, neck vein distention, and a dilated venous pattern over the upper chest. Superior sulcus (Pancoast) tumors may involve the brachial plexus and lower cervical and high thoracic vertebrae, causing shoulder or arm pain, muscle wasting, or weakness. Tumor extension into the paravertebral sympathetic chain and stellate ganglion may cause Horner syndrome, which is characterized by ipsilateral ptosis, miosis, enophthalmos, and anhidrosis. Although lymphadenopathy is usually asymptomatic, bulky mediastinal adenopathy can compress the esophagus, causing dysphagia. Tumor pericardial invasion may result in pericardial effusion, with risk of pericardial tamponade.

Symptoms and Signs Related to Metastatic Spread of Disease

Suspicion for metastatic disease should be prompted by constitutional symptoms such as weight loss, weakness, and fatigue; organ-specific symptoms or signs; and laboratory abnormalities including abnormal alkaline phosphatase or liver function tests, hypercalcemia, and unexplained anemia. Asymptomatic patients with limited chest radiographic abnormalities are much less likely to have metastatic disease.[9] Recognition of concerning symptoms and signs may allow a more efficient evaluation, particularly if diagnosis and staging can be accomplished with a single biopsy.

Lung cancer can spread to any organ, with the most common sites of metastasis being lymph nodes, pleura, bone, liver, adrenal glands, and brain. Malignant lymphadenopathy is usually asymptomatic, and a careful examination of the supraclavicular fossa should be performed. Pleural effusions may cause dyspnea, whereas bulky pleural or chest wall involvement typically generate discomfort. Bone metastases may be suggested by elevated alkaline phosphatase or hypercalcemia and typically present with pain. Liver lesions and adrenal metastases are most often asymptomatic. Lung cancer is the most common cause of metastatic disease in the brain, with presentations ranging from headache to focal neurologic deficits or seizures.

Lung cancer histology influences the likelihood of distant spread. The frequency of metastasis is higher with small-cell lung cancer (SCLC) than NSCLC. In patients with NSCLC, brain metastasis is more likely with adenocarcinoma than squamous cell carcinoma.[10]

Paraneoplastic Syndromes

Lung cancer can be associated with a wide range of paraneoplastic syndromes. These syndromes may be the presenting complaint, can appear during the course of disease, are not related to direct tumor invasion or distant spread, and do not preclude a curative-intent treatment approach.

Several well-defined paraneoplastic syndromes are hormonally driven. Hypercalcemia of malignancy is observed in 10% to 25% of patients with lung cancer, most commonly advanced squamous cell lung cancer.[11] Hypercalcemia in patients with lung cancer may result from extensive bone metastases or tumor production of parathyroid hormone–related protein. Symptoms include polyuria, polydipsia, nausea, and dehydration. Severe hypercalcemia can cause mental status changes, hypotension, and renal failure. The syndrome of inappropriate antidiuretic hormone presenting as hyperosmolar hyponatremia with inappropriately high urine osmolality is seen in approximately 11% of patients with SCLC.[12] It is unusual in NSCLC but can occur with many drugs, including cisplatin and vinca alkaloids, and a variety of benign conditions, including pneumonia.

Paraneoplastic syndromes seen with lung cancer may also present as hematologic abnormalities, hypercoagulability, hypertrophic osteoarthropathy, or inflammatory myopathy. Anemia is common, particularly in advanced disease. In a series of 1,201 newly diagnosed lung cancer cases, leukocytosis and thrombocytosis were described in 32% and 17% of patients, respectively.[13] Tumor-related neutrophil-predominant leukocytosis is more commonly seen in patients with advanced NSCLC, particularly in large-cell histology, and is thought to be related to elevated levels

of hematopoietic cytokines such as granulocyte colony-stimulating factor.[14] Hypercoagulability presenting as venous thromboembolism is well described in lung cancer.[15,16] In a study of 91,333 lung cancer patients in the California Cancer Registry, 3% developed venous thromboembolism within 1 year of diagnosis.[16] Significantly associated factors included younger age, increasing medical comorbidities, advanced cancer stage, and adenocarcinoma histology (particularly mucin producing). Hypercoagulability may reflect multiple etiologic elements, including tumor-specific procoagulant activity, tumor compression or direct invasion of vascular structures, patient-specific predisposition, and factors related to treatment.

Rheumatic paraneoplastic syndromes associated with lung cancer include hypertrophic pulmonary osteoarthropathy (HPOA) and inflammatory myopathies. HPOA is characterized by digital clubbing, arthropathy of large joints, and radiographic periostosis. HPOA associated with lung cancer is more commonly seen with adenocarcinoma.[17] HPOA can also be observed with other nonmalignant pulmonary disorders. An association between dermatomyositis, and to a lesser degree polymyositis, and cancer is recognized for all common solid tumors, including lung cancer.[18] The highest incidence of cancer diagnosis occurs within the first 5 years of identification of inflammatory myopathy.

Staging Evaluation

Staging is a critical component of the evaluation of every patient strongly suspected of or known to have lung cancer. Staging provides a common nomenclature essential for reliable communication of anatomic extent of disease, which then informs treatment decisions.

The initial step in staging should address the question of whether distant metastatic disease is present. This may be suggested by an abnormal clinical evaluation, focal symptoms or signs, or extensive radiographic intrathoracic disease. The next step would focus on the mediastinal nodes, which are often the critical determinant of whether surgery will be the first treatment approach.

Staging must be done rigorously and with as much precision as possible. Although staging should not provide a "menu" for treatment, it nonetheless drives many treatment decisions. High-quality and accurate staging requires awareness of the testing characteristics of both noninvasive and invasive modalities, as well as clinical acumen to implement the most efficient path to establishing diagnosis and stage.

Noninvasive Staging

Chest Computed Tomography: Every patient suspected of having lung cancer should have a chest computed tomography (CT), which provides the best anatomic assessment of the thorax. The appearance of the radiographic abnormality will confirm suspicion of lung cancer or shift concern to alternative diagnoses. Consideration of chest CT findings in the context of individual risk factors and symptoms will allow an experienced physician to reliably make a clinical diagnosis of lung cancer in most patients.[19] Chest CT findings will provide the information necessary to describe the characteristics of the primary tumor for staging purposes.

Chest CT is also the first step in evaluation of the mediastinum. Administration of contrast is not essential for this purpose but may help distinguish lymph nodes from vascular structures.[20] Mediastinal node involvement is the most common determinant of stage III disease. Accurate mediastinal assessment is thus critical, as it may inform the decision as to whether the initial treatment approach will be surgical or not. A systematic literature review assessing the accuracy of CT for noninvasive identification of mediastinal lymph node metastasis yielded pooled sensitivity and specificity of only 55% and 81%, respectively.[20] Chest CT is thus not sufficient to either rule in or rule out mediastinal metastases, as it may both overstage and understage the mediastinum. This highlights the importance of mediastinal evaluation with other noninvasive modalities such as positron emission tomography (PET) imaging, and of confirming that both enlarged mediastinal nodes and distant sites are truly malignant.

Positron Emission Tomography Imaging: PET imaging with 2-deoxy-2-[fluorine-18]fluoro-D-glucose (FDG) has assumed an important role in the evaluation of patients with lung cancer. PET provides a quantitative analysis of cellular metabolism. Tissues with increased rates of glycolysis and enhanced glucose utilization will be quantitatively identifiable by PET. This metabolic pattern is demonstrated by many malignancies, including lung cancer, but is not specific, as FDG uptake can also be seen in infections or inflammations. PET provides information about metabolic activity of the primary tumor, mediastinal nodes, and distant sites excluding the brain, and is both more sensitive and specific than CT. A systematic literature review of the accuracy of PET for noninvasive staging of the mediastinum demonstrated sensitivity and specificity of 80% and 88%, respectively, in a population with a 28% median prevalence of mediastinal metastasis.[20] It should be noted that PET is less sensitive in smaller lesions, including lymph nodes measuring <7 to 10 mm. Furthermore, since inflamed or infected tissues may be FDG-avid, PET is not sufficiently specific to obviate the need for tissue confirmation of an abnormality, particularly if the decision about potentially curative surgery depends on that site. These limitations were confirmed in randomized controlled trials (RCTs) evaluating PET imaging in lung cancer, where PET findings resulted in both correct and incorrect upstaging of patients.[21–24] PET may be particularly useful as a whole body examination in patients where suspicion of distant disease or mediastinal node involvement is high based on an abnormal clinical evaluation including baseline history, physical, and laboratory evaluation. In contrast, incorrect upstaging from false-positive PET findings may delay or even prevent an appropriate curative-intent approach in patients with early-stage disease. With the understanding that the clinical evaluation plays an important role in interpreting the findings of PET imaging, the current evidence base supports an overall benefit of PET in the evaluation of most patients with lung cancer.[20]

Brain Imaging: Patients with negative clinical evaluation have low likelihood of brain metastasis (median prevalence, 3%; range, 0%–21%).[25] NSCLC patients with larger primary tumors (>3 cm), evidence of mediastinal nodal involvement, or adenocarcinoma or large-cell histology have a higher likelihood of brain metastases.[20,26,27] Magnetic resonance imaging (MRI) is more sensitive than CT scanning of the brain.[28]

Abdominal Imaging: The adrenal glands, liver, and abdominal lymph nodes are common lung cancer metastatic sites. Incidental benign adrenal nodules are identified in 3% to 9% of adults and may mimic metastatic disease.[29] As with brain imaging, an abnormal clinical evaluation suggestive of a higher cancer stage should prompt thorough investigation of an adrenal abnormality. PET scan is highly accurate in distinguishing between a benign adenoma and a metastatic site.[30,31] In situations where suspicion is high that an adrenal finding on PET is a false positive, further evaluation should be performed with MRI or biopsy.[32] Benign liver abnormalities, typically cysts and hemangiomas, are also common. As with adrenal adenomas, PET is highly accurate in distinguishing liver metastases from benign lesions.[31,33]

Bone Imaging: PET is a reliable modality to evaluate for bone metastases. In direct comparative studies, the accuracy of PET exceeded that of radionuclide bone scanning.[34,35] Regardless of the imaging approach, including MRI, false-positive bone findings are often problematic. In an older lung cancer population, degenerative or post-traumatic bony changes are common findings and

are also often associated with symptoms. These may appear radiographically similar to metastatic disease and are often challenging to biopsy.

Invasive Staging

Consideration of invasive staging identifies four groups of patients. For patients with imaging evidence of metastatic disease, invasive evaluation should ideally achieve diagnosis and stage with a single biopsy, with the choice of procedure dictated by the site of metastasis. The exception to this would be a case with overwhelmingly convincing radiographic evidence of widespread disease, in which metastases can be clinically assumed and the most easily accessible site biopsied.

For patients with enlarged or FDG-avid hilar, mediastinal, or supraclavicular nodes and no radiographic evidence of distant metastases, biopsy of the lymph nodes should achieve diagnosis and stage with a single biopsy. For patients with a solitary lung lesion without radiographic evidence of mediastinal lymphadenopathy or distant disease but with radiographic features suggesting an increased likelihood of mediastinal nodal involvement such as a large primary tumor, central tumor, and hilar adenopathy, invasive evaluation of the mediastinal nodes should be pursued, with concurrent biopsy of the primary site if possible. For patients with solitary lung lesions without evidence of distant metastasis, regional lymphadenopathy, or features suggesting an increased likelihood of mediastinal nodal involvement, biopsy of the primary site should be pursued. In cases of peripheral stage IA lesions with strong suspicion of cancer, surgical resection to establish diagnosis and provide treatment at the same time may be reasonable. In all cases, the choice and accuracy of any invasive procedure will depend on local expertise. Procedural diagnostic accuracy reported by high-volume institutions may not be replicable in low-volume settings.

Invasive Mediastinal Staging: A variety of minimally invasive modalities are available for sampling of mediastinal nodes. The choice of procedure will be dictated by the anatomic accessibility of nodal stations to specific procedures and local expertise (Fig. 30.1

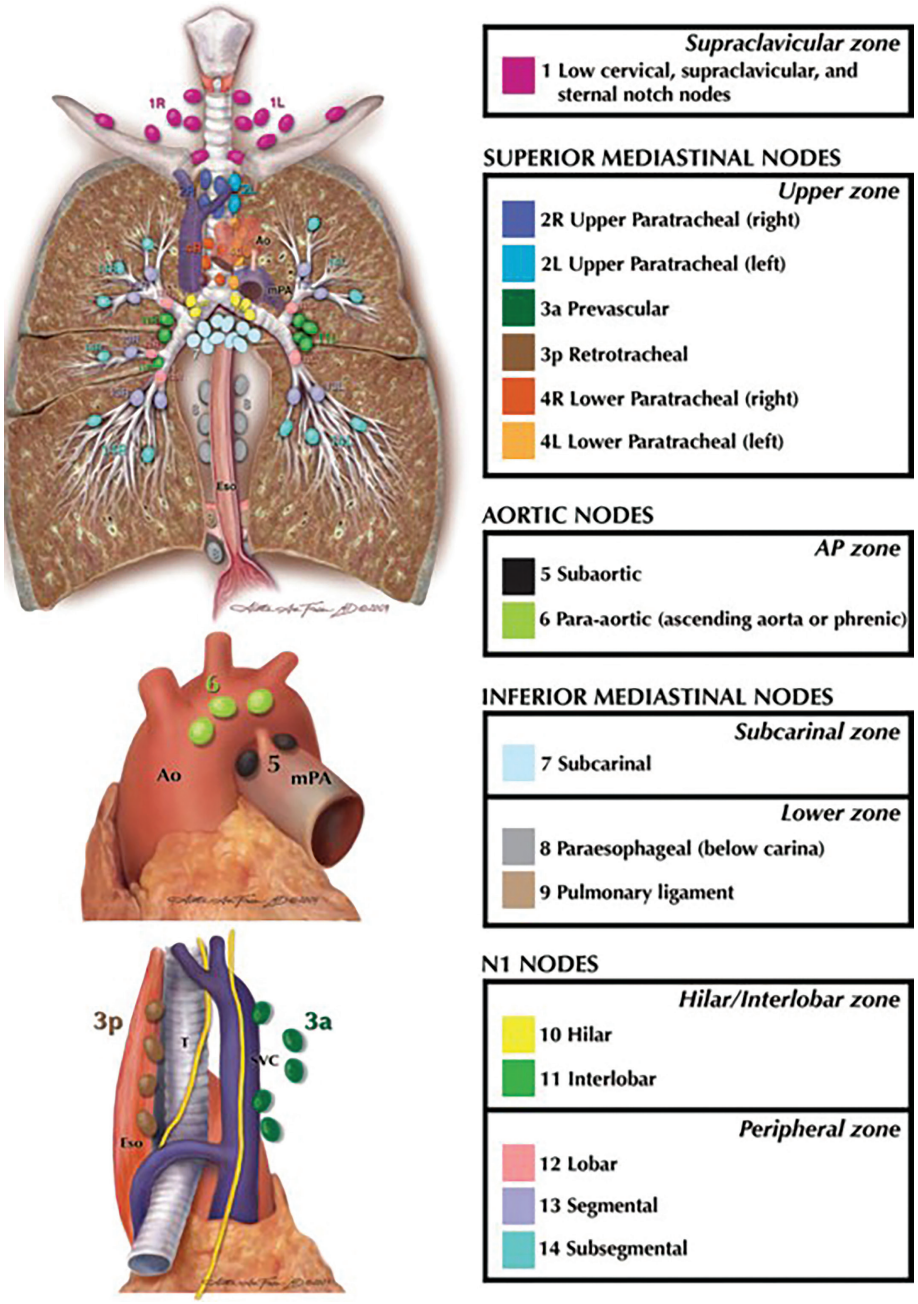


Figure 30.1 The International Association for the Study of Lung Cancer lymph node mapping system. (Reproduced with permission, from Rusch VW, Asamura H, Watanabe H, Giroux DJ, Rami-Porta R, Goldstraw P. The IASLC lung cancer staging project: a proposal for a new international lymph node map in the forthcoming seventh edition of the TNM classification for lung cancer. *J Thorac Oncol*. 2009;4(5):568–577.)

TABLE 30.2

Mediastinal Lymph Node Station Accessibility to Minimally Invasive Biopsy Modalities

Minimally Invasive Procedure	Accessible Lymph Node Stations
Mediastinoscopy	2, 4, 7
Anterior mediastinotomy (Chamberlain procedure)	3, 5, 6
Extended cervical mediastinotomy	5, 6
Bronchoscopy with endobronchial ultrasound (EBUS)	2, 4, 7, 10, 11
Endoscopy with endoscopic ultrasound (EUS)	4, 5, 7, 8, 9
Transthoracic needle biopsy	Most stations
Video-assisted Thoracoscopic Surgery (VATS)	Ipsilateral 2, 3, 4, 7, 8, 9, 10, 11 5, 6 with left-sided VATS

and Table 30.2).[36] Most studies report high accuracy of all procedures, in part attributable to the high prevalence of malignancy in study cohorts. Surgical procedures for mediastinal staging include standard cervical mediastinoscopy, extended cervical mediastinoscopy, parasternal mediastinotomy, and video-assisted thoracoscopic surgery (VATS). In a systematic literature review of the accuracy of standard cervical mediastinoscopy in 9,267 patients with lung cancer, the median sensitivity was 78% and the median negative predictive value (NPV) was 91%.[20] Sensitivity and NPV were somewhat better with video mediastinoscopy, at 89% and 92%, respectively. Achieving this level of accuracy assumes a thorough examination, with routine sampling of at least one node from stations 2R, 2L, 4R, 4L, and 7. Standard cervical mediastinoscopy is not able to access station 5 (subaortic) or station 6 (para-aortic) nodes, which are of particular concern in patients with left upper lobe cancers. Traditionally, the aortopulmonary window nodes have been invasively assessed by anterior mediastinotomy (Chamberlain procedure), with median sensitivity and NPV in patients with left upper lobe lung cancers reported at 71% and 91%, respectively.[20] Extended cervical mediastinoscopy has similar accuracy. Left-sided VATS provides access to the aortopulmonary window nodes, with sensitivity and NPV of 99% and 96%, respectively.[20] VATS also offers the opportunity to inspect and sample other sites in the ipsilateral thorax including pleural effusion and the pleural surface, the primary site, or other pulmonary lesions but is limited in that it can only access ipsilateral mediastinal nodes.

Needle techniques for mediastinal staging include transbronchial needle aspiration (TBNA), bronchoscopy with endobronchial ultrasound (EBUS), endoscopic ultrasound (EUS), and transthoracic needle biopsy (TTNB). Needle techniques are less invasive than surgical techniques, are increasingly available, and have become the procedure of first choice for patients with discrete mediastinal lymph node enlargement by CT or FDG uptake on PET. In considering the performance characteristics of these modalities, it is important to consider whether a limited procedure is performed for diagnostic purposes or as a comprehensive approach to achieve mediastinal staging. As is the case with surgical techniques, in the latter situation the accuracy of the procedure will depend on thoroughness of sampling of multiple nodal stations.

TBNA can be done with or without ultrasound. TBNA without ultrasound is typically performed to sample enlarged subcarinal nodes. In this case, the goal of the procedure is diagnostic, because although information regarding the nodal stage may be obtained, sampling of a single nodal station would be an incomplete staging of the mediastinum. EBUS has increasingly replaced unguided TBNA. In a pooled analysis evaluating the accuracy of EBUS in 2,756 patients undergoing mediastinal staging, overall median sensitivity was 89% with an NPV of 91%, with confirmation of actual nodal status achieved by mediastinoscopy, surgery, or radiographic follow-up.[20] As might be expected, sensitivity was highest when prevalence of disease was high (96% with median prevalence of disease >80%) and lowest when prevalence of disease was low (78% with median prevalence of disease <20%). Sensitivity was 91% in patients with radiographic involvement of hilar, mediastinal, or supraclavicular nodes, and was also high (89%) in patients without any nodal enlargement. EBUS presents an advantage over mediastinoscopy in its ability to concurrently sample the primary tumor as well as hilar nodes. Among the needle techniques, EUS has the unique ability to access lymph node stations 5, 8 and 9. The combination of EBUS and EUS offers the most comprehensive access to all mediastinal nodes except station 6.

There is ongoing debate as to whether surgical staging is necessary to confirm a negative EBUS in the setting of a radiographically normal mediastinum. This discussion hinges on the concern for noncurative surgery because of unidentified mediastinal lymph node involvement, as well as consideration of complications of a surgical staging intervention that is likely to be negative. In a meta-analysis of 13 studies evaluating EBUS for nodal staging in such patients, the overall probability of occult mediastinal metastasis with a negative EBUS was 7%.[37] The pooled prevalence of mediastinal lymph node involvement was 12.8%, with sensitivity and NPV of EBUS reported at 49.5% (range, 25% to 94.1%) and 93.0%, respectively. These test characteristics are similar to those reported for mediastinoscopy in the setting of a radiographically normal mediastinum, with median sensitivity and NPV of 47% and 91%, respectively, in the setting of a 16% pooled prevalence of mediastinal lymph node involvement.[20]

CT-guided TTNB can in theory sample all mediastinal nodal stations. TTNB would likely be performed only in patients with radiographically abnormal nodes, with sensitivity reported at 94%.[20] The purpose of TTNB would be predominantly diagnostic, as a thorough staging evaluation requiring sampling of multiple nodal stations by TTNB is impractical.

To put the invasive mediastinal staging procedures in a working perspective, the American College of Chest Physicians (ACCP) lung cancer guidelines provide a useful framework of four mediastinal CT patterns when considering invasive mediastinal staging, including mediastinal infiltration by the tumor, enlarged discrete mediastinal lymph nodes, a central tumor or tumor with enlarged N1 station lymph nodes, and a peripheral small tumor with normal-size lymph nodes (Fig. 30.2).[20] Direct mediastinal infiltration by the primary tumor may prevent delineation or measurement of discrete mediastinal nodes. In this case, tumor involvement of the adjacent nodes would be reasonably assumed. In case of discrete mediastinal node enlargement, defined by a shortest axis diameter >10 mm, the false-positive rates of both CT and PET are high. If staging depends on the status of the mediastinal nodes, they should be biopsied, preferably through a needle technique such as EBUS, EUS or combined EBUS/EUS.[20] An RCT of 241 patients with radiographic concern for mediastinal nodal involvement compared surgical staging to EBUS/EUS followed by surgical staging if EBUS/EUS were negative.[38] The sensitivities of surgery alone, endoscopic techniques alone, and endoscopic techniques followed by surgery were 79%, 85%, and 94%, respectively. This supports the recommendations that surgical staging be pursued if suspicion for mediastinal node involvement remains high after a negative needle technique.[20] When the mediastinal nodes are not enlarged but other radiographic findings such as a central tumor, large primary tumor, or enlarged hilar nodes suggest a high likelihood of mediastinal involvement, neither CT or PET is sufficiently reliable and invasive evaluation of the mediastinal nodes should be performed. In this case, a needle technique (EBUS, EUS, or combined EBUS/EUS) is recommended as the best first test, with subsequent surgical staging if the needle technique is negative and suspicion for

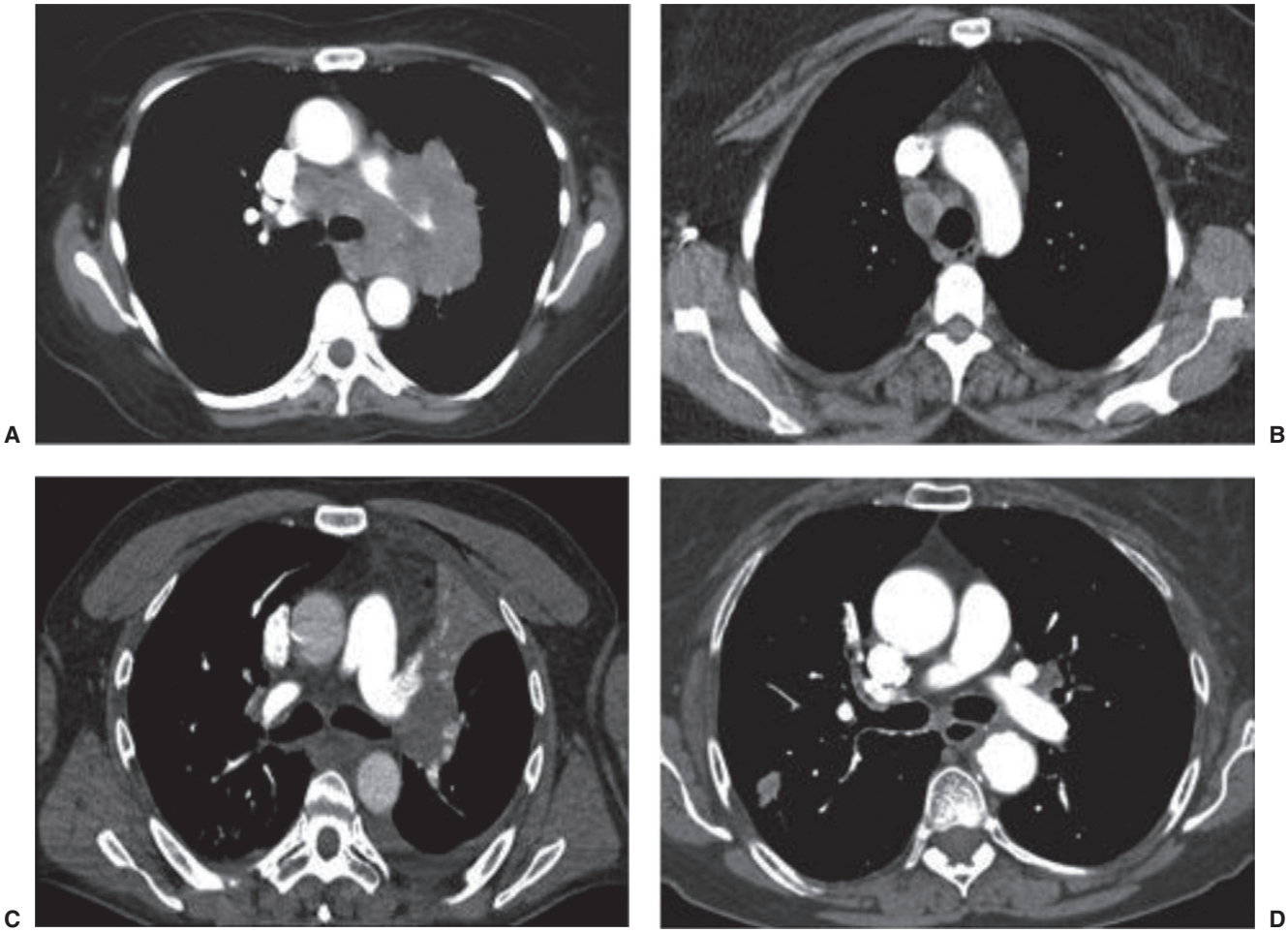


Figure 30.2 American College of Chest Physicians intrathoracic radiographic (CT scan) categories of lung cancer. **A:** Mediastinal infiltration by tumor. **B:** Enlarged discrete N2,3 nodes. **C:** A central tumor or a tumor with enlarged N1 nodes but a normal mediastinum. **D:** A peripheral small tumor (seen in the lower left corner of the image) with normal-sized lymph nodes. (Reproduced with permission from Silvestri GA, Gonzalez AV, Jantz MA, et al. Methods for staging non-small cell lung cancer: Diagnosis and management of lung cancer, 3rd ed: American College of Chest Physicians evidence-based clinical practice guidelines. *Chest*. 2013;143(5 Suppl):e211S–250S.)

mediastinal node involvement is high.[20] For patients with a small, peripheral primary tumor with normal mediastinum and no other radiographic abnormalities, the likelihood of mediastinal or distant disease involvement is very low, and pre-resection invasive mediastinal evaluation is typically not necessary.[25]

Stage Classification

The International Association for the Study of Lung Cancer (IASLC) recommends that both NSCLC and SCLC should be staged according to the tumor, node, metastasis (TNM) system as detailed in the current eighth edition staging system of the American Joint Committee on Cancer (AJCC) and the Union for International Cancer Control (UICC).[39] Staging is used to describe a malignancy through the course of treatment and evaluation. Clinical stage is determined by all available imaging and biopsy data. All patients should have a clinical stage assessment at presentation before treatment (indicated by the prefix "c"). Pathologic stage (indicated by the prefix "p") includes all information from clinical staging as well as analysis of tissues obtained from an intent-to-cure surgical resection. Pathologic stage is inherently more accurate than clinical stage, and the clinical stage generally is associated with worse outcomes than the pathologic stage.[39] However, only selected early-stage patients will achieve pathologic staging. Through the course of disease, restaging may be performed after treatment, at the time of a recurrence, or at autopsy indicated by the prefixes "y," "r," or "a," respectively.

In the TNM system, the T component describes the primary tumor, the N component describes the regional lymph nodes, and the M component describes the presence or absence of distant metastases (Table 30.3).[40] Each TNM combination is assigned to a stage. Stages are distinguished by significant differences in survival. The commonality of TNM groups within a given stage is that they share similar survival. The limitations of this system are immediately clear, as patients with dissimilar anatomic patterns but with similar survival will be grouped together. Furthermore, anatomic staging provides no information about biologic characteristics such as histology, targetable genetic mutations, expression of tumor antigens, or volume doubling time. Nonetheless, the staging system remains a cornerstone of lung cancer evaluation, as it establishes a reliable language for communication between practitioners and patients, provides important prognostic information, and identifies groups with similar outcomes for the purposes of clinical trials.

T Component

Although size is the predominant factor, additional T designation features include invasion into adjacent structures or additional tumor nodules in a lobe ipsilateral to the primary site. Several nuances of the T component should be noted.[41] If multiple T descriptors apply to a single tumor, the highest T descriptor should

TABLE 30.3

Lung Cancer Staging Definitions

T (Primary Tumor)		Label
T0	No primary tumor	
Tis	Carcinoma in situ (squamous or adenocarcinoma)	Tis
T1	Tumor ≤3 cm	
T1a(mi)	Minimally invasive adenocarcinoma	T1a*(mi)*
T1a	Superficial spreading tumor in central airways[a]	T1a *SS*
T1a	Tumor ≤1 cm	T1a *≤1*
T1b	Tumor >1 but ≤2 cm	T1b *>1–2*
T1c	Tumor >2 but ≤3 cm	T1c *>2–3*
T2	Tumor >3 but ≤5 cm or tumor involving:	
T2a	Visceral pleural[b]	T2 *Visc Pl*
T2b	Main bronchus (not carina), atelectasis to hilum[b]	T2 *Centr*
	Tumor >3 but ≤4 cm	T2a *>3–4*
	Tumor >4 but ≤5 cm	T2b *>4–5*
T3	Tumor >5 but ≤7 cm	T3 *>5–7*
	or invading chest wall, pericardium, phrenic nerve, or	T3 *Inv*
	separate tumor nodule(s) in the same lobe	T3 *Satell*
T4	Tumor >7 cm	T4 *>7*
	Or tumor invading: mediastinum, diaphragm, heart, great vessels, recurrent laryngeal nerve, carina, trachea, esophagus, spine; or tumor nodule(s) in a different ipsilateral lobe	T4 *Inv* T4 *Ipsi Nod*
N (Regional Lymph Nodes)		
N0	No regional node metastasis	
N1	Metastasis in ipsilateral pulmonary or hilar nodes	
N2	Metastasis in ipsilateral mediastinal/subcarinal nodes	
N3	Metastasis in contralateral mediastinal/hilar or supraclavicular nodes	
M (Distant Metastasis)		
	No distant metastasis	
M0	Malignant pleural/pericardial effusion[c] or pleural/pericardial nodules	M1a *Pl Dissem*
M1a	Or separate tumor nodule(s) in a contralateral lobe;	M1a *Contr Nod*
M1b	Single extrathoracic metastasis	M1b *Single*
M1c	Multiple extrathoracic metastases (1 or >1 organ)	M1c *Multi*
TX, NX: T or N status not able to be assessed.		

Reproduced with permission from Detterbeck FC, Boffa DJ, Kim AW, Tanoue LT. *The Eighth Edition Lung Cancer Stage Classification. Chest.* 2017;151(1):193–203.

be used. Minimally invasive adenocarcinoma (MIA; T1a[mi]) can only be defined in resected tumors, with a pathologic invasive component <5 mm and a lepidic component <3 cm.[42] The CT report should include both the long and short dimensions of the primary site that yield the greatest average dimension.[43] For solid and pure ground glass nodules, the long axis dimension should be used for the purposes of staging. For subsolid nodules, the long axis dimension of only the solid component should be used for the purposes of staging, as this is associated with a higher likelihood of local invasion.[44,45]

N Component

There are four N categories describe tumor status of the thoracic lymph nodes, in accordance with the IASLC lymph node staging map.[36] N0 indicates no regional lymph node involvement, N1 refers to tumor involvement of ipsilateral intrapulmonary or hilar nodes, N2 refers to involvement of ipsilateral mediastinal or subcarinal nodes, and N3 indicates involvement of contralateral mediastinal or hilar nodes or involvement of ipsilateral or contralateral supraclavicular nodes. Accurate staging of the mediastinal nodes is critical, as it is the most common factor distinguishing stage II from stage III lung cancer and for the determination of whether surgery will be the primary treatment approach.

M Component

The four M categories describe distant (metastatic) spread of tumor.[46] M0 indicates no evidence of distant metastasis. M1a refers to metastasis within the thorax, including malignant pleural or pericardial effusion or nodules, or the presence of tumor nodule(s) in the contralateral lung. M1b refers to a solitary extrathoracic metastatic focus. M1c describes multiple metastases in a single organ or metastases in multiple organs. The definition of oligometastatic disease in clinical trials and practice has varied widely, from the stringent definition of one metastatic site in one organ to a limited number of separate metastases in one or more organs for which local ablative therapy (surgery or radiation) of all lesions is feasible.

Stage Grouping

There are 64 possible combinations for lung cancer staging based on T, N and M variables (Table 30.4). Each stage, from IA through IVB, has a statistically different median survival. One important nuance relates to prediction of clinical outcomes based on stage. Although staging is inherently based on survival, multiple factors that are not related to stage influence the prognosis of an individual patient. These include patient-related factors such as comorbid medical conditions, performance status, and access to care; tumor-related factors including histology and molecular characteristics; and treatment-related factors including quality of care, treatment toxicity, and treatment response.

PATHOLOGY

Role and Principles of Histopathology and Molecular Diagnosis of Lung Cancer

Despite marked progress in understanding the biologic and molecular basis of lung cancer, the microscopic analysis of tumor samples remains the backbone of diagnosis and patient management. Histopathology is fast, inexpensive, and widely available, and provides information about critical biologic aspects, disease course expectations, prognosis, and treatment selection. The histologic diagnosis and pathologic staging are also used in practice to critically guide decisions on ancillary molecular testing for identification of disease features and predictive biomarkers. This is notorious in cases with diagnosis of lung adenocarcinoma or mixed tumors with an adenocarcinoma component where comprehensive molecular testing is warranted.[47] In contrast, other common histologic types of lung cancer such as squamous cell carcinoma and SCLC often lack actionable alterations and are therefore not systematically profiled molecularly.[47–49] Furthermore, there are specific histologic

TABLE 30.4
Lung Cancer Stage Grouping (Eighth Edition)

T/M	Label	N0	N1	N2	N3
T1	T1a ≤*1*	1A1	IIB	IIIA	IIIB
	T1b >*1–2*	1A2	IIB	IIIA	IIIB
	T1c >*2–3*	1A3	IIB	IIIA	IIIB
T2	T2a *Cent*, Visceral pleura	IB	IIB	IIIA	IIIB
	T2a >*3–4*	IB	IIB	IIIA	IIIB
	T2b >*4–5*	IIA	IIB	IIIA	IIIB
T3	T3 >*5–7*	IIB	IIIA	IIIB	IIIC
	T3 *Inv*	IIB	IIIA	IIIB	IIIC
	T3 *Satell*	IIB	IIIA	IIIB	IIIC
T4	T4 >*7*	IIIA	IIIA	IIIB	IIIC
	T4 *Inv*	IIIA	IIIA	IIIB	IIIC
	T4 *Ipsi Nod*	IIIA	IIIA	IIIB	IIIC
M1	M1a *Contr Nod*	IVA	IVA	IVA	IVA
	M1a *Pl Dissem*	IVA	IVA	IVA	IVA
	M1b *Single*	IVA	IVA	IVA	IVA
	M1c *Multi*	IVB	IVB	IVB	IVB

Reproduced with permission, from Detterbeck FC, Boffa DJ, Kim AW, Tanoue LT. *The Eighth Edition Lung Cancer Stage Classification. Chest.* 2017;151(1):193–203. Inv, invasion; Satell, satellite; Ipsi, ipsilateral; Nod, nodule; contr, contralateral; Dissem, dissemination.

subtypes prominently associated with distinct molecular features whose identification can only be guided by microscopic diagnosis.

The capacity to perform localized and spatially resolved identification of proteins using immunohistochemistry (IHC) or of nucleic acids using in situ hybridization on tumor sections is of strong complementary value and is frequently used to assess tumor-cell differentiation and detection of prognostic/predictive biomarkers. Histopathology and tumor-based molecular diagnosis are generally well integrated and genomic findings have inherently more value when interpreted in the context of morphology or specific sample characteristics such as sample type, fixation method, tumor location, type, and tumor-cell content. In general, oncogenic molecular alterations can overlap between early and advanced malignant lesions or across major histologic subsets.[50,51] Therefore, most lung cancer subtypes lack molecular alterations that can be used to support or confirm the diagnosis. Specific tumor types such as lymphoepithelial carcinoma, NUT carcinoma, and SMARCA4-deficient undifferentiated tumor are notable exceptions. Molecular analysis is more commonly used clinically for identification of actionable alterations associated with treatment-specific outcomes such as targeted therapies and immunostimulatory agents. The increasing use of circulating tumor DNA (ctDNA) analysis for noninvasive detection of molecular abnormalities in patients with lung cancer is also of high complementary value.

Limitations of the histopathologic analysis of lung cancer samples include the subjective nature of the microscopic interpretation that can affect the reproducibility of diagnosis.[52,53] In this regard, it is relevant to understand that histopathology patterns and diagnostic criteria reflect the extremes of a continuum that can vary substantially between patients, lesions, or sometimes even within a single tumor. Therefore, the salient characteristics present in the specimen are included in the reporting, but deviation from such patterns is common and expected. Another possible shortcoming of histopathology analysis is its potential for underperformance in small samples with low tumor content and/or underrepresentation of relevant tumor features. This is particularly notorious in borderline lesions, highly heterogeneous tumors including multiple mixed and complex histology patterns, or after anticancer treatments, where the definitive diagnosis relies on an excisional tumor specimen. Integration of clinical, morphologic, radiologic, and molecular markers is often sufficient to avoid a negative impact in patient management. In addition, novel strategies of histopathology analysis using digital microscopic images coupled to advanced analytics and deep learning–powered classification methods are expected to help in overcoming some of these issues and are progressively being incorporated in the clinical setting.[52,53]

Histopathologic Classification of Lung Cancer

Lung carcinomas include primary malignancies expected to originate from transformed epithelial cells lining different parts of the airway, and they are classified based on their cellular and architectural differentiation patterns under microscopic evaluation. Historically, primary lung malignancies have been subdivided into two major clinicopathologic groups: NSCLC (comprising adenocarcinomas, squamous cell carcinomas, and undifferentiated lung cell carcinomas) and SCLC.[54] Although this classification is conceptually useful and widely used in the literature, developments in the understanding of the biologic and clinical features of lung cancer subsets support the systematic use of more structured classifications both clinically and academically. Specifically, the progressive use of refined histologic criteria and molecular analysis tools has allowed to identify the differentiation in tumors previously recognized as "undifferentiated malignancies" or to establish novel tumor entities.[50] A widely used histologic classification is the *WHO Classification of Tumours*.[55] Its fifth edition was published recently and includes updated classification and terminology for both benign and malignant lung neoplasms (Table 30.5).[56] Additional changes in the classification of lung cancer are expected to occur in the future based on results from more detailed molecular studies, expansion of noninvasive diagnostic platforms, and incorporation of immune-related features. Based on their incidence in the population, the predominant histologic types of lung cancer are adenocarcinoma, squamous cell carcinoma, SCLC, and large-cell carcinoma.[56] However, there are other less common tumor types and subtypes that collectively account for a substantial fraction of lung cancers. These include preinvasive/precursor epithelial lesions, tumors originating in peribronchial salivary glands, primary nonepithelial malignancies such as sarcomas, lymphomas, neural proliferations, neoplasms with mixed or intermediate phenotypes/differentiation, and metastatic lesions. To achieve a comprehensive description of primary lung cancer subsets focusing on the salient histopathology and molecular features, they were organized into five general categories that partially overlap with the WHO classification including tumors with pneumocyte-like/glandular differentiation, squamous cell differentiation, poor and multilineage differentiation, neuroendocrine differentiation, salivary gland–type tumors, and nonepithelial lung malignancies.

Tumors with Pneumocyte-like/Glandular Differentiation

Adenocarcinomas are the most common form of lung cancer and comprise malignant lesions composed of transformed epithelial cells with pneumocyte-like differentiation and glandular type architectural growth patterns.[57] These lesions are expected to originate from type II pneumocytes or Clara cells in the bronchial epithelium and can be morphologically diverse.[58] Tumors with adenocarcinoma differentiation are subclassified based on the size of their invasive component and the dominant cytologic and architectural growth patterns. The pneumocyte-like differentiation, and absence of alternative differentiation, can also be recognized by the

TABLE 30.5

World Health Organization Classification of Malignant Lung Tumors (5th Edition)

Tumor Type	Tumor Subtype
1. Epithelial tumors	
1.1 Adenocarcinomas	1.1.1 Adenocarcinoma in situ of the lung
	1.1.2 Minimally invasive adenocarcinoma of the lung
	1.1.3 Invasive nonmucinous adenocarcinoma of the lung
	1.1.4 Invasive mucinous adenocarcinoma of the lung
	1.1.5 Colloid adenocarcinoma of the lung
	1.1.6 Fetal adenocarcinoma of the lung
	1.1.7 Enteric-type adenocarcinoma of the lung
1.2 Squamous cell carcinomas	1.2.1 Squamous carcinoma in situ of the lung
	1.2.2 Squamous cell carcinoma of the lung
	1.2.3 Lymphoepithelial carcinoma of the lung
1.3 Large cell carcinomas	1.3.1 Large cell carcinoma of the lung
1.4 Adenosquamous carcinoma	1.4.1 Adenosquamous carcinoma of the lung
1.5 Sarcomatoid carcinomas	1.5.1 Pleomorphic carcinoma of the lung
	1.5.2 Pulmonary blastoma
	1.5.3 Carcinosarcoma of the lung
1.6 Other epithelial tumors	1.6.1 NUT carcinoma of the lung
	1.6.2 Thoracic SMARCA4-deficient undifferentiated tumor
1.7 Salivary gland–type tumors	1.7.1 Pleomorphic adenoma of the lung
	1.7.2 Adenoid cystic carcinoma of the lung
	1.7.3 Epithelial-myoepithelial carcinoma of the lung
	1.7.4 Mucoepidermoid carcinoma of the lung
	1.7.5 Hyalinizing clear cell carcinoma of the lung
	1.7.6 Myoepithelioma and myoepithelial carcinoma of the lung
2. Lung neuroendocrine neoplasms	
2.1 Neuroendocrine tumors	2.1.1 Carcinoid/neuroendocrine tumor of the lung
2.2 Neuroendocrine carcinomas	2.1.2 Small-cell lung carcinoma
	2.1.3 Large-cell neuroendocrine carcinoma of the lung
3. Tumors of ectopic tissues	3.1 Melanoma of the lung
	3.2 Meningioma of the lung
4. Mesenchymal tumors specific to the lung	4.1 Pulmonary hamartoma
	4.2 Pulmonary chondroma
	4.3 Diffuse pulmonary lymphangiomatosis
	4.4 Pleuropulmonary blastoma
	4.5 Pulmonary artery intimal sarcoma
	4.6 Congenital peribronchial myofibroblastic tumor
	4.7 Primary pulmonary myxoid sarcoma with EWSR1-CREB1 fusion
5. PEComatous tumors	5.1 Lymphangioleiomyomatosis of the lung
	5.2 PEComa of the lung
6. Haematolymphoid tumors	6.1 MALT lymphoma of the lung
	6.2 Pulmonary diffuse large B-cell lymphoma
	6.3 Lymphomatoid granulomatosis of the lung
	6.4 Intravascular large B-cell lymphoma of the lung
	6.5 Pulmonary Langerhans cell histiocytosis
	6.6 Pulmonary Erdheim-Chester disease

Adapted with permission from WHO Classification of Tumours Editorial Board. Thoracic tumours [Internet]. Lyon (France): International Agency for Research on Cancer; 2021 (WHO classification of tumours series, 5th ed.; vol. 5). Available from: https://tumourclassification.iarc.who.int/chapters/35.

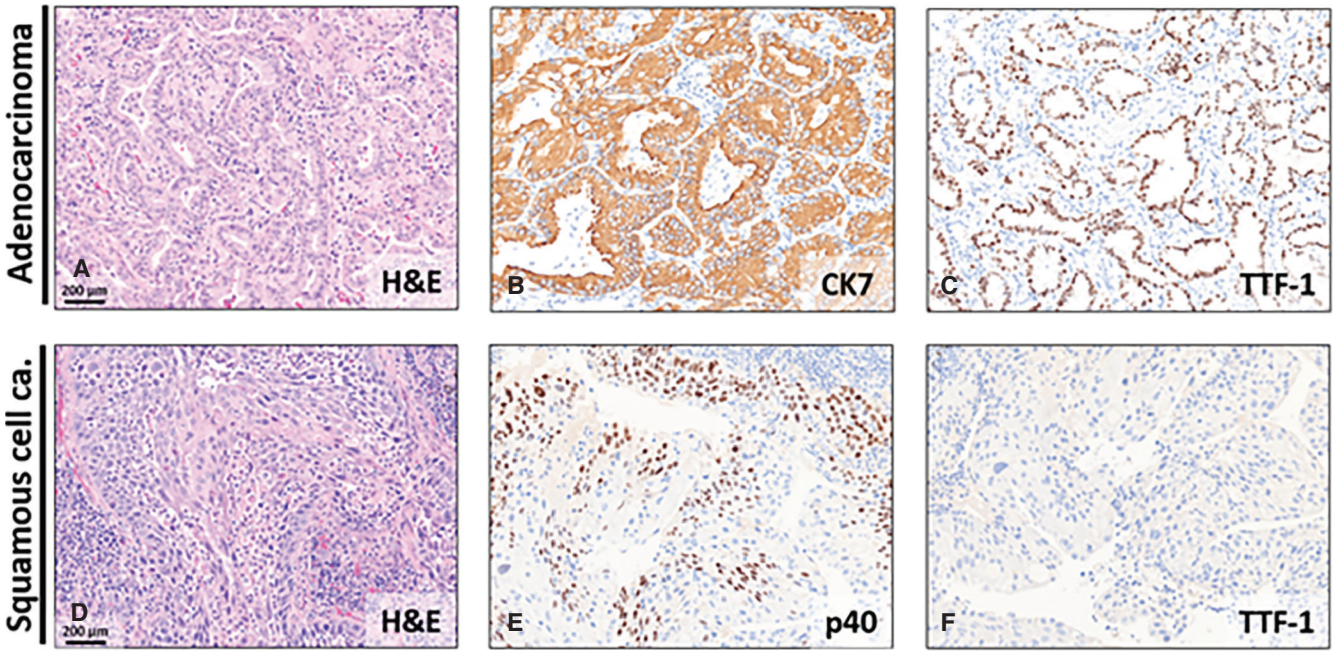


Figure 30.3 Histopathology and immunophenotype of major NSCLC subsets. **A–C:** Representative hematoxylin and eosin (H&E) stained preparation of a lung adenocarcinoma with glandular differentiation and acinar architectural growth pattern. The malignant cells show cytoplasmic immunoreactivity for CK7 **(B)** and nuclear positivity for TTF-1 **(C)**. **D–F:** Squamous cell carcinoma characterized by malignant epithelial cells with ample cytoplasm, nuclear atypia, intercellular bridges, and a solid growth pattern. Tumor cells express nuclear p40 **(E)** and lack TTF-1 immunoreactivity **(F)**.

expression of specific protein markers using IHC such as TTF-1 and Napsin-A (Fig. 30.3A–C).[56] Small, localized adenocarcinomas with lepidic growth pattern and lacking evidence of invasion or with a limited invasive component have excellent outcomes after complete resection. These lesions are assumed to be precursors to more advanced, extensively infiltrating, or disseminated tumors and are currently classified as adenocarcinoma in situ (AIC) and MIA.[56] Larger adenocarcinomas with more extensive infiltrating component are subclassified based on the tumor-cell morphology into nonmucinous and mucinous subtypes as well as by the presence of dominant histologic architectural growth patterns, which include lepidic, acinar, papillary, micropapillary, cribriform, or solid components. Lung adenocarcinomas display a higher frequency of actionable oncogenic driver mutations than nonadenocarcinomas including activating mutations or rearrangements in *EGFR*, *KRAS*, *ALK*, *ROS1*, *RET*, *MET*, *NTRK1-3*, *BRAF*, and *HER2*.[59,60] Therefore, these tumors are systematically evaluated for the presence of genomic variants, more commonly using comprehensive molecular panels. The expression of programmed cell death ligand 1 (PD-L1) protein using IHC is also evaluated to support treatment with immune checkpoint inhibitors (ICIs).

Adenocarcinoma In Situ: AIS is a preinvasive lung adenocarcinoma that usually presents as an incidental finding in imaging studies or screening. They can only be diagnosed in a resection specimen where the whole lesion has been sampled. Macroscopically, they are poorly defined, localized lesions ≤30 mm in size, with a tan or pale cut surface.[56] Microscopically, they consist of an epithelial lesion with lepidic growth pattern formed by continuous monolayers of nonmucinous neoplastic cells present along preexisting alveolar structures and with minimal nuclear atypia. They lack evidence of invasive features, necrosis, and papillary or micropapillary patterns, or presence of intra-alveolar tumor cells.[56] Mucinous AIS is uncommon, and these lesions are staged as TisN0M0 or stage 0 disease.[61,62]

Minimally Invasive Adenocarcinoma: By definition, MIA is a single nodular lesion on pathology examination and imaging studies. Like AIS, it is usually an incidental finding and can only be diagnosed in resection specimens where the whole lesion has been sampled. Macroscopically, they are small, localized, usually peripheral nodules and can contain a small, central, solid area.[56] Microscopically, MIA is a solitary epithelial lesion ≤30 mm in size, with a predominantly lepidic growth pattern and ≤5-mm invasion.[63] The invasive component can have diverse nonlepidic growth patterns and/or stromal infiltration. However, there must be no evidence of invasion to lymphatics, blood vessels, or pleura; necrosis; or tumor spread through airspaces. Mucinous differentiation is rare, and this lesion is staged as T1mi.[61]

Invasive Nonmucinous Adenocarcinoma: Invasive adenocarcinoma presents more commonly as solid nodules with associated nonsolid areas. Macroscopically, they are typically gray-white, poorly delimited nodules with central scarring fibrosis and can show anthracotic pigment deposition and focal pleural thickening.[56] Microscopically, they are malignant epithelial lesions with variable cytologic atypia, glandular-like differentiation by architecture, and one or more growth patterns: lepidic, acinar, papillary, micropapillary, cribriform, or solid.[44] Tumor cells show expression of pneumocyte differentiation markers on IHC and/or display histochemical demonstration of intracytoplasmic mucin. The tumors are classified based on their predominant histology patterns using 5% to 10% increments.[56] The solid and micropapillary patterns are associated with worse prognosis.[64]

Less common subtypes of adenocarcinoma include invasive mucinous, colloid, fetal, and enteric types. Invasive mucinous adenocarcinoma is less common than nonmucinous adenocarcinoma and can show multilobar and bilateral lung involvement. Tumors produce pulmonary consolidation and can be misdiagnosed as pneumonia during early stages. Macroscopically, they display poorly circumscribed lesions with a soft, gelatinous, or mucoid appearance.[56] Similar to nonmucinous adenocarcinoma, different invasive growth patterns can be identified. Tumors with a mixture of mucinous and nonmucinous components should be classified as mixed invasive adenocarcinoma if there is ≥10% of each component.[56] Mucinous adenocarcinomas show a distinct immunophenotype with positivity

for CK7, focal expression of CK20 or CDX2, and absence of pneumocyte differentiation markers. There is currently no established histologic grading system. Mucinous carcinomas have a higher frequency of mutations in *KRAS* and *ALK* rearrangements, and lower frequency of EGFR mutations than nonmucinous adenocarcinomas.[65,66] Colloid adenocarcinoma is a rare invasive adenocarcinoma with mucinous features characterized by extensive pools of extracellular mucin and effacement of underlying lung architecture. When present and associated with other histologic adenocarcinoma components, ≥50% of the colloid histologic pattern is required for its diagnosis.[56] The immunophenotype includes positivity for CK7; focal expression of CDX2, MUC2, and CK20; and negativity for pneumocyte differentiation markers. Nearly half of colloid lung adenocarcinomas harbor activating mutations in *KRAS*.[67] The histologic differential diagnosis includes metastases of mucinous carcinomas from extrapulmonary sites. The presence of signet-ring cells and a noncolloid component is associated with a worse prognosis. Fetal adenocarcinoma is an infrequent pulmonary adenocarcinoma resembling developing fetal lung in its pseudoglandular stage. Low-grade fetal adenocarcinoma occurs in younger individuals, and high-grade fetal adenocarcinoma is more common in elderly men who are heavy smokers.[68] Low-grade fetal adenocarcinomas are often discovered incidentally and are small, peripherally located tumors that stain positive for TTF-1, whereas high-grade fetal adenocarcinomas can be negative.[56] The differentiation between low-grade fetal adenocarcinoma and high-grade fetal adenocarcinoma is essential because of the marked difference in prognosis. Enteric-type adenocarcinoma is an extremely rare ADC resembling colorectal malignancies both morphologically and immunophenotypically. The immunophenotype includes expression of intestinal markers such as CDX2, villin, and HNF4α.[69] CK7 expression helps differentiate from colorectal adenocarcinoma, and TTF-1 can be absent. *KRAS* mutations are more frequent than in nonmucinous adenocarcinoma. The histologic differential diagnosis is a metastatic colorectal carcinoma, which is more common than primary enteric-type pulmonary adenocarcinoma.

Tumors with Squamous Cell Differentiation

Squamous cell carcinoma is the second most common form of lung cancer, arises from main or lobar bronchi, is typically located centrally, and comprises epithelial tumors with differentiation resembling squamous cells.[70] These tumors are strongly associated with tobacco consumption and span a spectrum of subtypes ranging from early-stage noninvasive precursors to high-grade poorly differentiated spindle-cell malignancies. They are characterized by common histologic features such as the presence of keratinization, intercellular bridges, and immunohistochemical positivity for markers of squamous cell differentiation such as p40 and p63 (see Fig. 30.3D–F).[56] Molecular analyses have identified multiple genomic alterations, and mRNA studies have been used to define transcriptomic profiles with prognostic significance.[71] Although actionable mutations in genes such as *EGFR* and *ALK* can occur in squamous cell carcinomas, most commonly in nonsmokers, their frequency is low and they are not always molecularly profiled. PD-L1 expression by IHC is usually evaluated to support treatment with immune checkpoint blockers.

In Situ Squamous Cell Carcinoma: In situ squamous cell carcinoma is an atypical squamous lesion lacking evidence of microscopic invasion and considered as a precursor lesion and usually diagnosed incidentally on bronchoscopy. Macroscopically, lesions are typically flat or nodular with mucosal thickening, increased vascularity, or marked surface irregularity.[56] Microscopically, there is usually a background of squamous metaplasia accompanied by areas with marked dysplastic cells characterized by defects in maturation, nuclear atypia, changes in cell orientation, and increased epithelial thickness. There is no evidence of tumor infiltration into the stroma, vascular structures, or peribronchial glands or necrosis.[56] The magnitude of histologic atypia and transformed cell layers are used to distinguish between in situ carcinoma from high-grade dysplasia. The immunophenotype is consistent with squamous differentiation with positivity for p40, p63, CK5/6, and high Ki-67. The distinction between in situ and invasive squamous cell carcinoma may sometimes be difficult on small biopsies.

Squamous Cell Carcinoma: Squamous cell carcinomas are invasive epithelial tumors with features of squamous cell differentiation. Macroscopically, they are most commonly irregular nodules with firm white, light-brown, or gray cut surface. In large lesions, central cavitation due to necrosis may occur. Microscopically, well-differentiated squamous cell carcinomas have distinctive morphologic features such as intercellular bridges and keratinization. Nonkeratinizing and poorly differentiated lesions may not display overt morphologic features and can be recognized by immunohistochemical evidence of squamous cell differentiation. The basaloid subtype shows prominent palisading of nuclei around the perimeter of cell groups in ≥50% of the lesion.[56] Less common histologic patterns include spindle-cell morphology; clear-cell cytologic changes; and papillary, pseudovascular, or alveolar-filling patterns. The immunophenotype of squamous cell carcinomas includes positivity for p40, p63, and high-molecular-weight keratins (CK5/6 and 34βE12), and absence of pneumocyte differentiation markers.[56] Lymphoepithelial carcinoma is a rare type of poorly differentiated squamous cell carcinoma admixed with variable amounts of lymphoplasmacytic infiltrate and association with Epstein-Barr virus.[72] The immunophenotype shows diffuse staining for pancytokeratin, p40, p63, and CK5/6, whereas the accompanying lymphoid infiltrate comprises a mixture of CD3-positive T cells and CD20-positive B cells.[56]

Adenosquamous carcinoma is an aggressive type of lung carcinoma showing components of both adenocarcinoma and squamous cell carcinoma. The two histologic components share the same driver gene mutations, suggesting an identical clonal origin rather than coexistence of two independent lesions. Macroscopically, they appear as gray-tan irregular masses arising either in the periphery or centrally.[56] Microscopically, these tumors are composed of a mixture of variable proportions of each histologic component, with each accounting for at least 10% of the lesion.[56] Definitive diagnosis and assessment of the proportion of histologic components requires a resection specimen with extensive sampling. The immunophenotype shows differentiation features consistent with each of the histologic patterns. These tumors can harbor actionable mutations present in lung adenocarcinomas, supporting the use of comprehensive molecular testing.

Tumors with Poor and Multilineage Differentiation

This group includes infrequent, high-grade invasive lung carcinomas, such as large-cell carcinoma, lacking a clear precursor lesion and with absence of features shown by more common histologic variants. In general, these tumors have complex morphologic features, and their incidence has decreased over time probably due to developments of more sensitive methods to determine the tumor-cell differentiation. In support of this notion, many large-cell carcinomas have been reclassified as adenocarcinomas or squamous cell carcinomas after careful phenotypic and molecular evaluation.[50,56] In addition, poorly differentiated lung carcinoma subsets have emerged from detailed molecular studies as novel tumor entities, and they can be recognized based on specific molecular alterations such as NUT carcinoma and thoracic SMARCA4-deficient undifferentiated tumor.[56] Additional tumors in this category include lesions with multilineage differentiation and biphasic patterns comprising concurrent epithelial and mesenchymal malignant elements such as sarcomatoid carcinomas. These tumors can display a poorly differentiated carcinoma component associated with one or more atypical sarcomatous differentiation elements in variable proportion. The relative frequency of these

group of tumors is low, and their diagnosis requires the exclusion of other forms of lung cancer and metastatic lesions. Therefore, they require the assessment of resection specimens with extensive sampling and use of ancillary studies to rule out or confirm the specific diagnosis. The clinical profile and risk factors are partially overlapping with other high-grade invasive lung carcinomas, but the prognosis is generally inferior. Actionable molecular alterations seen in adenocarcinomas can be present with variable frequency; therefore, comprehensive molecular profiling is recommended.

Large-cell carcinoma is an undifferentiated lung carcinoma lacking the cytologic, architectural, immunohistochemical, and histochemical features of adenocarcinoma, squamous cell carcinoma, and neuroendocrine and sarcomatous lung carcinoma. Macroscopically, they are usually large, circumscribed, solid masses with necrotic areas. Microscopically, they are formed by sheets or nests of large polygonal cells with vesicular nuclei, prominent nucleoli, and moderate amounts of cytoplasm. The immunophenotype shows positivity for cytokeratin and other epithelial markers but absence of markers of adenocarcinoma, squamous cell, and neuroendocrine differentiation.[56] The definitive diagnosis requires a thoroughly sampled resected tumor and ancillary markers, and cannot be made on nonsurgical biopsy or cytology specimens. Because it is a diagnosis of exclusion, the differential histologic diagnosis includes nearly all poorly differentiated lung carcinomas, primary nonepithelial tumors, metastatic lesions, and mesothelioma.

Poorly differentiated carcinomas with special features include rare, aggressive carcinomas with high-grade morphologic features and specific genomic alterations such as NUT carcinoma and thoracic SMARCA4-deficient undifferentiated tumor. These lesions have emerged from molecular studies of malignancies previously labeled as poorly differentiated carcinoma, and their recognition is challenging. NUT carcinoma of the thorax is an extremely rare, poorly differentiated, high-grade carcinoma most commonly arising in the mediastinum or thymus and genetically defined by the presence of NUTM1 gene rearrangement.[73] They can affect patients of any age, lack clear association with environmental factors, and are typically diagnosed in an advanced stage with invasion of local structures, pleura, and distant metastases. A characteristic feature is the lack of clear glandular differentiation and even spacing of cells reminiscent of squamous differentiation. The immunophenotype shows positivity for keratins, squamous cell markers, NUT protein, and CD34 in most cases. The identification of NUTM1 gene rearrangement is required for diagnosis and can be revealed by positive NUT immunostaining in >50% of nuclei or direct genomic analysis.[56] Thoracic SMARCA4-deficient undifferentiated tumors are high-grade malignant neoplasms of adults involving the mediastinum, pulmonary hilum, lung, or pleura with prominent association with smoking, male predominance, and biallelic somatic inactivation of the *SMARCA4* gene.[74] The immunophenotype shows complete loss or marked reduction of SMARCA4 staining. Positivity for CD34, TP53, SOX2, and SALL4 is common.[56] DNA sequencing can confirm the presence of *SMARCA4* mutation, but it is not necessary for the diagnosis.

Sarcomatoid carcinomas comprise an array of uncommon and poorly differentiated, high-grade invasive carcinomas with mesenchymal features. The main subtypes include pleomorphic carcinoma, pulmonary blastoma, and lung carcinosarcoma. Pleomorphic carcinomas are poorly differentiated, well-circumscribed tumors more commonly seen in smokers. Microscopically, they show epithelial tumor cells with any type of differentiation and an associated component ≥10% of malignant spindle and/or giant cells.[75] The immunophenotype shows positivity for keratins in the epithelial components with variable expression of epithelial markers in the spindle- or giant-cell areas.[56] Pulmonary blastomas are rare, typically large, well-delimited, biphasic, high-grade invasive lung tumors consisting of components of fetal adenocarcinoma and primitive mesenchymal stroma. The immunophenotype shows an epithelial component positive for keratins, EMA, and TTF-1.[76] The mesenchymal component expresses vimentin and tissue-specific antigens in the specific sarcoma foci. These tumors frequently harbor deleterious somatic mutations in *TP53*, *DICER1*, and *CTNNB1*, the latter leading to activation of the WNT pathway through aberrant nuclear localization of β-catenin protein. Pulmonary carcinosarcomas are rare, high-grade malignant tumors consisting of a malignant NSCLC component combined with at least one malignant sarcomatous heterologous element such as rhabdomyosarcoma, chondrosarcoma, or osteosarcoma. Comprehensive molecular profiling is recommended because actionable oncogenic alterations can occur. Differential histologic diagnosis includes other biphasic lung tumors, metastatic lesions, and mesothelioma.

Tumors with Neuroendocrine Differentiation

These tumors span a range of entities with dissimilar grade, morphologic features, molecular characteristics, and clinical behavior comprising low-grade typical carcinoid, intermediate-grade atypical carcinoid, and high-grade neuroendocrine carcinomas (large-cell neuroendocrine carcinoma and SCLC). Poorly differentiated lesions occur predominantly in older patients and show a strong association with smoking. A relatively small fraction of patients treated with EGFR inhibitors experience recurrence with SCLC histology, suggesting neuroendocrine differentiation as an acquired resistance mechanism.[77] Most primary lung neuroendocrine tumors are high-grade, poorly differentiated lesions, with the majority being SCLC, followed by large-cell neuroendocrine carcinomas. Carcinoids occur in younger patients, have lower association with tobacco consumption, and account for only a small proportion of all cases. The immunophenotype of these malignancies shows expression of neuroendocrine markers such as chromogranin-A, synaptophysin, and CD56.[56]

Salivary Gland–type Tumors

These malignancies include the spectrum of primary carcinomas more commonly seen in the salivary glands such as pleomorphic adenoma, adenoid cystic carcinoma, epithelial-myoepithelial carcinoma, mucoepidermoid carcinoma, hyalinizing clear-cell carcinoma, and myoepithelial carcinoma. In general, these tumors are uncommon, arise in proximal bronchi or the trachea, can show biphasic epithelial-mesenchymal features, and show morphologic and molecular properties like their salivary gland counterparts.

Nonepithelial Lung Tumors

These are rare diseases and include primary mesenchymal tumors, malignant neoplasms developed in ectopic lung tissues, PEComatous neoplasms, and hematolymphoid malignancies.[56] In general, these tumors show variable biologic, clinical, and morphologic features.

Molecular Diagnostics and Immune-related Predictive Biomarkers

Somatic alterations in multiple genes are associated with increased oncogenic signaling and enhanced sensitivity to molecularly targeted therapies blocking these pathways. These include genes most commonly mutated in lung adenocarcinomas such as *EGFR*, *KRAS*, *ALK*, *ROS1*, *BRAF*, *RET*, *MET*, *NTRK1*-3, and *HER2*.[59,60] Additional oncogenic alterations occur recurrently in lung carcinomas, but they lack approved targeted therapies showing prominent clinical activity. The spectrum of genomic variants seen in lung cancer includes different structural defects ranging from relatively simple single nucleotide substitutions or short indels to more complex copy number changes and chromosomal rearrangements. These alterations are more commonly assessed clinically using sequential testing with different assays or using comprehensive multigene panels. In addition, prediction of clinical benefit to immune checkpoint blockers in patients with advanced NSCLC has relied

TABLE 30.6

Approved PD-L1 Immunohistochemistry Assays for Lung Cancer

IHC Assay	28-8	28-8	22C3	SP142	SP263
Therapy	Nivolumab	Nivolumab + ipilimumab	Pembrolizumab	Atezolizumab	Atezolizumab
Type of Therapy	Anti-PD-1	Anti-PD-1 + anti-CTLA-4	Anti-PD-1	Anti-PD-L1	Anti-PD-L1
Diagnostic Designation	Complementary	Companion	Companion	Companion	Companion
Scoring and Interpretation	TC staining: ≥1%, ≥5%, ≥10%	≥1% TC expression	TPS ≥1%	IC ≥10% or TC ≥50%	≥1% TC expression

PD-L1, programmed cell death ligand 1; IHC, immunohistochemistry; PD-1, programmed cell death 1; TC, tumor cell; TPS, tumor proportion score; IC, immune cell.

mostly in the semiquantitative expression of PD-L1 protein in baseline tumor sections using IHC. Multiple PD-L1 IHC assays have been developed with different analytical performance, sample scoring methodologies, and marker stratification cut points (Table 30.6).[78] More recently, genomic tests have also been incorporated to assess the potential of benefit to immune checkpoint blockers such as measurement of tumor mutational burden (TMB) and identification of mismatch repair deficiency/microsatellite instability (MSI-H). However, the predictive value of TMB is limited, and its standardization has been challenging.[79] Detection of MSI-H status is widely available and better standardized, but it is extremely rare in lung carcinomas.[79] The identification of genomic alterations in tumor biopsy samples can be performed using diverse methodologies, with the most widely used including direct dideoxynucleotide-based DNA sequencing such as Sanger sequencing, real-time polymerase chain reaction (PCR) with quantitative output (qPCR), in situ DNA hybridization, and massive parallel DNA sequencing with next generation sequencing (NGS). Due to the increasing availability of small biopsies and the rapid expansion of the number of genomic alterations being analyzed, many laboratories are moving from sequential single-gene tests to multimarker panels including simultaneous evaluation of tens to hundreds of individual genes and comprehensive genomic metrics.

Direct Dideoxynucleotide-based DNA Sequencing

This method is based on the random incorporation of fluorescently labeled chain-terminating dideoxynucleotides by DNA polymerase during in vitro DNA replication followed by sequence reading using capillary electrophoresis.[80,81] It is widely available, is inexpensive, can analyze relatively large DNA fragments (>500 bp) with a low error rate, and is still considered as the standard reference. However, the technique is laborious with a relatively long turnaround time of 7 to 10 days, has limited throughput with the capacity to study only discrete regions of single genes at a time, requires elevated DNA input, and the sensitivity for detection of low allelic frequency variants or analysis of samples with low tumor proportion is suboptimal.[82] Furthermore, Sanger sequencing is not able to study complex gene rearrangements or change in gene copy numbers. To date, this method has been used clinically to identify single nucleotide substitutions or short indels in well-annotated regions of oncogenes with mutation "hotspots" such as exons 18 to 21 of *EGFR*; codons 12, 13, 59, and 61 of *KRAS*; codon 600 of *BRAF*; exon 20 of *HER2*; and exons 9 and 20 of *PIK3CA*.

Real-time Polymerase Chain Reaction

This method, also referred to as qPCR, is based on continuous quantification of the accumulation of DNA amplification product during the PCR process using nonspecific fluorescent dyes that intercalate with any double-stranded DNA or with sequence-specific DNA probes consisting of oligonucleotides labeled with a fluorescent reporter on one end and a signal quencher in the other end of the probe.[83] The latter strategy permits the detection of signal only after hybridization of the probe with its complementary sequence such as a mutant allele sequence, increasing the specificity, and allowing for the simultaneous detection of multiple sequences using different fluorophores. This method is sensitive, experimentally simple, requires low sample input, has a short turnaround time of 1 to 3 days, and can be applied to ctDNA samples.[83] However, it requires access to specialized equipment, expensive commercial kits, and reagents, and is not amenable to study complex gene rearrangements or copy number changes. Examples of the use of real-time PCR for clinical detection of actionable oncogenic mutations include regulatory-approved allele-specific identification of variants in exons 18 to 21 of *EGFR*; exons 12, 13, and 61 of *KRAS*; and codon 600 of *BRAF*.

In Situ DNA Hybridization

This method is based on the localized binding of chromogenic (CISH) or fluorescently (FISH) labeled complementary DNA, RNA, or modified nucleic acids strand probes to a specific nucleic acid sequence in a tissue section or cell preparation.[84] DNA FISH is most commonly used in oncology to determine the structure of chromosomes, to achieve spatial gene mapping, and for the identification of discrete oncogene rearrangements such as fusions and translocations, and copy number changes. This method is technically simple, widely available, inexpensive, sensitive, and specific; requires low sample input; has a relatively short turnaround time of 1 to 3 days; and is considered as the reference standard for structural genomic defects. However, analysis of samples requires the use of dark-field epifluorescence microscopy and trained personnel to interpret the morphologic context, and is not suitable for interrogation of single nucleotide variants and short indels.[84] Examples of clinical uses of FISH in oncology include the identification of *ALK*, *ROS1*, *RET*, and *NTRK1-3* gene rearrangements using break-apart probes and increased copy number of *MET* and *HER2* genes using dual color probes targeting the gene of interest and the centromeric region of the chromosome.

Massive Parallel Sequencing

This method includes several high-throughput DNA sequencing platforms and is based on the generation of ligation-based barcoded sequencing libraries by in vitro clonal amplification of templates followed by parallel synthesis of multiple complementary strands and continuous registration of the incorporated nucleotides using different sensors such as fluorescence labels or nucleotide electrochemical properties.[81] Different commercial platforms also use distinct DNA enrichment strategies to develop targeted sequencing panels or incorporate mRNA/cDNA analysis, which can affect the performance for detection of certain variant types and the assay throughput. The use of high-throughput sequencing in the form of predefined targeted gene panels has rapidly expanded and is commonly used in oncology for simultaneous study of multiple alterations in several cancer-related genes.[85]

Massive parallel DNA sequencing is highly sensitive; requires relatively low sample input to accommodate small specimens; and can identify a wide array of genomic alterations such as single nucleotide variants, short indels, translocations, and copy number changes, as well as TMB and MSI status. However, it is not widely available, can have longer turnaround time of 10 to 14 days, and requires access to sophisticated instrumentation and expertise including the use of advanced bioinformatics for results interpretation.

Liquid Biopsy and Circulating Tumor DNA Analysis

The elevated sensitivity of massive parallel sequencing has been exploited to study small fractions of ctDNA present in the plasma from patients with cancer, which is commonly referred to as liquid biopsy.[86] Highly multiplexed panels covering diverse oncogenic variants ranging from 55 to 311 genes are commercially available and regulatory approved for the identification of discrete gene variants. These tests can also report TMB and MSI status. The major advantages of ctDNA analysis include the noninvasive nature of the sampling, a relatively short turnaround time of 5 to 8 days, and the possibility to capture molecular alterations present in multiple tumor regions or even multiple synchronous tumors within the same patient. However, the sensitivity of variant detection using ctDNA is prominently lower than in the analysis of primary tumors, and high false-negative rates have been reported.[87] Furthermore, the detection of ctDNA may be affected by multiple factors including intrinsic tumor properties, disease stage, and treatment effect. False-positive results have also been recorded in the presence of clonal hematopoiesis of indeterminate potential (CHIP), particularly in older patients or after cancer treatment including chemotherapy or radiotherapy.[88] Analysis of large DNA segments such as whole exomes and the study of additional analytes beyond ctDNA obtained from blood are also expected in the future to expand the use of liquid biopsies.

Biomarkers for Immunotherapy

Multiple studies have revealed the potential of immune-related metrics obtained from the tumor microenvironment to predict sensitivity and resistance to immune checkpoint blockers in patients with lung cancer. These include direct measurement of tumor infiltrating lymphocytes or other immune-cell populations, as well as integrated RNA signatures able to capture signal from diverse cell types and immune activation and suppression pathways.[89,90]

MOLECULAR LANDSCAPE OF NON–SMALL-CELL LUNG CANCER

Studies of the genomic landscape of lung cancer have provided evidence that different histological classes of the disease have distinct genetic profiles. Lung adenocarcinomas are characterized by the frequent presence of mutations in genes encoding proteins in receptor tyrosine kinase (RTK) and mitogen-activated protein kinase (MAPK) signaling that the cancer cells depend on for their survival and that, in many cases, have been found to be actionable therapeutic targets that have transformed the diagnosis and treatment of the disease. This contrasts with squamous cell lung carcinomas in which genetically defined subsets with clinical ramifications have not been clearly identified. A pan-lung cancer analysis of the Cancer Genome Atlas data from adenocarcinomas and squamous cell carcinomas found that there was an approximately 12% overlap in significantly mutated genes between the two tumor types.[91] Indeed, the profile of recurrently altered genes is more similar to that of squamous cell carcinomas originating in other tissues. SCLC also has a distinct genetic profile that is characterized by a high frequency of co-occurring genetic alterations in *TP53* and *RB1*.[92] Despite these differences, the somatic mutation rates in these three histologic classes of lung cancer are among the highest of all solid tumors with eight to nine mutations per megabase probably due to the high frequency of lung cancers associated with exposure to tobacco carcinogens.[93]

Lung Adenocarcinomas

Mutations in oncogenes in the RTK-RAS-MAPK signaling pathway have been identified in approximately two-thirds of lung adenocarcinomas, are generally mutually exclusive with each other, and have been shown to be bona fide drivers. Importantly, many of these represent actionable targets with matching investigational or U.S. Food and Drug Administration (FDA)-approved targeted therapies.

Mutations

KRAS is the most frequently mutated RAS family member in lung adenocarcinomas.[94] Mutations in *KRAS* are observed in approximately 30% of lung adenocarcinomas in people of European or African descent and are linked to a history of smoking.[95] The prevalence of *KRAS* mutations in people of Asian or Native American ancestry is lower.[96,97] *KRAS* G12C/D/V mutations are the most prevalent and account for approximately 85% of lung adenocarcinoma–associated *KRAS* mutations.[98] Additional mutations in *KRAS* at the G12 residue and at residues G13 and Q61 are also found, albeit at low frequencies. KRAS is a small GTPase that activates downstream progrowth and survival pathways when it is bound to GTP, and oncogenic *KRAS* mutations commonly lead to impaired GTPase activity.

Most *EGFR* mutations are either small in-frame deletions in exon 19 or the L858R point mutation, which represent approximately 50% and 40% of the cases, respectively. Additional less frequently observed *EGFR* mutations include exon 20 insertion mutations (4%–10% of cases) and EGFR G719X and L861Q mutations. EGFR is a member of the ERBB family of RTKs, which includes EGFR, ERBB2 (also known as HER2), ERBB3, and ERBB4. Lung cancer–associated *EGFR* mutations are found in exons encoding the tyrosine-kinase domain of the receptor. The mutations lead to constitutive activation of the RTK that causes uncontrolled proliferation and survival via activation of MAPK and PI3K signaling. Exon 19 deletion and L858R mutations were initially found to exhibit sensitivity to the first-generation adenosine triphosphate–competitive reversible EGFR tyrosine-kinase inhibitors (TKIs) gefitinib and erlotinib.[99–101] The initial observation in 2004 of this connection between *EGFR* mutations and sensitivity to TKIs transformed the diagnostic and treatment landscape for NSCLC and stimulated the search for additional targetable oncogenic drivers and drugs for this disease.

ERBB2 mutations, specifically in-frame exon 20 3-12 base pair insertion mutations have been identified in 1% to 4% of NSCLCs, most frequently in adenocarcinomas and never-smokers.[102,103] A 12 base pair insertion mutation in *ERBB2* (A775_G776insYVMA) that leads to duplication of amino acids YVMA accounts for most lung cancer–associated mutations in this gene. Point mutations (L755S and G776C) can also occur and account for approximately 10% of ERBB2 mutations in NSCLC. ERBB2 is overexpressed in 2% to 30% of NSCLCs and amplified in 2% to 5% of cases.

Mutations that affect exon 14 of *MET* are found in approximately 3% to 4% of NSCLCs.[104] Exon 14 encodes the site for binding of the E3 ubiquitin ligase CBL, which is important for MET degradation. The mutations interfere with CBL binding leading to sustained levels of MET activity. *MET* amplification is observed in 1% to 5% of NSCLCs at baseline and in 5% to 20% of *EGFR* mutant tumors with acquired resistance to TKIs.

Mutations in *BRAF* are observed in approximately 3% of lung adenocarcinomas and are associated with a history of smoking.[105] The *BRAF* V600E mutation accounts for 50% of cases, followed by the *BRAF* G469A/V (35%–40%) and *BRAF* D594F (6%–11%) mutations.

Fusions

ALK rearrangements juxtapose the *ALK* sequence encoding its tyrosine-kinase domain with a 5′ fusion partner, most commonly *EML4* in lung cancer, leading to the production of a fusion in which ALK is constitutively active, thus engaging pathways that lead to uncontrolled cell proliferation and survival.[106] The discovery of *ALK* fusions paved the way for the identification of additional gene fusions involving RTKs in NSCLC. Gene fusions involving *ROS*, *RET*, and *NTRK1-3* are found in 1% to 2%, 1% to 2%, and 0.2% of NSCLCs, respectively.[107–109] Similar to *ALK* fusions, *ROS*, *RET*, and *NTRK* fusions lead to constitutive activation of the tyrosine kinase and activation of downstream pathways that promote cell proliferation and survival. In addition, the clinical characteristics of patients whose tumors harbor *ROS* and *RET* gene fusions are similar to those observed for *ALK*, whereas there is more variability in *NTRK* fusion cases.

NRG1 gene fusions mainly involving CD74 as the 5′ partner is found in 0.14% to 0.4% of lung adenocarcinomas and are enriched in invasive mucinous adenocarcinomas.[110] NRG1 is an ERBB3/4 ligand, and targeting the ERBB signaling is being investigated as an approach to treat this subset of tumors.

Lung Squamous Cell Carcinomas

With the exception of several genes that are commonly mutated or show copy number alterations in both histologies, the mutational landscape of lung squamous cell carcinomas is quite different from that of lung adenocarcinomas.[91,111] Whereas in lung adenocarcinomas the mutations in genes in RTK-MAPK signaling prevail and define genetic subgroups of the disease, the genetic alterations in the PI3K pathway, most frequently *PIK3CA*, *PTEN*, and *AKT3*, predominate in squamous histology, being observed in up to 50% cases. Alterations in the genes from the *KEAP1-NFE2L2-CUL3* axis, which regulate the cellular response to oxidative stress, are present in approximately 30% of squamous cell lung carcinomas. These mutations lead to increased expression of antioxidant genes that allow cancer cells to better withstand environmental stress. Amplification of genes involved in squamous cell differentiation, *SOX2* and *TP63*, has been found in approximately 40% of cases. Of note, both *SOX2* and *TP63* are located on chromosome 3q, which is selectively amplified in squamous cell lung cancer.

Tumor Suppressor Genes

Tumor suppressor gene alterations are quite common in NSCLCs. Of the six significantly mutated genes in both adenocarcinomas and squamous tumors, five (*TP53*, *RB1*, *CDKN2A*, *NF1*, and *ARID1A*) are tumor suppressor genes.[91] *TP53* is by far the most commonly altered tumor suppressor gene occurring in 46% to 81% of all NSCLCs.[94,111] Additional commonly mutated tumor suppressor genes in lung adenocarcinomas are *APC*, *ARID1A*, *ATM*, *KEAP1*, *NF1*, *RB1*, *RBM10*, *SETD2*, *SMAD4*, *SMARCA4*, and *STK11*. In squamous tumors, commonly mutated tumor suppressor genes include *ARID1A*, *CDKN2A*, *CUL3*, *FBXW7*, *KDM6A*, *MLL2*, *NF1*, *PTEN*, and *RB1*. The chromatin regulators *EP300* and *CREBBP* are mutated in both subtypes of NSCLC. Increasing evidence from preclinical studies indicates that the specific configuration of tumor suppressor gene alterations in a tumor can influence tumor growth and progression, as well as response to therapy.[112,113]

Integrative "Omics" Profiling of Non–small-cell Lung Cancer

Whole transcriptome profiling was performed as part of the Cancer Genome Atlas analysis of lung adenocarcinomas and squamous carcinomas.[94] The three expression subtypes that could be distinguished in lung adenocarcinomas included proximal proliferative, which was enriched for tumors with *KRAS* and *STK11* mutations; proximal inflammatory, which was enriched in tumors harboring *TP53* and *NF1* alterations and a solid histopathology; and terminal respiratory unit tumors, which were enriched for *EGFR* mutant and *ALK/ROS/RET* fusion. In squamous cell carcinomas, four subtypes have been identified.[111,114] The classical type has a high degree of chromosomal instability and methylation and is enriched for tumors with *KEAP1*, *NFE2L2*, and *PTEN* mutations, as well as the 3q26 amplicon. The primitive type is characterized by *RB1* and *PTEN* mutations, whereas the basal is enriched for NF1 mutations. The fourth type was secretory. More recent efforts have centered on integrating multiple types of omic data to further refine tumor subsets and identify links between genomics, transcriptome, and the proteome in both histologies.[115–118]

TREATMENT OF PATIENTS WITH STAGE I TO III

General Aspects of Treatment

Structural Aspects of Patient Care

Optimal care for lung cancer requires that major patient management decisions are made with involvement of all relevant specialties. Collaborative decision making defines multidisciplinary care, and this approach is recommended or mandated in many countries for the treatment of lung cancer patients.[119–121] Studies show that this is associated with better timeliness of care, appropriate treatment, and survival.[119,121–125] There is also extensive evidence that structural aspects of care have a marked effect on outcomes.[126] Studies have quite consistently found lower perioperative mortality and better long-term survival in high-volume teaching facilities and treatment by specialists compared to low-volume centers, nonteaching facilities, and generalists, respectively.[126] However, the relationship between volume and outcomes is not consistent on an individual institution level. It is likely that volume is only a marker for an organized process of care delivery.

General Aspects of Surgery

The most prominent advance in surgery for lung cancer is the markedly reduced perioperative morbidity and mortality of a minimally invasive approach such as Video-Assisted Thoracoscopic Surgery (VATS) when compared to thoracotomy. Following VATS lobectomy, the average hospital stay is approximately 4 days and patients return to normal function within about 1 to 3 months.[127] Furthermore, long-term outcomes are similar for VATS or open resection in RCTs and adjusted nonrandomized comparisons (NRCs).[126,127] However, adoption of VATS is variable in different centers, likely reflecting available expertise. Many guidelines recommend that surgical care be delivered by a thoracic surgeon with a focused interest in noncardiac thoracic surgery.[120,126]

The improved perioperative outcomes allow older and more frail patients to undergo surgery.[126] Patient selection is also altered by the recognition that previous criteria had been quite conservative. Series of VATS lobectomy in patients with severe pulmonary compromise report a 30-day mortality of only 2% to 3% and pulmonary complications in 10% to 20%.[128,129] There has also been a major advance in the ability to safely carry out extensive resections for tumors that appeared to be locally invasive but subsequently were found to invade the thoracic inlet, vertebral column, or vena cava.

General Aspects of Radiotherapy

Advances in Radiation Technique

In the 1990s, 3D conformal radiation therapy was implemented. The greater ability to manipulate and contour beams from multiple

angles permits better coverage of the tumor target and decreases the dose to surrounding normal tissue. Better targeting is also facilitated by technology that enables fusion of complementary imaging modalities, such as CT, PET, and MRI. Intensity-modulated radiation therapy is another advance that uses a dynamic multileaf collimator to vary the intensity across the beam aperture during treatment, resulting in a more conformal plan further reducing surrounding normal tissue exposure. Tumor motion during breathing can be measured during inhalation and exhalation using 4D technology. Various motion management techniques can decrease the expansion margin around the target. Image-guided radiation therapy directly images the tumor target and/or critical organs immediately before treatment, allowing adjustment and reduction of normal tissue exposure.

Proton therapy exploits a distinct physical characteristic of protons that deposits most of the dose abruptly near the end of the proton's path. This could reduce the dose to the esophagus, heart, or uninvolved lung just beyond the target. Although phase II trials of fractionated proton therapy with concurrent chemotherapy in lung cancer have shown encouraging survival and low toxicity,[130,131] an RCT comparing proton therapy to intensity-modulated radiation therapy for locally advanced NSCLC showed no difference in toxicity or local failure.[132] Additional trials are ongoing.

Radiation Toxicity

The risk and severity of radiation toxicity are related to the dose and volume of exposed normal tissue, underlying comorbidities, and functional aspects of the organ at risk. Resources that summarize the evidence and provide practical guidance for dose limits include the Quantitative Analysis of Normal Tissue Effects in the Clinic (QUANTEC) for 3D conformal radiation therapy,[133] high dose per fraction, and Hypofractionated Treatment Effects in the Clinic (HyTEC) for stereotactic body radiotherapy (SBRT).[134]

Radiation Pneumonitis and Pulmonary Fibrosis

Clinically significant radiation pneumonitis occurs in 5% to 50% of lung cancer patients, occurring any time between 0 to 18 months with a peak incidence at 2 months. The typical presentation includes persistent, nonproductive cough; dyspnea; low-grade fever; and fatigue. Chest radiograph or CT scans may be normal. Imaging findings include ground glass opacification (within 1 month to 6 months), patchy consolidation (4 months to 12 months), or fibrosis (6 months to several years). Early radiographic changes occur within the intermediate- to high-dose region, and later changes may extend into lower-dose regions. The risk and severity is lower for patients treated with smaller-volume SBRT compared to larger-volume conventionally fractionated radiation.[135] The risk for radiation pneumonitis may be increased for lower lung field tumors and for patients who are older or receiving concurrent chemotherapy such as docetaxel, gemcitabine, and especially carboplatin plus paclitaxel.[136]

Radiation pneumonitis causing moderate to severe symptoms is typically treated with glucocorticoids, tapering slowly over weeks to months to minimize the recurrence of symptoms. An infectious etiology should be ruled out before starting steroids. Prophylactic antibiotics or anticoagulants do not appear to affect the development of radiation pneumonitis. Pulmonary parenchymal fibrosis after thoracic radiation is associated with long-term dyspnea and treated with supportive care such as an oxygen supplement and pulmonary rehabilitation because no specific intervention has demonstrated efficacy to slow or reverse the progression of fibrosis, including glucocorticoids.

Esophagitis and Esophageal Stenosis

Acute esophagitis is often the dose-limiting toxicity of fractionated radiation therapy for thoracic malignancies. Grade ≥3 esophagitis with severely impaired eating, tube feeding, or hospitalization occurs in approximately 18% of patients undergoing chemoradiation.[136] Candidiasis or gastroesophageal reflux may contribute to the symptoms. Risk factors include older age, concurrent chemotherapy, and accelerated fractionation. Treatment is primarily supportive with dietary manipulation, analgesics, antifungal and proton pump inhibitors, topical agents such as sucralfate, or local anesthetics. Clinically significant late toxicity such as stenosis and fistula formation is uncommon, occurring in <5% of patients.[137]

Heart Toxicity

The risk of cardiac toxicity is variable, depending on the tumor location and treatment technique. Pericarditis occasionally develops during treatment or shortly thereafter. Chronically, progressive fibrosis may lead to constrictive pericarditis, cardiomyopathy, heart failure, ischemic heart disease, valvular stenosis, and arrhythmia.[138] The risk of radiation therapy–induced cardiac toxicity may be increased by older age, smoking, hypertension, diabetes, obesity, genetic predisposition, and possibly concurrent paclitaxel.[139] Minimization of the dose to the heart may reduce the risk of major adverse cardiac events.

Treatment of Clinical Stage I

Definitive local therapy options for clinical stage I NSCLC include lobectomy, segmentectomy, wedge resection, SBRT, and thermal ablation. The treatment choice is based on patient and tumor characteristics.

Perioperative mortality and morbidity are low in healthy patients and without differences regarding the extent of resection in RCTs[140,141] and large database studies.[142–144] Forced expiratory volume in 1 second decreases approximately 15% after lobectomy and approximately 10% after segmentectomy without affecting the exercise capacity.[145] The loss is greater in multisegment resections (approximately 12%) compared to single-segment (5%) or wedge resection (4%).[145] After VATS resection, quality of life (QoL) returns to normal after 3 months to 6 months, whereas thoracotomy and larger resections such as pneumonectomy have a lasting impact.

Lobectomy has been associated with improved overall survival (OS) and lower recurrence rates compared to sublobar resection such as segmentectomy or wedge resections in the 1980s.[146] NRCs that adjusted for confounders also show better 5-year OS after lobectomy compared to wedge resection (10%–15% absolute difference) or segmentectomy (5% to 10% absolute difference).[142,145,147] CALGB-140503, an RCT comparing lobectomy to sublobar resection in solid tumors ≤2 cm, showed no differences in perioperative mortality between lobar and sublobar resections.[140] Additional outcomes from this trial, including survival and recurrence patterns, may help clarify the role for sublobar resections in patients with early-stage NSCLC.

SBRT is another option for patients with stage I NSCLC. Multiple studies show no average QoL impairment after SBRT.[148] Furthermore, short-term outcomes have been better after SBRT compared to surgery in operable patients with 90-day mortality of <1% compared to approximately 3% respectively.[148,149] SRBT is also associated with decreased morbidity from <1% to 5–15%. Pooled results from two small RCTs comparing SBRT to lobectomy in[58] patients demonstrated better OS and disease-free survival (DFS) after SBRT.[150] Results of an ongoing RCT—VALOR—are anticipated in 2026. Adjusted NRCs have shown worse 5-year survival for SBRT compared to lobectomy or sublobar resection.[148,151,152]

Surgery is fairly well tolerated in selected older patients, with sublobar resection offering marginal short-term benefit with a detriment in long-term outcomes compared to lobectomy.[128] Short-term morbidity and mortality in older patients increases slightly with age.[128] Sublobar resection has little impact on perioperative mortality compared to lobectomy in older patients, with a possible minor

decrease in morbidity. VATS decreases morbidity and mortality in older patients.[128] In this population, SBRT demonstrates short-term advantages and long-term disadvantages compared to surgery. Mortality and morbidity increases for both SBRT and surgery, with the difference widening with increasing age (1% to 2% difference for those 71 years to—75 years of age and 2% to 4% for age >80 years).[153] Grade ≥3 morbidity is lower after SBRT than surgery in older cohorts (5%–10% vs 10%–20%).[148] Adjusted NRCs in older cohorts demonstrate meaningfully worse 5-year outcomes after SBRT compared to surgery with a 5% to 15% absolute decrease in OS. The difference is not clearly modified by age, tumor size, or resection extent.[148] If standard SBRT is not an option in medically inoperable patients with a tumor in close proximity to critical organs, alternative hypofractionated radiation therapy regimens or chemoradiotherapy can be used.

In patients with severely compromised pulmonary function, below criteria commonly cited as contraindications to surgery, the reported 30-day mortality for lobectomy has been surprisingly low for both VATS (2% to 3%) and open lobectomy (3%–8%) in carefully selected cases.[128] VATS markedly decreases perioperative morbidity and mortality in severely compromised patients.[128,129]

Adjusted NRCs found a 10% to 20% worse 5-year OS after SBRT compared to lobectomy or sublobar resection in patients with compromised pulmonary function.[148] Results of the ongoing STABLE-MATES trial comparing SBRT to sublobar resection are anticipated in 2024.

Percutaneous thermal ablation has no impact on QoL but is associated with a 10% to 20% risk of pneumothorax and grade ≥3 toxicity.[148] Furthermore, adjusted NRCs have demonstrated an approximately 20% decrease in 5-year OS compared to wedge resection. When compared to SBRT, the OS is slightly decreased.[148]

Lung cancers with a prominent ground glass/lepidic (GG/L) component exhibit relatively indolent behavior.[154–156] Prospective studies of observation demonstrate that most GG/L tumors do not progress over 5 to 10 years, and in case of tumor progression, there was no stage progression or loss of curability.[155,156] Prospective studies suggest that sublobar resection is a reasonable option for GG/L tumors, usually without OS detriment compared to lobectomy despite the increase in local recurrences.[128,157]

Treatment of Clinical Stage II

Lobectomy is the standard for resection of clinical stage II NSCLC, preferably through a minimally invasive approach.[120,126] Clinical guidelines suggest that systematic sampling should be performed on at least three mediastinal node stations.[120] Survival and recurrence appears to be better after a complete node dissection from randomized and nonrandomized studies.[126] Whenever possible, a sleeve resection should be performed if needed rather than pneumonectomy.[126] A sleeve resection involves resection of a portion of the bronchus to the lung with an end-to-end anastomosis of the main bronchus to that of the uninvolved lobes of the lung. Several NRCs demonstrate lower operative mortality and complications for sleeve resection compared to pneumonectomy, with equivalent or better long-term survival, lower recurrence rates, and better QoL.[126,158,159]

Despite appropriate preoperative stage evaluation, some patients with clinical stage N0 or N1 are found to have N2 involvement intraoperatively, changing the stage to III. In these cases, the treatment follows the same principle applied to patients with stage II disease.

Adjuvant Therapy

Despite complete surgical resection, patients with NSCLC remain at risk for tumor recurrence and death. Adjuvant chemotherapy offers a small but statistically significant survival improvement in patients with completely resected NSCLC compared to observation.[160–162] In the Lung Adjuvant Cisplatin Evaluation (LACE) pooled analysis, which included 4,584 patients from five RCTs, the 5-year OS absolute benefit was 5.4% with the use of platinum-based adjuvant chemotherapy.[163] The 5-year OS from adjuvant chemotherapy improved from 64% to 67% in stage IB, 39% to 49% in stage II, and 26 to 39% in stage IIIA, leading to the American Society of Clinical Oncology guidelines recommendation for adjuvant cisplatin-based chemotherapy in patients with stage II to IIIA disease but not routinely for stage IB.[164] The recommendation for adjuvant chemotherapy in patients with stage IB was based on exploratory analyses of RCTs, most notably the Cancer and Leukemia Group B (CALGB) 9633 trial, where 344 patients with completely resected stage IB NSCLC were randomized to adjuvant chemotherapy with carboplatin plus paclitaxel or observation.[165] Although there was no survival advantage from adjuvant chemotherapy across the entire cohort, an exploratory analysis showed a statistically significant survival improvement in patients with tumors ≥4 cm (hazard ratio [HR] 0.69; 90% confidence interval [CI], 0.48 to 0.99; $P = 0.043$). This cutoff of 4 cm was used in subsequent adjuvant trials. Of note, with the newer staging system classifications, lymph node–negative tumors measuring >4 to 5 cm have been classified as T2b with the stage changing to IIA.[40]

With the established benefit from adjuvant cisplatin-based chemotherapy, the next step was to evaluate whether the addition of a third drug would improve the outcomes in this patient population. In the ECOG 1505 study, patients with completely resected stage IB ≥4 cm to stage IIIA NSCLC were randomized to four cycles of chemotherapy alone or in combination with bevacizumab, a monoclonal antibody against vascular endothelial growth factor (VEGF).[166] The chemotherapy consisted of cisplatin 75 mg/m^2 on day 1 plus an investigator choice of docetaxel 75 mg/m^2 on day 1, pemetrexed 500 mg/m^2 on day 1, gemcitabine 1,200 mg/m^2 on days 1 and 8, or vinorelbine 30 mg/m^2 on days 1 and 8. Patients randomized to the experimental group received bevacizumab 15 mg/kg every 21 days starting on the first day of chemotherapy and continuing for 1 year. The addition of bevacizumab did not improve the median OS (85.8 months for bevacizumab vs not reached in the control arm) or the median DFS (40.6 months vs 42.9 months). Although the study was not designed to compare the chemotherapy regimens, there were no significant differences in outcomes according to the chemotherapy backbone. Cisplatin plus pemetrexed was the most commonly used regimen, being selected in 33% of patients. With adjuvant randomized trials focused in demonstrating a benefit from chemotherapy over best supportive care, there is limited data on the optimal adjuvant regimen.

The JIPANG study was a randomized phase III study where 804 patients with completely resected pathologic stage II to IIIA nonsquamous NSCLC were randomized to cisplatin 75 mg/m^2 on day 1 plus pemetrexed 500 mg/m^2 on day 1 or cisplatin 80 mg/m^2 on day 1 plus vinorelbine 25 mg/m^2 on days 1 and 8 with cycles every 21 days.[167] The median DFS, the primary end point of the study, was 37.3 months and 38.9 months respectively in the vinorelbine and pemetrexed arms (HR, 0.98; 95% CI, 0.81 to 1.2; $P = 0.47$). There was also no difference in the OS (HR, 0.98). The cisplatin plus vinorelbine arm was associate with increased incidence of grade 4 adverse events (57.3% vs 5.1%) and grade 3 febrile neutropenia (11.6% vs 0.3%). In the TREAT study, 132 patients were randomized to cisplatin plus vinorelbine or pemetrexed.[168] There were no significant differences in OS or rates of distant metastases.

Several studies evaluated the role of adjuvant EGFR TKIs in patients with resected NSCLC. In a meta-analysis of five publications including three RCTs and two retrospective studies, adjuvant gefitinib, erlotinb, or icotinib was associated with a significant improvement DFS (HR, 0.48; 95% CI, 0.36 to 0.65) among patients with *EGFR* mutation.[169] Nevertheless, although the OS was also improved, it did not reach statistical significance (HR, 0.72; 95% CI, 0.49 to 1.06). In the CTONG1104 study, patients with completely resected stage II to IIIA NSCLC and activating *EGFR* mutations were randomized to gefitinib for 24 months or four cycles of cisplatin plus vinorelbine.[170] Although the gefitinib

TABLE 30.7

Selected Randomized Clinical Trials Evaluating Adjuvant Immunotherapy in NSCLC

Trial (NCT No.)	Experimental Arm[a]	Primary End point	Status
IMpower-010 (02486718)	Atezolizumab	DFS	Results available
ANVIL (02595944)	Nivolumab	DFS and OS	Completed accrual
BR.31 (02273375)	Durvalumab	DFS in PD-L1 positive	Completed accrual
PEARLS (02504372)	Pembrolizumab	DFS	Completed accrual
ACCIO (04267848)	Pembrolizumab	DFS and OS	Ongoing accrual

[a]All trials include surgery followed by adjuvant chemotherapy. After completion of chemotherapy, patients are randomized to immunotherapy or control with placebo or best supportive care.
NSCLC, non–small-cell lung cancer; DFS, disease-free survival; OS, overall survival; PD-L1, programmed cell death ligand 1.

arm was associated with increased 3-year DFS (39.6% vs 32.5%), there was no improvement in the 5-year OS (43.2% vs 51.2%; $P = 0.78$). Furthermore, in an updated analysis, the 5-year DFS was not improved in the gefitinib arm (22.6% vs 23.2%). In the ADAURA study, patients with completely resected stage IB to IIIA NSCLC harboring *EGFR* exon 19 deletion or L858R mutations alone or in combination with other *EGFR* mutations were randomized to osimertinib 80 mg orally per day or placebo for 3 years.[171] The primary end point was DFS according to the investigator assessment among patients with stage II to IIIA disease. There were 339 and 343 patients enrolled in the osimertinib and placebo arms, respectively. Adjuvant chemotherapy was not required and was administered in 60% of patients in both arms. The DFS at 24 months in patients with stage II or IIIA was 90% in the osimertinib group and 44% in the placebo group (HR, 0.17; 99% CI, 0.11 to 0.26; $P < 0.001$). The median DFS was not reached in the osimertinib group, and the OS was immature. The DFS benefit from osimertinib was also seen in patients with stage IB (HR, 0.39) and irrespective of the adjuvant chemotherapy used.

Several trials are currently evaluating the role for adjuvant ICIs (Table 30.7).[172] The first study to reports results was IMpower010.[173] In this trial, 1,280 patients with completely resected stage IB ≥4 cm to IIIA NSCLC were enrolled to the initial phase, which consisted of adjuvant platinum-based chemotherapy with cisplatin plus docetaxel, pemetrexed, gemcitabine, or vinorelbine. Patients who received at least one of the four planned cycles of chemotherapy and did not have disease recurrence were eligible for the randomized phase. The primary end points of the study were investigator-assessed DFS in patients with stage II or IIIA disease and tumor expression of PD-L1 on ≥1% tumor cells by the SP263 IHC assay, then all patients with stage II or IIIA disease and all patients randomly assigned with stage IB to IIIA. Despite its use only in patients with nonsquamous histology, the combination of cisplatin and pemetrexed was the most commonly regimen administered, accounting for approximately 40% of the treatments. In the randomized phase, 507 patients were assigned to atezolizumab 1,200 mg every 3 weeks for up to 16 cycles and 498 to best supportive care. The sequential use of atezolizumab was associated with increased 3-year DFS in patient with PD-L1–positive stage II to IIIA (60% vs 48%; stratified HR, 0.66; 95% CI, 0.5 to 0.88; $P = 0.039$), all tumors with stage II to IIIA (56% vs 49%; stratified HR, 0.79; 95% CI, 0.64 to 0.96; $P = 0.02$), and all tumor stage IB to IIIA (58% vs 53%; stratified HR, 0.81; 95% CI, 0.67 to 0.99; $P = 0.04$). The secondary analysis for the stage II to IIIA population with PD-L1 ≥50% showed improved DFS for patients receiving sequential atezolizumab (HR, 0.43; 95% CI, 0.27 to 0.68). Nevertheless, the exploratory analysis for patients with stage II to IIIA showed no improvement in the DFS for those with PD-L1 of 1% to 49% (HR, 0.87; 95% CI, 0.6 to 1.26) or PD-L1 <1% (HR, 0.97; 95% CI, 0.72 to 1.31). Among the 43 patients with PD-L1–positive stage II to IIIA disease who had tumors harboring *EGFR* mutation, including 23 randomized to atezolizumab and 20 randomized to best supportive care, there was no benefit from sequential atezolizumab (HR, 0.57; 95% CI, 0.26 to 1.24). Based on the results from IMpower010, sequential adjuvant atezolizumab was approved by the FDA in October 2021 for completely resected stage II to IIIA NSCLC with PD-L1 expression on ≥1%.

Neoadjuvant Therapy

Neoadjuvant chemotherapy may reduce the tumor size leading to an increased resection rate and provide early treatment of subclinical micrometastatic disease that may be better tolerated than if administered after the surgery. However, this approach may be associated with a delay in surgery, and the tumor may become unresectable in case of tumor growth. In a meta-analysis including 2,385 patients from 15 randomized trials, neoadjuvant chemotherapy increased the 5-year survival from 40% to 45% when compared to surgery alone (HR, 0.87; 95% CI, 0.78 to 0.96; $P = 0.007$).[174] In a systematic review including >10,000 patients accrued to 22 adjuvant and 10 neoadjuvant trials, the HRs for adjuvant and neoadjuvant therapy compared to surgery alone were 0.80 and 0.81, respectively, suggesting that there are no differences in survival according to the timing of chemotherapy administration.[175] Among patients treated with neoadjuvant chemotherapy, the histopathologic response in the resected specimen may predict the outcomes. In a retrospective study including 192 NSCLC patients treated with neoadjuvant chemotherapy followed by complete response and 166 control NSCLC patients who did not receive neoadjuvant therapy, the percentage of residual viable tumor cells was a significant predictor for survival only in those who received neoadjuvant therapy.[176] In the neoadjuvant group, 36 patients (19%) had ≤10% viable cells and achieved a significant improvement in 5-year OS (85% vs 40%; $P < 0.0001$) and 5-year DFS (78% vs 35%; $P < 0.001$) compared to those with >10% viable cells. The presence of ≤10% residual viable tumor cells following neoadjuvant therapy is termed *major pathologic response* and has been proposed as a surrogate for improved OS in patients treated with neoadjuvant therapy.

The use of immunotherapy in the neoadjuvant setting is postulated to be more effective than after tumor resection due a higher antigen load present in the body prior to surgery, which may result in increased antigen presentation and priming of tumor-specific T cells.[177] The initial studies evaluating the use of neoadjuvant immunotherapy in patients with resectable NSCLC, including ICIs alone or in combination with chemotherapy, showed promising results.[178–181] These results led to the launching of several randomized phase III clinical trials comparing neoadjuvant chemotherapy alone or in combination with immune checkpoint blockade (Table 30.8).[172] The first randomized phase III study to report preliminary results was CheckMate-816, where 358 patients with resectable NSCLC stage IB ≥4 cm to IIIA were ran-

TABLE 30.8

Selected Randomized Clinical Trials Evaluating Neoadjuvant Immunotherapy in NSCLC

Trial (NCT No.)	Experimental Arm[a]	Primary Endpoint
CheckMate-816 (02998528)	Chemotherapy plus nivolumab	pCR and DFS
KEYNOTE-671 (03425643)	Chemotherapy plus atezolizumab	EFS and OS
IMpower-030 (03456063)	Chemotherapy plus atezolizumab	EFS
AEGEAN (03800134)	Chemotherapy plus durvalumab	pCR and EFS

[a]Patients in the control arm received chemotherapy with platinum doublet alone.
NSCLC, non–small-cell lung cancer; pCR, Pathologic complete response; DFS, disease-free survival; EFS, event-free survival; OS, overall survival.

domized to three cycles of neoadjuvant chemotherapy alone or in combination with nivolumab 360 mg every 3 weeks.[182] The primary end points were pathological complete response (pCR) and event-free survival. The nivolumab arm was associated with increased rates of pCR (24% vs 2.2%; odds ratio, 13.9; $P < 0.0001$), with the higher efficacy observed for all stages. Furthermore, there were improvements in major pathologic response (36.9% vs 8.9%; odds ratio, 5.7) and surgical completion (83% vs 75%), without increasing the risks for postsurgical complications. Based on the results from CheckMate-816, neoadjuvant platinum doublet plus nivolumab was approved by the FDA in March 2022 for the treatment of patients with resectable NSCLC. There are several ongoing studies evaluating the role of neoadjuvant targeted therapy with a general design that also includes adjuvant chemotherapy followed by targeted therapy for 2 years to 3 years.[183]

In summary, cisplatin-based adjuvant chemotherapy is the current standard of care for patients with resected stage II to IIIA NSCLC. Patients with *EGFR* exon 19 deletion or L858R mutations and those with PD-L1 ≥1% may benefit from sequential osimertinib or atezolizumab, respectively, with the former also being an option for those who did not receive adjuvant chemotherapy. Neoadjuvant chemotherapy appears to be as effective as adjuvant chemotherapy, and there have been promising early results from neoadjuvant ICIs alone or in combination with chemotherapy.

Postoperative Radiation Therapy

Postoperative radiation therapy (PORT) is not recommended for completely resected early-stage NSCLC based on several metaanalyses and subset analyses of RCTs.[126] Current evidence also argues against routine use of PORT in patients with completely resected stage IIIA (N2) NSCLC based on the results of two RCTs: Lung ART and PORT-C. In Lung ART, the 3-year DFS was 47% with the PORT arm and 44% without PORT, with median DFS of 30.5 months and 22.8 months, respectively (HR, 0/86; 95% CI, 0.68–1.08; $P = 0.18$).[184] In the PORT-C study, PORT improved the 3-year DFS (42.8% vs 30.6%; HR, 0/75; 95% CI, 0.57 to 1; $P = 0.05$) but not the 3-year OS (82.6% vs 83.1%; HR, 0.83; 95% CI, 0.53 to 1.3; $P = 0.41$).[185]

Treatment of Clinical Stage III

Single-modality therapy using surgery or radiotherapy alone for patients with stage IIIA (N2) is suboptimal, and the recommended treatment is with either neoadjuvant therapy followed by surgery or chemoradiotherapy. Patients with N2 disease who are considered for surgical resection should receive neoadjuvant chemotherapy or chemoradiotherapy.[186]

RCTs have shown no improvement in OS for preoperative therapy followed by surgery compared to chemoradiotherapy alone.[187–190] In Intergroup-0139, the largest study comparing chemoradiotherapy with or without surgical resection for stage III NSCLC, patients with stage T1-3N2 NSCLC were treated with concurrent chemoradiotherapy including cisplatin plus etoposide and 45 Gy of thoracic radiotherapy.[187] In the absence of tumor progression, patients were randomized to surgery or continued radiation therapy until 61 Gy. The primary end point was OS. Surgical resection was associated with a numerically higher but not statistically significant improvement in median OS (23.6 vs 22.2 months; HR, 0.87; 95% CI, 0.7 to 1.1; $P = 0.24$) and 5-year OS (27% vs 20%). In an unplanned exploratory analysis matching the groups for age, performance status and clinical T stage, the median OS for surgical resection was increased in patients undergoing lobectomy (33.6 months vs 21.7 months; $P = 0.002$) but not those undergoing pneumonectomy (18.9 months vs 29.4 months). In a meta-analysis including three randomized trials including 334 patients comparing neoadjuvant chemotherapy or chemoradiotherapy, the latter was associated with a significant increase in mediastinal downstaging and complete response, as well as higher rates of complete resection.[191] Nevertheless, there was no improvement in progression-free survival (PFS) or OS at 2, 4, or 6 years.

In reasonably healthy patients with discrete N2 involvement, multimodality treatment with or without surgery offers a reasonable treatment approach. There are no clearly defined subsets of patients in whom surgery is predicted to improve the outcomes. Although there may be a slight oncologic benefit with resection, this may be eclipsed by the increased risk of perioperative mortality. Both patient preferences and local factors are particularly important in selecting the best approach for an individual patient. Guidelines recommend that the entire treatment strategy be defined at the outset by a multidisciplinary discussion.

The standard therapy for patients with good performance who are medically or surgically inoperable is with concurrent platinum-based doublet chemotherapy plus 2 Gy of daily thoracic radiation for a total of 60 Gy over 6 weeks followed by consolidation durvalumab.[186] Commonly used chemotherapy during radiation therapy includes carboplatin plus paclitaxel, cisplatin plus etoposide, cisplatin plus pemetrexed, and carboplatin plus pemetrexed.[192–195] There have been two large RCTs comparing the chemotherapy doublets. In the PROCLAIM study, 599 patients with unresectable stage III NSCLC were randomized to three cycles of cisplatin 75 mg/m^2 plus pemetrexed 500 mg/m^2 every 3 weeks during concurrent thoracic radiation followed by three cycles of consolidation pemetrexed or to cisplatin 50 mg/m^2 on days 1 and 8 plus etoposide 50 mg/m^2 on days 1 to 5 every 4 weeks during the thoracic radiation therapy followed by two cycles of consolidation chemotherapy.[194] Thoracic radiation therapy was administered with 2 Gy per fraction daily for 5 days per week in 30 to 33 fractions for a target of 60 to 66 Gy. The primary end point was OS. The median and 3-year OS were 26.8 months and 40%, respectively, for the pemetrexed arm and 25 months and 37%, respectively, for the etoposide arm (HR, 0.98; $P = 0.83$). In the second trial, 200 patients were randomized to cisplatin plus etoposide to weekly carboplatin at area under the curve (AUC) 2 plus paclitaxel 45 mg/m^2 during concurrent thoracic radiotherapy with 60 to 66 Gy.[196] The primary end point was OS at 3 years. Among the 200 randomized patients, 191 were treated—95 in the cisplatin plus etoposide arm and 96 in the carboplatin plus paclitaxel arm. The 3-year OS was significantly higher in the cisplatin plus etoposide arm (41.1% vs 26%; $P = 0.024$). Cisplatin plus etoposide was associated with a numerically increased median OS (23.3 vs 20.7 months; HR, 0.76; 95% CI, 0.55 to 1.05; $P = 0.09$) and median PFS (14 vs 12 months; $P = 0.1$).

The role for an increased dose of radiation was evaluated by the RTOG-0617 trial, where 544 patients with unresectable stage III

NSCLC were randomized to standard 60 Gy or 74 Gy of thoracic radiotherapy with concurrent weekly carboplatin and paclitaxel with or without cetuximab.[192] The study showed a detriment in the median OS from the increased radiation dose (20.3 months vs 28.7 months; HR, 1.38; 95% CI, 1.09–1.76; $P = 0.004$) and no benefit from the addition of cetuximab (25 months vs 24 months). With the updated results, the 5-year PFS and OS for the standard radiotherapy arm were 18.3% and 32.1%.[197]

The current standard was defined by the PACIFIC study, where 713 patients with unresectable stage III NSCLC who did not have tumor progression after chemoradiotherapy including at least two cycles of a platinum-based chemotherapy doublet defined by the local practice were randomized in a 2:1 ratio to durvalumab 10 mg/kg or placebo every 2 weeks for up to 12 months within 42 days from completion of the chemoradiation.[198] The coprimary end points were PFS and OS. In the initial report, the median PFS was significantly improved with consolidation durvalumab (16.8 vs 5.6 months; 95% CI, 0.42 to 0.65; $P < 0.001$). The rates of pneumonitis of any grade and grade 3 to 4 were 33.9% and 3.4%, respectively, in the durvalumab arm and 24.8% and 2.6%, respectively, in the placebo arm. In the latest update, both the median O (47.5 vs 29.1 months; HR, 0.72; 95% CI, 0.59–0.89) and 5-year OS (42.9% vs 33.4%) were significantly higher in the durvalumab arm.[199] The median PFS (16.9 vs 5.6 months; HR, 0.55; 95% CI, 0.45 to 0.68) and 5-year PFS (33.1% vs 19%) were also significantly improved with durvalumab. Subset analyses showed no OS improvement for patients with *EGFR* mutation, prespecified PD-L1 <25%, or post hoc analysis of PD-L1 <1%, with the latter associated with an unstratified HR of 1.15 (0.75–1.75).

With the establishment of consolidation durvalumab as the standard of care, the next generation of studies is incorporating ICIs during the concurrent chemoradiotherapy.[200] Another new approach is to include immunotherapy doublets in the consolidation phase, with CheckMate-73L (NCT04026412) evaluating nivolumab plus ipilimumab and the SKYSCRAPER-03 (NCT04513925) evaluating tiragolumab plus atezolizumab.

TREATMENT OF STAGE IV

Treatment of Oncogene-driven Non–small-cell Lung Cancer

Lung cancers, particularly lung adenocarcinomas, are archetypes of tumors that are enriched for actionable drivers.[201] An actionable driver is a genomic alteration that is both central to oncogenesis and matches patients to targeted therapy that is active in the clinic. *KRAS* mutations, including KRAS G12C, were first identified in the 1980s. For many years, these mutations were the only known driver alterations in NSCLCs. *EGFR* mutations were discovered in late 2003 and early 2004,[202] and they were the first alterations in lung cancer identified as actionable with targeted therapy.[203] Clinical trials of EGFR TKIs achieved high objective response rates (ORRs) and prolonged PFS compared to platinum doublet chemotherapy.[204] This resulted in the approval of EGFR TKIs for *EGFR*-mutant lung cancers by health-care agencies around the world.

The first actionable fusions in lung cancer were found a few years after the discovery of *EGFR* mutations, and included *ALK*[205], *ROS1*[206] and *RET*.[207] *ALK* fusion-positive lung cancers were the first fusion-driven lung cancers to be extensively explored in clinical trials. Similar to targeted therapy for *EGFR*-mutant lung cancers, ALK TKI therapy achieved high ORRs and prolonged PFS compared to platinum doublet chemotherapy in patients with *ALK* fusion-positive lung cancers.[205] The recognition of amplification as an actionable genomic alteration in lung cancer largely began with the finding in the mid-2000s that *MET* amplification drives resistance to EGFR TKI therapy in *EGFR*-mutant lung cancers. *MET* amplification was found to be a primary lung cancer driver shortly after that.[208] MET TKIs have since been shown to be active in both contexts.[209,210]

The sucess with targeted therapy in other oncogene-driven cancers also influenced the exploration of TKI therapy in patients with lung cancer. The *BRAF* V600E mutation was a recognized melanoma driver for which BRAF inhibitor therapy was active. Echoing these observations, targeted therapy was explored and found to be active, and later approved for BRAF V600E mutant lung cancers.[211] The successes with antibody drug conjugate (ADC) therapy in ERBB2/HER2-dependent breast cancers led to the eventual demonstration that ADCs are active in *ERBB2*-mutant lung cancers.[212]

NTRK1/2/3 fusions were found to be highly enriched in rare cancers and sarcomas, and they later demonstrated to be much less frequently found in lung cancers and other malignancies across adult and pediatric patients. The development of TRKA/B/C (encoded by *NTRK1/2/3*) inhibitors for these fusions thus took a tumor- and age-agnostic development path that resulted in the approval of these TKIs for any cancer, including lung cancer, with an *NTRK1/2/3* fusion across patients of all ages.[213]

Some mutations and fusion-driven lung cancers are associated with characteristic clinical features including age, histology, smoking history, and typical pathologic findings (Table 30.9).

Mutation-driven Lung Cancers

Molecular Heterogeneity

Mutations that drive lung cancer take several forms, including point mutations, insertions, and deletions. Mutations that recur at a high frequency are called *hotspots*. Oncogenic point mutations in *KRAS* (e.g., KRAS G12C or KRAS G12D), *EGFR* (e.g., EGFR L858R), and *BRAF* (e.g., BRAF V600E) lead to amino acid changes that affect the function of the oncoprotein encoded by the mutant gene. Insertions commonly involve *EGFR* (e.g., SVD, NPH, or FQEA exon 20 insertions)[217] and *ERBB2* (e.g., the exon 20 YVMA insertion).[102] These insertions can be of various lengths. The EGFR FQEA (phenylalanine, glutamine, glutamic acid, alanine) insertion and the ERBB2 YVMA (tyrosine, valine, methionine, alanine) insertion are 12 base pair insertions, whereas the EGFR NPH (asparagine, proline, histidine) insertion is a 9 base pair insertion. Deletions are also of varying lengths and commonly involve *EGFR* exon 19 or *MET* exon 14.[208]

Oncogenic mutations can affect various regions within a gene. In lung cancers, *EGFR* mutations occur at exon 18 (e.g., EGFR G719X), exon 19 (e.g., *EGFR* exon 19 deletions), exon 20 (e.g., EGFR S768I or EGFR exon 20 insertions), or exon 21 (EGFR L858R).[202] In contrast, *ERBB2* mutations predominantly occur at exon 20.[102] Point mutations result in changes at specific codons of the mutant protein. Indels may target a particular stretch of amino acid residues. *EGFR* exon 20 insertions typically occur between residues 769 and 775,[102] whereas *ERBB2* exon 20 insertions typically occur between codons 775 and 881.[102] Finally, mutations can affect specific protein domains. *EGFR* exon 20 insertions, for example, can affect the C-helix and the near and far loop after the C-helix of the EGFR protein.[102]

Oncogenesis: Thematically, oncogenic mutations in lung cancer result in constitutive activation of the mutant protein. As evidence of their oncogenic nature, all actionable mutations discussed in this section typically occur in a mutually exclusive fashion and can transform cell lines in vitro and result in the formation of tumors in vivo.[218] As has been demonstrated with *EGFR* mutations,[219] mouse models genetically engineered to contain these drivers in normal lung epithelia develop lung adenocarcinomas.[220]

Classical sensitizing *EGFR* mutations are thought to shift the equilibrium of EGFR to a constitutively active state that allows ligand-independent receptor dimerization.[221] This leads

TABLE 30.9

Actionable Genomic Alterations

Gene	Alteration	Molecular Features	Clinicopathologic Features
EGFR	Mutation	"Classical sensitizing" mutations *EGFR* exon 19 deletions *EGFR* L858R "Uncommon/Atypical" mutations Include *EGFR* S768I, *EGFR* L861Q, and *EGFR* G719X (the latter two are most common) *EGFR* exon 20 insertions including the SVD, NPH, and FQEA insertions	10%–15% (Caucasian) and 50%–60% (Asian) adenocarcinomas; most commonly in never or former light smokers with a younger median age than smoking-associated lung cancers[202] ≤1% of NSCLC adenocarcinomas in Caucasians (the most common mutations make up 7%–10% of all *EGFR* mutations); similar phenotype to classical sensitizing mutations [214] ≤1% of NSCLC adenocarcinomas in Caucasians (4%–10% of all *EGFR* mutations); similar phenotype to classical sensitizing mutations[102]
HER2	Mutation	*ERBB2* exon 20 insertions (most common) or point mutations Includes the YVMA insertion	1% of adenocarcinomas, similar phenotype to *EGFR*-mutant lung cancers[102]
KRAS	Mutation	*KRAS* G12C	13%–14% of adenocarcinomas for *KRAS* G12C; for *KRAS* mutations of any type, transitions are more common in never smokers and transversions are more common in smokers[215]
MET	Mutation	*MET* exon 14 alterations Splice site alterations (most common, predominantly indels) Point mutations Large deletions	4% of adenocarcinomas; common phenotype includes lung adenocarcinomas from older patients who have had a more substantial cigarette smoking history; these mutations are also enriched in sarcomatoid carcinomas[208]
	Amplification	*MET* copy number increases High-level *MET* amplification: >5-fold increase in gene copy number or *MET/CEP7* ratio >5	<1% of adenocarcinomas when found de novo, clinicopathologic features not well established[208]
BRAF	Mutation	*BRAF* V600E	1%–2% of adenocarcinomas for *BRAF* V600E[216]; many patients have a prior or current history of smoking, most are adenocarcinomas, and many non-V600E class 2 and 3 *BRAF* mutations are found in lung cancers
ALK	Fusion	Molecularly heterogeneous with different 5′ partners of varying lengths and more conserved 3′ breakpoints *EML4-ALK* (most common) has several variants	4% of NSCLCs; common phenotype includes lung adenocarcinomas from younger patients who are never or former light smokers[205]
ROS1	Fusion	Molecularly heterogeneous *CD74-ROS1* (most common)	1%–2% of NSCLCs; phenotype like *ALK* fusion-positive lung cancers[206]
RET	Fusion	Molecularly heterogeneous *KIF5B-RET* (most common)	1%–2% of NSCLCs; phenotype like *ALK* fusion-positive lung cancers[207]
NTRK1/2/3	Fusion	Molecularly heterogeneous No dominant fusion type *ETV6-NTRK3* is common for *NTRK3* fusions	<1% of NSCLCs; phenotype like *ALK* fusion-positive lung cancers[213]

NSCLC, non–small-cell lung cancer.

to downstream signaling through the MAPK pathway that promotes cancer growth. Oncoprotein activation and increased downstream effector signaling are likewise observed with other *EGFR* mutations,[102] *ERBB2* mutations,[102] and BRAF V600E.[216] KRAS G12C mutations are unique in that activation occurs via disruption of the guanine exchange cycle, locking the KRAS G12C mutant protein in the inactive GDP-bound form.[222]

MET exon 14 alterations are exemplary of how aberrant splicing can drive oncogenesis in solid tumors.[223] Most of these alterations are mutations that occur at intronic splice sites flanking exon 14 of *MET*. Aberrant splicing leads to exon 14 exclusion in mRNA. The final protein product is a truncated MET receptor that lacks the juxtamembrane domain encoded by *MET* exon 14, which encodes the site for binding of the E3 ubiquitin ligase CBL that is important for MET degradation. The absence of this domain in *MET* exon 14–altered lung cancers can result in decreased degradation and increased MET on the cell surface that drives oncogenesis.[104]

Clinicopathologic Features: The frequency of *EGFR* mutations varies by ethnicity, being detected in 10% to 15% of lung adenocarcinomas in people of European or African descent.[95] *EGFR* mutations are more prevalent in East Asians, where they may be present in 40% to 50% of adenocarcinomas and are also increased in those with Native American ancestry,[96,224,225] leading to a high prevalence of *EGFR* mutations in Latin America.[96] The oncogenic mutations described thus far, including *EGFR*, *ERBB2*, *KRAS*, *MET*, and *BRAF* mutations, are identified most commonly in adenocarcinomas. Whereas many mutation-driven cancers are nonmucinous

adenocarcinomas, diverse *KRAS* mutations are enriched in invasive mucinous adenocarcinomas.[226] Patients with mutation-driven lung cancers tend to be never-smokers or have a current or former history of limited cigarette smoking exposure.[202] The median age of patients diagnosed with these lung cancers tends to be lower than that of the historical median age of unselected NSCLCs.[227] These clinicopathologic features, however, do not characterize all patients with oncogene-driven lung cancers. Oncogenic mutations have less commonly been identified in patients with squamous cell lung cancers, mixed histology lung cancers, lung cancers with neuroendocrine features, and SCLCs.[228] Patients with mutation-driven lung cancers can also present at various ages and with a substantial prior or current history of cigarette smoking.

Variations in smoking history have been associated with specific mutations. As an example, *KRAS* transition mutations with G to A nucleotide change such as GGT to GAT that results in *KRAS* G12D are more common in never-smokers, whereas transversion mutations with G to T nucleotide change such as GGT to TGT that results in *KRAS* G12C are more frequent in smokers.[215] Notably, *MET* exon 14–altered lung cancers present with unique clinicopathologic features compared to other oncogene-driven lung cancers, including a higher median age and a more substantial smoking history. *MET* exon 14 alterations are also enriched in patients with sarcomatoid carcinomas of the lung.[208]

Fusion-driven Lung Cancers

Structure: Many fusions/rearrangements that drive oncogenesis in lung cancer share structural features.[229] Fusions involving *ALK*, *ROS1*, *RET*, and *NTRK1/2/3* typically harbor these genes in the downstream or 3′ position with an intact kinase domain and are classified as classical 3′ fusions. When breakpoints are conserved, the downstream gene commonly begins with a particular exon (e.g., exon 20 for *ALK*) and ends with the portion encoding the C terminus. Occasionally, the downstream gene may harbor the transmembrane domain, putatively allowing localization and tethering to the tumor-cell surface.[230] Other fusions have unique structural features. *FGFR1/2/3* fusions, for example, can harbor the kinase-containing gene in the 5′ position, as is the case for *FGFR3-TACC*, classified as a 5′ fusion.[229] *ERBB2* or *ERBB4* have been found in the 5′ position in cancers that contain fusions involving either gene.[231] *NRG1* fusions do not typically contain a kinase. The *NRG1* gene encodes the EGF-like domain that binds ERBB3.[232]

Partner Gene: A wide variety of gene partners have been identified.[230] The most common 5′ partners are *EML4* for *ALK* fusions, *CD74* for *ROS1* fusions, *KIF5B* for *RET* fusions, *ETV6* for *NTRK3* fusions, and *CD74* for *NRG1* fusions. When the upstream partner is found on the same chromosome, the fusion is intrachromosomal. This commonly occurs with *ALK* fusions, where in *EML4-ALK*, both genes are on chromosome 2, as well as with *RET* fusions, where in *KIF5B-RET*, both genes are on chromosome 10. Intrachromosomal fusions are formed via inversion (paracentric or pericentric), deletion, or duplication. In contrast, when the upstream partner is on a separate chromosome, the fusion is interchromosomal and considered a translocation. This frequently occurs with *ROS1* fusions, many of which, including *CD74-ROS1*, are translocations.[206] Fusion variants have different upstream or downstream gene partner lengths. This has been best studied in *ALK* fusions where several *EML4-ALK* variants have been identified, some of which are expressed as multiple isoforms. Approximately 90% of *EML4-ALK* fusions are either variant 1 (*EML4* exon 13 fused to *ALK* exon 20), variant 2 (*EML4* exon 20 fused to *ALK* exon 20), or variant 3a/b (*EML4* exon 6a fused to *ALK* exon 20).[233]

Oncogenesis: Gene partners can contribute motifs that result in ligand-independent dimerization, a common mechanism of chimeric oncoprotein activation. *RET* fusions, for example, have 5′ partners that harbor coiled-coil, LisH, WDR, or SAM domains that mediate oncoprotein homodimerization.[207] In contrast, most *ROS1* fusion partners lack dimerization motifs, suggesting that other mechanisms such as increased chimeric oncoprotein expression, mediated by a promoter in the 5′ element or exclusion of an inhibitory domain, may be involved.[206] The biology of *NRG* fusions is unique.[234] The chimeric oncoprotein encoded by *NRG1* fusions is localized to the tumor-cell surface, presents the EGF-like domain, and activates ERBB2-ERBB3 heterodimers in a paracrine or autocrine fashion. *NRG2* fusions, however, may preferentially bind ERBB4.[235]

ALK, *ROS1*, and *RET* fusions, and presumably an activating in-frame fusion, result in the transformation of NIH/3T3 fibroblasts and anchorage-independent growth in vitro.[205–207] The expression of these fusions leads to tumor formation when engrafted subcutaneously in nude mice in vivo. Adenomas or adenocarcinomas develop in the lungs of transgenic mice that harbor these fusions in respiratory epithelial cells. When found de novo, oncogenic fusions tend to be mutually exclusive with other oncogenes.[229]

Pathologic Features: Oncogenic fusions are most commonly found in NSCLCs, particularly lung adenocarcinomas.[230] Some of these adenocarcinomas have signet ring cell features, as has been described for those with *ALK*, *ROS1*, or *RET* fusions. In addition, the unique variant of invasive mucinous adenocarcinoma that is *KRAS* wild type is known to harbor a variety of activating fusions.[226] Gene fusions are less commonly found in other histologies, such as squamous cell lung carcinomas, adenosquamous lung carcinomas, NSCLC not otherwise specified, or lung cancers with neuroendocrine features. Therefore, fusion-inclusive diagnostic testing should thus be considered across all lung cancer histologies when feasible.

Clinical Features: Most patients with fusion-positive lung cancers are diagnosed at a younger median age (e.g., 52 years for *ALK* fusion-positive lung cancers) compared to patients with unselected NSCLCs (70 years).[227] Fusion-positive lung cancers are also associated with never-smokers or patients with a former light smoking history defined as <10 pack-years. Nevertheless, these features are not present in all patients with fusion-positive lung cancers, and activating oncogenic fusions have been identified in lung cancers from older patients or those with a substantial prior history of cigarette smoking.[205–207]

Fusions can be missed, even by comprehensive DNA-based NGS, for various reasons. *ROS1* fusions have repetitive elements where breakpoints occur, and *NTRK3* fusion breakpoints occur in large intronic regions that cannot effectively be interrogated by DNA-based assays. Complementary RNA-based NGS can surmount these challenges and maximize the likelihood of detecting gene fusions in lung cancer.[236]

Amplification-driven Lung Cancers

Increases in gene copy number can drive oncogenesis. Unlike mutations and fusions where an alteration is either present or absent in a binary fashion, copy number increases are a continuous variable.[104] Identifying appropriate cutoff points for "positivity" are thus critical. Another major point is that copy number gains can arise from two different processes, polysomy or amplification. Polysomy occurs when multiple copies are present of a chromosome that harbors the gene of interest. This may occur from chromosomal or whole genome duplication. In contrast, amplification occurs when the gene of interest is amplified in a more focal setting. Amplification is thus thought to more truly depend on the gene of interest for oncogenesis in comparison to polysomy.[208]

MET amplification is an actionable alteration in lung cancer. Much of the early literature on *MET* amplification was based on FISH testing. Calculating the ratio of *MET* to *CEP7* (the centromere of chromosome 7 that also harbors *MET*) can identify true amplification, whereas in polysomy, *MET* would not be

selectively increased relative to *CEP7*. Although no consensus exists, MET amplification has been classified via the following *MET*/*CEP7* ratios into low (≥1.8 to ≤2.2), intermediate (>2.2 to <5), and high (≥5).[209] NGS is now more commonly used to identify *MET* amplification. Most assays will rely on the read depth approach to determine the copy number, assuming that the read depth signal is proportional to the number of chromosomal segment copies in the specimen.[104] Although a greater than fivefold increase in the gene copy number has been used to define MET dependence by some investigators, there is currently no consensus on cutoffs for amplification via NGS.

De novo *MET* amplification is found in a small proportion of NSCLCs. High amplification defined by more than a sevenfold increase is thought to be more indicative of oncogenic dependence on MET,[104] particularly since high-level *MET* amplification is less likely to overlap with other oncogenic drivers.[237] The clinicopathologic features of patients with *MET*-amplified cancers are not well established, although many patients with these cancers have lung adenocarcinomas.

Targeted Therapy

Targeted therapies involve the use of small molecules or antibodies. The small molecule may be a TKI or direct inhibitor, and selective or nonselective (Fig. 30.4).

Small Molecules: Small molecules are the most commonly approved targeted therapies for oncogene-driven lung cancers.[238] Most are kinase inhibitors, including osimertinib, afatinib, dacomitinib, erlotinib, gefitinib, mobocertinib, capmatinib, tepotinib, crizotinib, dabrafenib, trametinib, alectinib, ceritinib, brigatinib, lorlatinib, selpercatinib, pralsetinib, larotrectinib, and entrectinib. Other targets for small-molecule inhibitors include the serine/threonine kinase BRAF, which is targeted by dabrafenib. The KRAS G12C inhibitors, including sotorasib and adagrasib, irreversibly inhibit the inactive state of KRAS G12C, interfering with signal transduction and oncogenic signaling.[222]

TKIs have been classified by generation.[239] The first-generation TKIs erlotinib and gefitinib are reversible inhibitors of active in classical sensitizing *EGFR* mutations including exon 19 deletion or L858R. Second-generation TKIs, including afatinib and dacomitinib, are more potent covalent inhibitors of mutant *EGFR* with a narrower therapeutic window between mutant and wild-type *EGFR*. The third-generation TKI osimertinib is an irreversible inhibitor of mutant EGFR with both improved central nervous system (CNS) coverage and increased EGFR wild-type sparing leading to a wider therapeutic window.[240] The term *next-generation TKI* has often been loosely used to refer to drugs that improve on prior generations without referring to a particular TKI generation.

TKI generation classifications are not, however, standard across targeted therapies. In *ALK* fusion-positive lung cancers, crizotinib

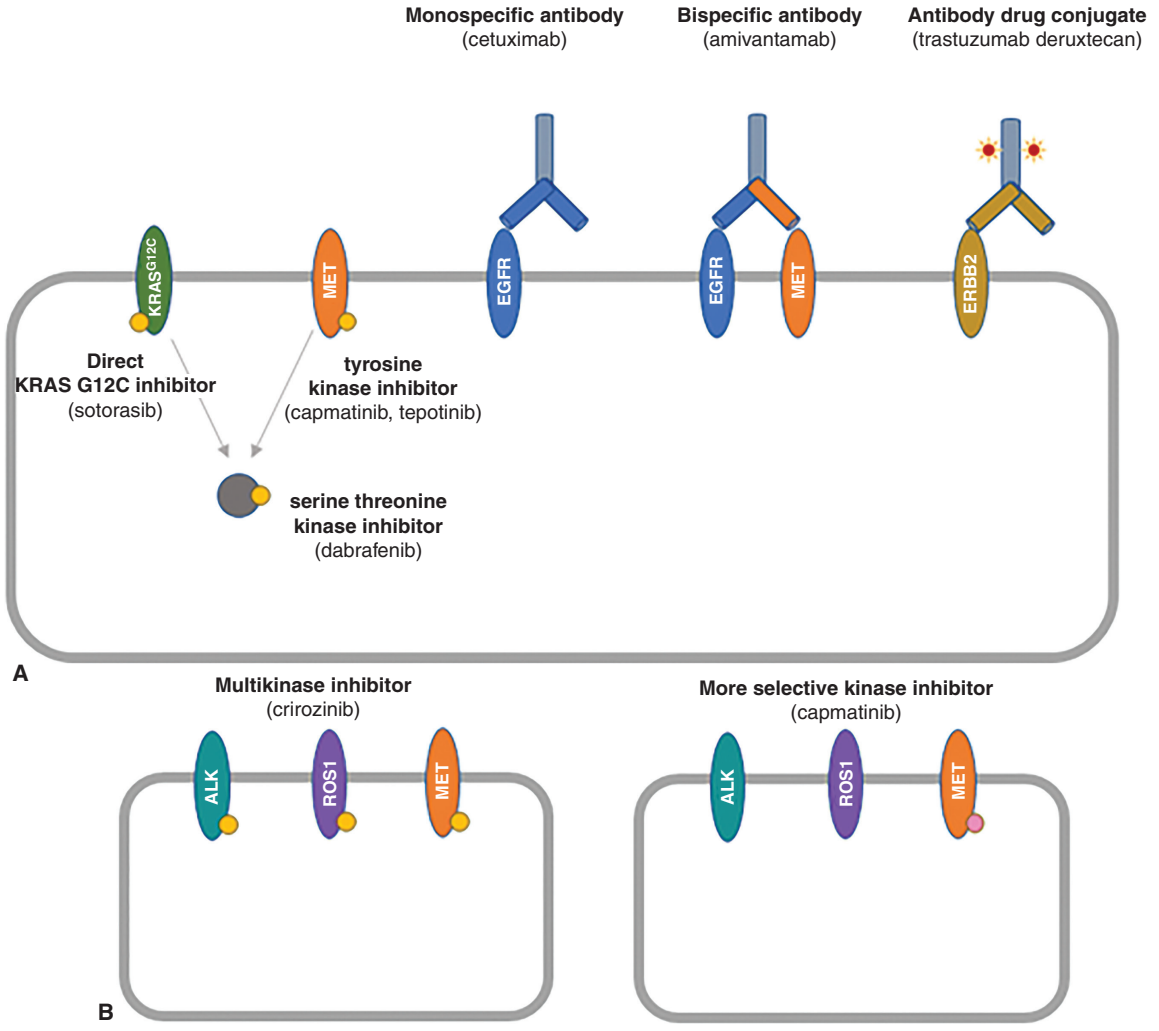


Figure 30.4 Modalities of targeted therapy. **A.** Small molecule targeted drugs may be broadly classified as direct inhibitors, tyrosine kinase inhibitors or serine-threonine kinase inhibitors. Monoclonal antibodies may be monospecific, bispecific or antibody-drug conjugates. **B.** Tyrosine kinase inhibitors may selective or non-selective (multikinase inhibitors).

is a first-generation TKI. Second-generation ALK TKIs, including alectinib, ceritinib, and brigatinib, have improved potency against ALK compared to crizotinib, better CNS coverage, and activity against selected *ALK* resistance mutations that can emerge with crizotinib use. The third-generation ALK TKI lorlatinib has the same features as second-generation TKIs but with expanded *ALK* resistance mutation coverage.[106] Unlike EGFR TKIs, all generations of ALK TKIs are reversible inhibitors of ALK and require inhibition of the wild-type ALK kinase domain that is present de novo in *ALK* fusion-positive lung cancers. Finally, the generation of a particular agent may be target specific. As an example, ceritinib is considered a second-generation ALK TKI but a first-generation ROS1 TKI.[206]

TKIs can also be classified in terms of selectivity (Table 30.10). In *RET* fusion-positive lung cancers, older RET inhibitors were multikinase agents with inhibitory activity against many non-RET targets, including vascular endothelial growth factor receptor 2 (VEGFR2), MET, ROS1, EGFR, and KIT. In contrast, newer selective RET inhibitors such as selpercatinib and pralsetinib were rationally designed to inhibit RET and avoid many of the non-RET targets.[207] Other oncogene-driven lung cancers for which selective TKIs are available include MET-dependent lung cancers (capmatinib, tepotinib)[208] and *NTRK* fusion-positive lung cancers (larotrectinib).[213]

TKIs can also be classified by mechanism of binding to the targets.[241] Type I inhibitors bind the active (DFG-in) protein kinase conformation. Most TKIs developed for oncogene-driven lung cancers are type I TKIs. Type I inhibitor subtypes exist within targeted therapy families. For MET TKIs, crizotinib is a type Ia drug that inhibits MET in addition to other kinases. Capmatinib, tepotinib, and savolitinib are examples of type Ib drugs that are more selective for MET than type Ia inhibitors. Type II inhibitors bind the inactive (DFG-out) protein kinase conformation, and many of the available agents are multikinase inhibitors. Cabozantinib is an example of a type II inhibitor.[208]

Large Molecules: Antibody-based therapies are classified as large molecules. Similar to natural antibodies, many of the therapeutic antibodies that were initially explored preclinically or clinically in lung cancer were monoclonal and monospecific.[208] Monospecific antibodies bind a singular antigen or epitope. Monospecific EGFR antibodies include cetuximab (a recombinant chimeric [human/mouse] antibody) and necitumumab (a fully humanized IgG1 antibody). Bevacizumab is a humanized anti-VEGF antibody.[242] In contrast, bispecific antibodies recognize two different epitopes.[243] These epitopes can be on the same antigen or different antigens. Examples of the latter include amivantamab (a fully human bispecific anti-EGFR and anti-MET antibody)[244] and zenocutuzumab (a bispecific anti-ERBB2 and anti-ERBB3 antibody).[245]

ADCs are composed of a monoclonal antibody linked to a cytotoxic drug.[246] The modular components of ADCs include the antibody, the linker, and the payload, which is also known as the warhead. The antibody backbone can be one of the four human IgG subclasses (IgG1–4). Most approved ADCs have an IgG1 backbone. For trastuzumab deruxtecan (T-DXd) and trastuzumab emtansine (T-DM1), the trastuzumab backbone is a humanized anti-HER2 IgG1 antibody. The patritumab backbone of patritumab deruxtecan is a fully human anti-HER3 IgG1 antibody.[247] Linkers can be classified into cleavable and noncleavable. Cleavable linkers are designed to release a payload in a tumor-specific environment. T-DXd and sacituzumab govitecan have enzyme-cleavable linkers. Noncleavable linkers are reliant on lysosomal degradation for payload release. T-DM1 has a noncleavable linker.[246] Payloads are typically highly potent chemotherapeutic agents delivered at subnanomolar concentrations. Common payloads include the auristatins (monomethyl auristatin E [MMAE] and monomethyl auristatin F [MMAF]), calicheamicins, maytansinoids (emtansine/DM1), and camptothecins (deruxtecan/DXd). The auristatins and maytansinoids disrupt microtubule dynamics. Calicheamicins and camptothecins cause DNA strand breaks. The camtothecin analog and exatecan derivative DXd lead to DNA strand breaks by inhibiting topoisomerase I. The drug-to-antibody ratio reflects the average number of payloads attached to each monoclonal antibody.[246]

Rational Drug Design: Early-generation TKI therapy was first explored for many oncogene-driven lung cancers. In classical sensitizing *EGFR* mutations, several pivotal studies elucidated the activity of first-generation agents (gefitinib[203] and erlotinib[248]) and second-generation agents (afatinib[249] and dacomitinib). In *ALK* fusion-positive lung cancers, the first-generation ALK TKI crizotinib was likewise the first agent to be explored. With high ORRs and durable PFS, these early-generation TKIs established prior care standards for advanced, treatment-naïve patients with these cancers.

History has demonstrated, however, that improved drug design can change care standards for patients with oncogene-driven lung cancers.[250] Next-generation TKI therapy—agents that featured increased CNS and resistance coverage—improves outcomes compared to earlier-generation TKIs.[250] In the phase III trial that randomized patients with *EGFR*-mutant lung cancer to the third-generation EGFR TKI osimertinib or first-generation EGFR TKIs (gefitinib or erlotinib), an improvement in the primary end point of PFS was achieved, with the former being more effective.[251] Significant improvement in PFS was also observed in three separate phase III trials that randomized patients to a second-generation alectinib[252] or brigatinib,[253] or the third-generation lorlatinib[254] ALK TKI or the first-generation ALK TKI crizotinib. As such, osimertinib and alectinib, brigatinib, or lorlatinib are preferred choices for treatment-naïve *EGFR*-mutant and *ALK* fusion-positive lung cancers, respectively.

Combination TKI therapy may improve outcomes compared to single-agent TKI therapy. In *BRAF* V600E-mutant lung cancers, single-agent BRAF TKIs such as dabrafenib or vemurafenib were first explored. Although these drugs were active with an ORR of 30% to 40% and PFS of 5 months to 7 months, the combination of the BRAF TKI dabrafenib with an MEK TKI trametinib was associated with improved results.[211] This echoed the improved activity achieved in *BRAF* V600E-mutant melanomas when moving from single-agent to combination therapy.[255]

Beyond improvements in TKI design and the use of TKI combinations, a class shift from small to large molecules has also been observed. The bispecific antibody amivantamab was the first large-molecule targeted therapy approved for an oncogene-driven lung cancer subset, the *EGFR* exon 20 mutant lung cancers.[244] Anti-ERBB3 therapy has been explored and is active in *NRG1* fusion-positive lung cancers.[256] Furthermore, the ADC trastuzumab deruxtecan showed substantial activity in *ERBB2*-mutant lung cancers when compared to the historical activity of anti-ERBB2 TKIs such as afatinib and dacomitinib.[212]

Activity: Most targeted therapies are associated with high response rates in molecularly selected patients (Table 30.11). In patients with advanced oncogene-driven lung cancer who are targeted therapy naïve, the overall activity of systemic therapy can be broadly grouped into two categories, recognizing that not all driver-treatment pairs fall neatly within these groups. The first category is associated with ORRs of 60% to 80% and a median PFS >1 year and includes (1) osimertinib or erlotinib plus bevacizumab for classical sensitizing EGFR mutations, (2) alectinib, brigatinib, ceritinib, or lorlatinib for *ALK* fusions, (3) entrectinib or ceritinib for *ROS1* fusions, (4) selpercatinib or pralsetinib for *RET* fusions, and (5) larotrectinib or entrectinib for *NTRK1/2/3* fusions. The second category is associated with ORRs of 30% to 50% and a median PFS <1 year and includes (1) amivantamab or mobocertinib for *EGFR* exon 20 mutations, (2) trastuzumab deruxtecan for *ERBB2* mutations, (3) sotorasib for *KRAS* G12C mutations, (4) tepotinib or crizotinib for *MET* exon 14 alterations, and (5) capmatinib or crizotinib for *MET* amplification.

TABLE 30.10

Therapeutic Targets

	EGFR	ERBB2	ERBB3	KRAS^{G12C}	MET	BRAF	MEK	ALK	RET	ROS1	TRKA/B/C
Multikinase inhibitors											
Afatinib	X	X									
Dacomitinib	X	X									
Mobocertinib	X	X									
Alectinib								X	X		
Ceritinib								X		X	
Brigatinib	X							X		X	
Lorlatinib								X		X	
Crizotinib					X			X		X	
Cabozantinib					X				X	X	
Entrectinib								X		X	X
Repotrectinib								X		X	X
More selective kinase inhibitors											
Osimertinib	X										
Erlotinib	X										
Gefitinib	X										
Capmatinib					X						
Tepotinib					X						
Dabrafenib						X					
Trametinib							X				
Selpercatinib									X		
Pralsetinib									X		
Larotrectinib											X
Direct KRAS G12C inhibitors											
Sotorasib				X							
Adagrasib				X							
Monospecific monoclonal antibodies											
Cetuximab	X										
Necitumumab	X										
Bispecific antibodies											
Amivantamab	X				X						
Zenocutuzumab		X	X								
Antibody drug conjugates											
Trastuzumab emtansine		X									
Trastuzumab deruxtecan		X									
Patritumab deruxtecan			X								

Agents are grouped by class. Note that although some multikinase inhibitors may harbor activity against a particular target, they may not be effective inhibitors of that target. For example, alectinib can technically inhibit RET but is not highly active in *RET* fusion-positive lung cancers.

Many targeted therapies, and TKIs in particular, are active in the CNS. In *EGFR*-mutant and *ALK* fusion-positive lung cancers, second- or third-generation TKIs including osimertinib,[251] alectinib,[252] brigatinib,[253] and lorlatinib[254] have improved CNS outcomes, defined as intracranial PFS or time to the acquisition of CNS disease in patients without baseline brain metastases compared to earlier-generation agents in phase III trials. Other TKIs with high intracranial activity include entrectinib,[269] selpercatinib,[272] pralsetinib,[273] and larotrectinib.[276] Beyond parenchymal brain metastases, TKIs can be active against leptomeningeal disease. This has been most extensively explored for osimertinib in *EGFR*-mutant lung cancers in a prospective trial,[277] but other TKIs, including lorlatinib[267] or selpercatinib,[278] have also demonstrated proof-of-concept clinical activity against leptomeningeal disease.

Safety: Selected toxicity from targeted therapy can be predicted by understanding the consequences of inhibiting the primary kinase of interest in non-neoplastic tissues. These are called *on-target side effects*. EGFR inhibition is associated with rash, usually maculopapular; diarrhea; changes in the hair such as hypertrichosis; and

TABLE 30.11

Targeted Therapy Activity

Drug	Class	Patients/Tumors	ORR	Median PFS	Year Approved[a]
"Classical sensitizing" *EGFR* mutation					
Osimertinib	3rd Gen EGFR TKI	Treatment naïve[251] EGFR TKI treated (T790M+)[251]	80% 59%	18.9 mo 10.1 mo	2018 2015
Erlotinib and ramucirumab	1st Gen EGFR TKI, VEGF-A mAb	Treatment naïve[257]	76%	19.4 mo	2020
"Atypical/uncommon/nonresistant" *EGFR* mutation					
Afatinib	2nd Gen EGFR TKI	Treatment naïve and pretreated[258]	71%	10.7 mo	2018
***EGFR* exon 20 mutation**					
Amivantamab	EGFR and MET BsAb	Post platinum-based chemotherapy[244]	40%	8.3 mo	2021
Mobocertinib	EGFR TKI	Post platinum-based chemotherapy[259]	28%	7.3 mo	2021
***ERBB2* mutation**					
Trastuzumab deruxtecan	ERBB2 ADC	Systemic therapy pretreated[212]	55%	8.2 mo	
KRAS G12C mutation					
Sotorasib	KRAS G12C inhibitor	Systemic therapy pretreated[260]	37%	6.8 mo	2021
***MET* exon 14 alteration**					
Capmatinib	Type Ib MET TKI	Treatment naïve[210] Systemic therapy pretreated[210]	68% 41%	12.4 mo 5.4 mo	2020 2020
Tepotinib	Type Ib MET TKI	Treatment naïve and pretreated [261]	46%	8.5 mo	2021
Crizotinib	Type Ia MET TKI	Treatment naïve and pretreated[262]	32%	7.3 mo	
***MET* amplification**					
Capmatinib	Type Ib MET TKI	Treatment naïve, GCN ≥10[210] Pretreated, GCN ≥10[210]	40% 29%	4.2 mo 4.1 mo	
Crizotinib	Type Ia MET TKI	Treatment naïve and pretreated, MET/CEP7 ≥4[209]	38%	6.7 mo	
BRAF V600E mutation					
Dabrafenib and trametinib	BRAF TKI and MET TKI	Treatment naïve[211]	64%	10.9 mo	2017
***ALK* fusion**					
Alectinib	2nd Gen ALK TKI	Treatment naïve[252,263] Crizotinib pretreated[264]	83%[254] 50%	34.8 mo[265] 8.9 mo	2017 2015
Brigatinib	2nd Gen ALK TKI	TKI naïve[253] Crizotinib pretreated	74% 53%	24.0 mo —	2020 2017
Ceritinib	2nd Gen ALK TKI	TKI naïve[265] Crizotinib pretreated[266]	73% 56%	16.6 mo 6.9 mo	2017 2014
Lorlatinib	3rd Gen ALK TKI	Treatment naïve[254] TKI pretreated[b,267,268]	76% 39%	Not reached 6.9 mo	2021 2018
***ROS1* fusion**					
Entrectinib	ROS1 TKI	TKI naïve[269]	77%	19.0 mo	2019
Ceritinib	ROS1 TKI	TKI naïve[270]	67%	19.3 mo	
Crizotinib	ROS1 TKI	TKI naïve[271]	72%	19.3 mo	2016
Lorlatinib	ROS1 TKI	TKI pretreated[c,267]	35%	8.5 mo	
***RET* fusion**					
Selpercatinib	RET TKI	Treatment naïve[272] Platinum doublet pretreated[272]	85% 64%	Not reached 19.3 mo	2020
Pralsetinib	RET TKI	Treatment naïve[273] Platinum doublet pretreated[273]	79% 62%	13.0 mo 16.5 mo	2020
NTRK1/2/3 fusion					
Larotrectinib	TRK TKI	TKI naïve[274]	73%	35.0 mo	2018
Entrectinib	TRK TKI	TKI naïve[275]	57%	11.2 mo	2019

[a]Accelerated/full approval by the US FDA when applicable.
[b]Only outcomes in patients treated with ≥2 ALK TKIs shown.
[c]Only crizotinib pretreated outcomes shown.
The objective response rate (ORR) and median progression-free survival (PFS) of various targeted therapy agents in patients with metastatic or locally advanced oncogene-driven lung cancers are summarized based on data from the initial approval of these targeted therapies or from seminal clinical trials, including those with regulatory intent. A selection of approved drugs is shown. Beyond approved agents, drugs that are listed in one or more national or societal treatment guidelines are included.
Gen, generation; GCN, gene copy number. Drug classes are as follows: TKI, tyrosine-kinase inhibitor; mAb, monoclonal antibody, BsAB, bispecific antibody; ADC, antibody-drug conjugate.

paronychia.[279] These toxicities have been observed with small- or large-molecule targeted therapies with anti-EGFR activity such as osimertinib, afatinib, erlotinib, and amivantamab.[244] A wider therapeutic index or window can help decrease toxicity. Osimertinib, for example, has a more substantial window between the inhibition of mutant EGFR and wild-type EGFR compared to earlier-generation TKIs, leading to a decreased frequency of toxicities due to wild-type EGFR sparing.[251] MET inhibitors are associated with peripheral edema.[210,261,262] Single-agent BRAF inhibition can result in keratosis and squamous cell carcinomas of the skin most likely due to a paradoxical activation of the MAPK pathway in non-neoplastic tissues[280] and as such can be attenuated with the concurrent administration of an MEK inhibitor such as trametinib. MEK inhibition can cause ocular toxicity such as central serous retinopathy, neurosensory retinal detachment, and rarely central retinal vein occlusion.[281] TRKA/B/C inhibition with larotrectinib, entrectinib, and repotrectinib has been associated with dizziness, ataxia, orthostasis, peripheral or perioral neuropathy, increased appetite with weight gain, cognitive changes, and withdrawal pain with temporary or permanent TRK TKI discontinuation.[282] Not every kinase is associated with a unique toxicity profile when inhibited. As an example, the clinical consequences of the inhibition of ALK and ROS1 are not well understood.[206]

In contrast, off-target toxicities result from the inhibition of other cell surface or intracellular proteins apart from the oncoprotein of interest. When the activity of the multikinase RET inhibitor cabozantinib was explored in *RET* fusion-positive lung cancers, hypertension and palmar-plantar erythrodysesthesias (hand-foot syndrome) were observed, likely due to the inhibition of VEGFR2.[283] Hypopigmentation may also observed with cabozantinib due to KIT inhibition. The tolerability was increased with the development of selective RET inhibitors such as selpercatinib and pralsetinib due to a reduced activity in non-RET targets.[207]

Some toxicities are not clearly related to targets. KRAS G12C, for example, is not thought to exist in non-neoplastic tissues, and a direct and selective KRAS G12C inhibitor is not conceptually expected to result in toxicity. Nevertheless, treatment-related gastrointestinal toxicities have been reported for both adagrasib and sotorasib.[260] Other examples of non–target-related toxicities include nausea, vomiting, elevated liver enzymes, interstitial lung disease or pneumonitis, dyslipidemia with lorlatinib,[284] hypogonadism with crizotinib,[285] and early-onset pulmonary events with brigatinib.[253]

Resistance to Targeted Therapy

Primary Resistance: Resistance to targeted therapy may be primary or acquired (Table 30.12). Primary resistance occurs when little to no benefit is observed with targeted therapy in a specific genomic context. Structure-function dynamics can mediate primary resistance,[286] creating a mismatch between a particular mutation type and a targeted therapy. Examples include resistance of *EGFR* exon 20 insertions to first-generation EGFR TKIs[286] and resistance to the type I BRAF inhibitors vemurafenib and dabrafenib in class 2 or 3 non-V600E-mutant lung cancers.[287] Although the type I BRAF inhibitors work well against the sensitizing *BRAF* V600E mutation that encodes an oncoprotein that functions as a monomer, they are not active against class 2 *BRAF* alterations (e.g., *BRAF* K601E/N/T, L597Q/V, G469A/V/R, and G464V/E mutations, or *BRAF* fusions) that function as homodimers. When a type I BRAF inhibitor binds the first dimer, it decreases the likelihood of further drug binding to the second dimer, thus rendering the drug ineffective. Class 3 *BRAF* alterations (e.g., *BRAF* G466V/E/A and D594N/G/A/H mutations) can encode crippled oncoproteins with low or no activity with dimer dependency that are further complicated by dependence on RAS.[288] These factors likewise impart primary resistance to type I BRAF inhibitors.

Acquired Resistance: Acquired resistance to targeted therapy occurs when oncogene-driven tumors develop resistance after an initial period of benefit. Although acquired resistance definitions can vary, one of the first sets of criteria was developed in classical sensitizing *EGFR*-mutant lung cancers.[289] Patients who met criteria for acquired resistance to an EGFR TKI included those who had previously received EGFR TKI therapy, had objective clinical benefit from TKI therapy defined as objective response or stable disease for at least 6 months, and developed systemic progression of disease while on continuous TKI therapy.

Several broad mechanisms may mediate acquired resistance. The first major mechanism is called *on-target* or *kinase-intrinsic resistance*, whereby the cancer acquires an alteration or alterations affecting the primary oncogene under the selective pressure of targeted therapy.[229] These cancers maintain dependence on the primary oncogene for growth. Acquired on-target resistance mechanisms to TKI therapy commonly take the form of new or secondary mutations affecting the kinase domain of the primary oncogene. These mutations can result in steric hindrance that prevents effective TKI binding, changes in affinity for substrates such as ATP, and changes in the preferred conformation of the mutant kinase.[206] The classification of these mutations can be based on the regions of the oncoprotein with resultant amino acid substitutions including gatekeeper mutations, solvent front mutations, or xDFG mutations (see Table 30.12). Many of these mutations are paralogous to each other. The *EGFR* T790M mutation was the first gatekeeper mutation described in lung cancer, being acquired in approximately 60% of cases of classical sensitizing *EGFR* mutations treated with first- or second-generation EGFR TKIs.[290,291] Solvent front mutations are acquired in a proportion of *ALK*, *ROS1*, *RET*, or *NTRK1/2/3* fusion-positive lung cancers in response to TKI therapy.[229] xDFG mutations have similarly emerged with targeted therapy in fusion-positive lung cancers.[292]

The second major mechanism is *off-target*, *kinase-extrinsic*, or *bypass resistance*, whereby the cancer acquires an alteration or alterations affecting genes other than the primary oncogene in response to targeted therapy. The acquired alterations are typically known oncogenic drivers that are found de novo. In *EGFR*-mutant[291] and *RET* fusion-positive[293] lung cancers, for example, *MET* amplification can emerge with TKI therapy. In *MET* exon 14–altered lung cancers, hotspot *KRAS* mutation or *KRAS* amplification can emerge with TKI therapy.[294] Fusions involving *ALK*, *ROS1*, *RET*, or *NTRK1/2/3* have also emerged as mechanisms of TKI resistance, particularly in *EGFR*-mutant lung cancers.[295]

The third major resistance mechanism is *histologic transformation*.[296] By leveraging lineage plasticity, tumors are able to adapt to resist targeted therapy by transitioning from one committed developmental pathway to another. More specifically, an oncogene-driven lung adenocarcinoma may transform to a squamous cell lung cancer, lung cancer with neuroendocrine features, or SCLC in response to TKI therapy. Transformations were initially described in *EGFR*-mutant lung cancers but have since been observed in many other oncogene-driven cancers.[77] Beyond morphologic changes, transformed cancers can express proteins associated with the transformed morphology such as p40 for squamous cell lung cancers and chromogranin or synaptophysin for SCLCs, which can be detected by IHC. Mutations in *RB1* and *TP53* have likewise been implicated in neuroendocrine transformation.[297]

Of note, the wealth of resistance mechanisms described here have been described in the context of TKI use. For large molecules such as monoclonal antibodies, bispecific antibodies, and ADCs, mechanisms of resistance remain poorly understood. These presumably include antigen downregulation or loss, alteration of intracellular trafficking or lysosomal drug breakdown, and payload tolerance or efflux.[246]

Targeting Resistance

Next-generation TKIs have been designed to have activity against kinase domain mutations that mediate on-target resistance, originating the concept of sequential TKI therapy in oncogene-driven tumors. *EGFR*-mutant lung cancers that acquire *EGFR* T790M in

TABLE 30.12

Resistance to Targeted Therapy

Resistance	Associated Alteration(s)	Examples/Notes
Primary Resistance		
Mutation mismatch	*EGFR* exon 20 insertions and first-generation EGFR TKI therapy	Exon 20 insertions lock the EGFR kinase in a rigid C-helix-in configuration, do not diminish ATP affinity, and create steric hindrance to first-generation EGFR TKI binding due to P loop shifting into the drug binding pocket; consequently, gefitinib and erlotinib are not highly active in patients with *EGFR* exon 20-mutant lung cancers
Class mismatch	*BRAF* class 2 or 3 alterations and type I BRAF TKIs	Class 2 *BRAF* mutations (e.g., *BRAF* G469A) encode oncoproteins that function as homodimers Vemurafenib or dabrafenib are ineffective against homodimers due to negative cooperativity; binding one protomer reduces affinity for the second protomer
Acquired Resistance		
On-target genomic resistance	Kinase domain mutations Gatekeeper Solvent front xDFG Other	Involve the primary oncogene driving the cancer *EGFR* T790M, *ALK* L1196M, *ROS1* L2026M, *RET* V804M *ALK* G1202R, *ROS1* G2032R, *NTRK1* G595R, *NTRK3* G623R *NTRK1* G667C, *NTRK3* G696A *EGFR* C797S, *MET* D1228N
Off-target genomic resistance	Heterogeneous alterations Mutations Fusions Amplifications	Alterations are typically oncogenic when found de novo *KRAS* G12A, *BRAF* V600E, *MET* exon 14 alteration Involving *ALK, ROS1, RET, NTRK1/2/3* *MET* amplification
Histologic transformation	*TP53* and/or *RB1* mutations	Manifests as transdifferentiation from lung adenocarcinoma to small-cell or squamous cell lung cancer

TKI, tyrosine-kinase inhibitor.

response to the first-generation EGFR TKI erlotinib, for example, are amenable to therapeutic targeting with the third-generation EGFR TKI osimertinib.[298] In turn, an *EGFR* exon 19 deletion-mutant and *EGFR* T790M co-mutant lung cancer that acquires an *EGFR* C797S mutation may be amenable to targeting with fourth-generation EGFR TKIs that harbor activity against EGFR C797S.[299] *ALK* fusion-positive lung cancers that acquire an *ALK* L1196M gatekeeper mutation with crizotinib may respond to alectinib, and a subsequent acquisition of an *ALK* G1202R mutation with alectinib may be amenable to treatment with lorlatinib.[267] *ROS1* fusion-positive lung cancers that acquire a *ROS1* G2032R mutation with crizotinib may be treated with repotrectinib.[206] Notably, the use of sequential TKI therapy may be associated with the development of compound mutations defined as two or more mutations that presumably occur in cis on the same allele such as co-mutant *ALK* L1196M and *ALK* G1202R in an *ALK* fusion-positive lung cancer. Newer drugs that target compound mutations are being explored.[300]

Beyond the use of sequential type I TKI therapies, TKI class switching may be useful in selected situations. In *MET* exon 14–altered lung cancers that acquire a MET D1228N mutation with a type I MET TKI such as capmatinib or tepotinib, switching to the type II MET TKI cabozantinib has been shown to re-establish disease control.[301] The class switch can be particularly important when a mutation changes the conformation of the mutant kinase. In a *ROS1* fusion-positive lung cancer that acquires a *ROS1* D2033N mutation with the type I ROS1 TKI crizotinib, the mutation makes the kinase prefer the DFG-out or inactive conformation that is amenable to inhibition by cabozantinib, which also inhibits ROS1.[302] The same effect has been demonstrated in *NTRK1/2/3* fusion-positive cancers.[303]

Selected off-target resistance mechanisms can be addressed with targeted therapy. The first major strategy is combination therapy. In *EGFR*-mutant lung cancers that acquire *MET* amplification in response to EGFR TKI therapy, prospective trials have demonstrated the activity with the combination of an EGFR TKI and a MET TKI, such as osimertinib and savolitinib.[304] The second major strategy is switching to a targeted therapeutic with a different mechanism of action, such as switching from a TKI to antibody-based targeted therapy. Many *EGFR*-mutant lung cancers develop EGFR TKI resistance via ERBB3 (HER3) activation, and ERBB3 protein expression is found on the surface of many of these cancers. Patritumab deruxtecan, an anti-ERBB3 ADC, is active in EGFR TKI refractory *EGFR*-mutant lung cancers.[247] Interestingly, response was observed across cancers with diverse on- and off-target resistance mechanisms developed in response to EGFR TKI therapy.

Histologic transformation may require transitioning patients to histology-specific chemotherapy. Patients whose oncogene-driven lung cancers transform from adenocarcinoma to SCLC may benefit from a platinum and etoposide doublet.[77] This underscores the utility of repeat tumor biopsies in patients at the onset of resistance.[296] Beyond identifying genomic changes that may match patients to subsequent standard or investigational targeted therapies, an examination of morphologic changes can inform the choice of subsequent cytotoxic chemotherapies.

Approach to Management

In advanced oncogene-driven lung cancers for which targeted therapy is highly active, guidelines recommend that targeted therapy be given in the treatment-naïve setting. This includes *EGFR*-mutant (classical sensitizing), BRAF V600E-mutant, *MET* exon 14–altered, *ALK* fusion-positive, *ROS1* fusion-positive, *RET* fusion-positive, and *NTRK1/2/3* fusion-positive lung cancers. For patients with *EGFR*-mutant[203] and *ALK* fusion-positive[305] lung cancers, randomized trials have clearly shown that TKI therapy improves outcomes compared to platinum doublet chemotherapy. Notably, when supported by randomized phase III trials, the use of a later-generation agent is preferred, including osimertinib[251] for

classical sensitizing *EGFR*-mutant lung cancers, and alectinib,[252] brigatinib,[253] or lorlatinib[254] for *ALK* fusion-positive lung cancers. It is not uncommon, however, for diagnostic testing to identify one of these oncogenes after first-line systemic therapy with platinum doublet has already been initiated. In these cases, the initial systemic therapy regimen may be continued if response is optimal with targeted therapy reserved for progression. Switching the treatment to targeted therapy may occur immediately after the detection of an actionable gene alteration, in case of toxicity, or suboptimal response to chemotherapy.

In other oncogene-driven lung cancers for which targeted therapy was predominantly explored in chemotherapy pretreated patients and resulted in more modest clinical outcomes compared to other oncogene-drug pairs such as amivantamab[244] and mobocertinib[259] in *EGFR* exon 20 mutant lung cancers and sotorasib for *KRAS* G12C mutant lung cancers,[260] guidelines recommend targeted therapy after progression on a platinum doublet.

The approach to the choice of treatment after progression on targeted therapy is influenced by several factors. The first factor is the pattern of disease progression. In patients who develop solitary site or oligoprogression, local therapy with surgery, radiation, or ablation to the progressing sites with continued targeted therapy use can be considered.[306] The second factor is the mechanism of resistance. Although sequential TKI therapy may be appropriate for on-target resistance, patients whose cancers develop bypass resistance can be transitioned to the standard-of-care platinum doublet or a clinical trial when available. It is important to note that platinum doublet chemotherapy is highly active in oncogene-driven lung cancers, as evidenced by the ORR and median PFS achieved in phase III trials with a chemotherapy control arm in *EGFR*-mutant[203] and *ALK* fusion-positive[305] lung cancers.

Caution should be exercised when considering single-agent immunotherapy in oncogene-driven lung cancers. Although these express a wide range of PD-L1 levels (0%, 1%–49%, ≥50%), TMB tends to be low for many oncogene-positive tumors. Several studies[307] have demonstrated poor outcomes with low ORRs and short median PFS with single-agent checkpoint inhibition with *EGFR*-mutant,[308] *MET* exon 14–altered,[309] *ALK* fusion-positive,[310] *ROS1* fusion-positive,[311] and *RET* fusion-positive[312] lung cancers. In several of these series, outcomes were poor even in selected patients with cancers with PD-L1 ≥50% or high TMB. This does not mean, however, that single-agent immunotherapy is inactive in all patients with oncogene-driven lung cancers. Response rates are reported to be numerically higher for particular oncogenes, such as *KRAS*-mutant, *MET* exon 14–altered, and *BRAF*-mutant lung cancers.[307]

The use of chemoimmunotherapy may circumvent the liabilities of single-agent checkpoint inhibition, although less is known about clinical outcomes in this context. A subset analysis of the registrational data set that resulted in the approval of carboplatin, paclitaxel, atezolizumab, and bevacizumab showed that patients with *EGFR*-mutant lung cancers may have derived benefit, although benefit was less clear for patients with *ALK* fusion-positive lung cancers.[313] *KRAS*-mutant lung cancers, in general, can be associated with a more substantial prior smoking history and higher TMB, making the administration of chemoimmunotherapy in this context appropriate.

Prior immunotherapy exposure may modulate the toxicity of targeted therapy. Increased immune-related adverse events are observed with osimertinib when given after prior PD-L1 inhibition.[314] With crizotinib, an increase in liver enzymes was reported for patients who previously received ICIs.[315] Similarly, the frequency of hypersensitivity to selpercatinib, which includes rash, elevated liver enzymes, and thrombocytopenia, appeared to be increased in patients previously treated with ICIs.[316]

Treatment according to Oncogene

EGFR Mutations: The preferred treatment for patients with tumors harboring the classical sensitizing *EGFR* exon 19 deletion or L858R mutations is with osimertinib. This recommendation is based on the FLAURA study, which randomized 556 patients with advanced-stage NSCLC harboring one of the sensitizing *EGFR* mutations to osimetinib 80 mg per day or a first-generation EGFR TKI including a choice between erlotinib 150 mg per day or gefitinib 250 mg per day.[251] The primary end point of PFS was met with a significant improvement observed in the osimertinib arm (18.9 months vs 10.2 months; HR, 0.46; 95% CI, 0.37 to 0.47; $P < 0.001$). In the updated analysis, the median OS was also significantly improved with osimertinib (36.8 months vs 31.8 months; HR, 0.8; 95% CI, 0.64 to 1; $P = 0.046$).[317] Tumor progression on osimertinib may be treated according to the clinical setting.[318] Patients with oligoprogression may be treated with local ablative therapy, whereas those with slow and asymptomatic progression may continue the treatment without the need for a change in therapy.

Two additional options that are less commonly used include the combination of erlotinib with ramucirumab 10 mg/kg, based on the RELAY study, and dacomitinib, based on the ARCHER-1050 trials, both showing improved outcomes compared to first-generation EGFR TKIs.[257,319]

For patients initially treated with an earlier generation EGFR TKI who develop T790M mutation as the mechanism of resistance, AURA3 comparing osimertinib to platinum plus pemetrexed showed better outcomes for the former, with higher median PFS (10.1 vs 4.4 months; HR, 0.3; 95% CI, 0.23 to 0.41; $P < 0.001$) and ORR (71% vs 31%; $P < 0.001$).[298]

Uncommon mutations, including S768I, L861Q, G719X, and exon 20 insertions, account for approximately 10% to 20% of *EGFR* mutations and have variable sensitivities to EGFR TKIs.[214] In a post hoc analysis of 75 patients with uncommon mutations included in three clinical trials, afatinib was associated with 71% ORR and median PFS of 10.7 months in the 38 patients with uncommon mutations outside of exon 20.[258] Among the 23 patients with exon 20 insertion, the ORR and median PFS were 8.7% and 2.7 months, respectively. Osimertinib has also been shown to be active in uncommon *EGFR* mutations other than exon 20 insertions with an ORR and median PFS of 50% and 8.2 months, respectively, in a prospective study including 37 patients.[320] When evaluated by the subtype of mutation, the ORR was higher in L861Q (78%) than in G719X (53%) and S768I (38%), with a median PFS of 15.2 months, 8.2 months, and 12.3 months, respectively.

Two new drugs have been approved for the treatment of *EGFR* exon 20 insertions. In the phase I CHRYSALIS study, amivantamab 1,050 mg (1,400 mg in patients ≥80 kg) weekly for 4 weeks followed by every 2 weeks starting on week 5, was associated with an overall response rate and median PFS of 40% and 8.3 months, respectively, in patients with NSCLC harboring *EGFR* exon 20 insertions and previously treated with platinum doublet.[244] The response rate was 41% for the 54 patients with mutations in the near loop region (A767 to P772) and 25% for the 8 patients with mutations in the far loop (H773 to C775). The only patient with mutation in the helical region (E762 to M766) responded to treatment. The median OS was still immature at the time of the initial publication but was reported as 22.8 months (95% CI, 14.6 months–not reached). Another treatment option is with mobocertib 160 mg per day, which was associated with an ORR, median PFS, and median OS of 28%, 7.3 months, and 24 months, respectively, in a phase I/II study involving patients with previously treated NSCLC.[321] Responses were seen in all subtypes of *EGFR* exon 20 mutations.

ALK Rearrangement: The standard therapy for patients with ALK rearrangement is with a next-generation ALK inhibitor including alectinib, brigatinib, lorlatinib, or ceritinib. Alectinib had shown activity in patients previously treated with crizotinib[264] and was further evaluated as first-line therapy in the ALEX study, where 303 patients with stage III or IV *ALK*-positive NSCLC were randomized to alectinib 600 mg twice daily or crizotinib 250 mg

twice daily.[252] The primary end point of PFS was met in the initial report (HR, 0.47; 95% CI, 0.34 to 0.65; $P < 0.001$). The updated analysis showed a significant increase in the median PFS (34.8 months vs 10.9 months; HR, 0.43; 95% CI, 0.32 to 0.58).[263] Furthermore, alectinib was also associated with a significant increase in median OS (not reached vs 57.4 months) and 5-year OS (62.5% vs 45.5%). In the ALTA-1L study, 275 patients without previous treatment with ALK inhibitors were randomized to brigatinib 180 mg per day (with a 7-day lead-in period of 90 mg per day) or crizotinib with PFS as the primary end point.[322] Brigatinib was associated with increased ORR (71% vs 60%) and 6-month PFS (67% vs 43%). In the updated analysis, the median PFS was significantly higher with brigatinib (24 vs 11 months; HR, 0.49; 95% CI, 0.35–0.68; $P < 0.0001$).[253] The third-generation ALK inhibitor lorlatinib was effective in patients previously treated with first- or second-generation ALK TKIs.[268] In the CROWN trial, 296 patients with no prior ALK therapies were randomized to lorlatinib 100 mg per day or crizotinib with PFS as the primary end point.[254] Lorlatinib was associated with a significant increase in 12-month PFS (78% vs 39%; HR, 0.28; 95% CI, 0.19 to 9.41; $P < 0.001$) and ORR (76% vs 58%). Ceritinib was effective in patients previously treated with crizotinib[266] and was more effective than platinum plus pemetrexed in the first-line setting with a significant improvement in median PFS (16.6 months vs 8.1 months).[265] Nevertheless, ceritinib is less commonly used due to the gastrointestinal toxicities.

Other Targets: The standard treatment for patients with *BRAF* V600E mutation is a combination of dabrafenib 150 mg twice per day plus trametinib 2 mg per day. In the first-line setting, this combination was associated with an ORR of 54% and median PFS of 10.9 months.[211] In the updated analysis, the median and 5-year OS were 18.2 months and 22%, respectively.[323]

Treatment options for patients with *ROS1* rearrangements include crizotinib, entrectinib, and ceritinib. In the PROFILE 1001 study, which included 53 patients, the ORR, median PFS, and OS were 72%, 19.3 months, and 51.4 months, respectively.[271] In a combined analyses of three studies including 161 patients treated with entrectinib 600 mg or more per day, the ORR, median PFS, and 12-month OS were 67.1%, 15.7 months, and 81%, respectively.[324] Another option is ceritinib, which was associated with an ORR, median PFS, and median OS of 62%, 9.3 months, and 24 months, respectively, in 32 patients.[270] Lorlatinib has shown activity in patients who are TKI-naïve and those previously treated with crizotinib with an ORR of 62% and 35%, respectively, although none of the patients with G2032R responded.[267] Promising treatment options for patients who had tumor progression after the initial TKI include repotrectinib and taletrectinib, which are active against the *ROS1* G2032R mutation.[325,326]

The main treatment options for *MET* alterations include capmatinib, tepotinib, and crizotinib. In the GEOMETRY study, the ORR and median PFS were 41% and 5.4 months, respectively, for the 69 previously treated patients harboring *MET* exon 14 skipping mutation receiving capmatinib 400 mg twice per day.[210] For 28 previously untreated patients, the ORR and median PFS were 68% and 12.4 months, respectively. In contrast, three cohorts of *MET* amplification with a gene copy number <9 were closed for futility, with an ORR between 7% and 12%. For patients with gene copy number of at least 10, the ORR and median PFS were 29% and 4.1 months, respectively. In the VISION trial, the ORR for the 99 evaluable patients with *MET* exon 14 skipping mutation treated with tepotinib 500 mg once daily was 46%, with a median PFS of 8.5 months.[261] In the expansion cohort of the PROFILE 1001 study, the ORR, median PFS, and median OS for the 69 patients with *MET* exon 14 alterations treated with crizotinib were 32%, 7.3 months, and 20.5 months, respectively.[262] In the AcSe study, the ORR, median PFS, and median OS were 16%, 3.2 months, and 7.7 months, respectively, for patients with >6 *MET* copies treated with crizotinib.[327] For patients with *MET* exon 14 skipping mutation, the ORR, median PFS, and median OS were 12%, 3.6 months, and 9.5 months, respectively.

The main treatment options for patients with *RET* rearrangement are selpercatinib and pralsetinib. In the LIBRETTO-001 trial, the ORR, median PFS, and 1-year PFS were 64%, 16.5 months, and 66%, respectively, by independent review for patients previously treated with a platinum doublet chemotherapy.[272] For previously untreated patients, the ORR, median PFS, and 1-year PFS were 90%, not reached, and 75%. The recommended dose is 160 mg twice per day. In the ARROW study, the ORR was 61% in the 87 patients previously treated with platinum-based chemotherapy and 70% in the 27 patients who were treatment naïve.[273] The recommended dose is 400 mg per day. Another option is cabozantinib, which was associated with an ORR of 28% and median PFS of 5.5 months in a small study involving 26 patients.[283] New-generation RET inhibitors under development include TPX-0046 and BOS17238.[328]

Patients with NTRK fusions may be treated with larotrectinib 100 mg twice per day or entrectinib, with an ORR and median PFS of 73% and 35.4 months, respectively, in the former and 57% and 11.2 months, respectively, in the latter.[274,275]

The only currently approved treatment for *KRAS* G12C mutation is sotorasib. In the CodeBreak-100 study, 126 patients with previously treated NSCLC harboring *KRAS* G12C mutation received sotorasib 960 mg per day with the primary end point of ORR.[260] The ORR, median PFS, and OS were 37.1%, 6.8 months, and 12.5 months, respectively. Ongoing studies include the CodeBreak-101 evaluating sotorasib combinations and the CodeBreak-200 comparing sotorasib with docetaxel in previously treated patients. Another KRAS G12C inhibitor in the advanced stage of development is adagrasib, which was well tolerated in the first-in-human study.[329] Among the 15 patients with NSCLC treated with the recommended phase II dose of 600 mg twice daily and evaluable for response, the ORR was 53%, with a median PFS of 11 months.

For patients with *HER2* mutations, promising results were seen in the DESTINY-Lung1 study, where 91 patients with previously treated NSCLC received trastuzumab deruxtecan 6.4 mg/kg every 3 weeks with ORR as the primary end point.[212] The ORR, median PFS, and median OS were 55%, 8.2 months, and 17.8 months, respectively. In a study evaluating poziotinib in 90 patients with previously treated NSCLC harboring *HER2* exon 20 insertions, the ORR and median PFS were 27.8% and 5.5 months, respectively.[330]

STAGE IV NON–SMALL-CELL LUNG CANCER WITHOUT DRIVER ALTERATIONS

First-line Therapy

Chemotherapy

Platinum-based chemotherapy regimens were the standard first-line treatment for stage IV NSCLC until immunotherapy became part of the treatment of lung cancer. Studies have attempted to determine the optimal platinum agent—cisplatin or carboplatin—in advanced disease. Although cisplatin may be slightly more active than carboplatin, particularly with respect to response rate, the impact on OS is minimal, and the toxicity, especially nonhematologic toxicitiy, is worse.[331]

The choice of agents that are typically combined with a platinum analog differ between histologic types and include pemetrexed, paclitaxel, docetaxel, nab-paclitaxel, gemcitabine, vinorelbine, irinotecan, and etoposide. Early studies failed to demonstrate superiority of one regimen over another, and decisions regarding which agent to use are based on the toxicity profile, ease and convenience of administration, and cost.[332] The notable exception is

with the use of pemetrexed. In an RCT comparing cisplatin plus pemetrexed to cisplatin plus gemcitabine, the former was more effective in nonsquamous histology, whereas the latter was more effective in squamous tumors.[333] The use of nonplatinum doublets in first-line therapy is not encouraged but may be an alternative for patients who have contraindications to both platinum analogs.[334] The addition of a third cytotoxic agent does not improve outcomes and is not recommended.[335]

Following completion of the four to six cycles of a platinum-based chemotherapy doublet, multiple randomized trials showed improved PFS with maintenance chemotherapy.[336] Nevertheless, OS benefit was observed only in trials using maintenance pemetrexed, either following a platinum plus pemetrexed (continuation maintenance) or a different platinum-based combination (switch maintenance).[337,338] Maintenance chemotherapy is not indicated for patients with squamous histology due to decreased efficacy from pemetrexed in this patient population.

Anti–Vascular Endothelial Growth Factor Therapy

Bevacizumab is a recombinant humanized monoclonal antibody that targets VEGF ligand and blocks its interaction with VEGFR, resulting in antiangiogenic activity. In a randomized phase II study comparing carboplatin plus paclitaxel with or without bevacizumab at 7.5 mg/kg (low dose) or 15 mg/kg (high dose), the higher dose was associated with improved outcomes.[339] However, among the 66 patients who received bevacizumab in either arm, there were six life-threatening hemorrhages (9.1%), of which four were fatal. The serious hemorrhagic events occurred predominantly in patients with squamous histology, which were excluded in subsequent trials with bevacizumab. Two large, randomized trials were subsequently conducted to confirm the benefits of adding bevacizumab to chemotherapy in patients with nonsquamous NSCLC. In the ECOG 4599 trial, 878 patients with stage IIIB or IV NSCLC were randomized to carboplatin plus paclitaxel with or without bevacizumab at 15 mg/kg every 3 weeks followed by maintenance placebo or bevacizumab until tumor progression.[340] The primary end point of OS was met with improvement in the bevacizumab arm compared to chemotherapy alone (12.3 months vs 10.3 months; HR, 0.79; $P = 0.03$). Bevacizumab was also associated with increased response rates (35% vs 15%) and median PFS (6.2 months vs 4.5 months). In the AVAiL trial, 1,043 patients were randomized to cisplatin plus gemcitabine alone, with bevacizumab at 7.5 mg/kg (low dose) or 15 mg/kg (high dose) followed by maintenance placebo or bevacizumab until tumor progression.[341] The primary end point of the study was amended and changed from OS to PFS. In the initial report, the PFS was 6.7 months in the low-dose bevacizumab arm, 6.5 months in the high-dose bevacizumab arm, and 6.1 months in the chemotherapy alone arm, with a statistically significant improvement in both bevacizumab arms, although the 0.6-month improvement did not clearly represent a clinical benefit. In the updated analysis, the OS was 13.4 months in the high-dose bevacizumab arm, 13.6 months in the low-dose bevacizumab arm, and 13.1 months in the chemotherapy alone arm, with no significant improvement in OS observed.[342]

In a meta-analysis including ECOG 4599, AVAiL, and two randomized phase II studies, the OS was significantly improved with the addition of bevacizumab to chemotherapy with an estimated HR of 0.90 (95% CI, 0.81 to 0.99; $P = 0.03$).[343] The use of bevacizumab was also associated with increased risk of grade ≥3 proteinuria, hypertension, hemorrhagic events, neutropenia, and febrile neutropenia compared to chemotherapy alone.

With data showing benefit from both maintenance pemetrexed and bevacizumab, ECOG-ACRIN 5508 was conducted to determine the optimal strategy.[344] In this trial, 1,516 patients were treated with up to four cycles of carboplatin plus paclitaxel and bevacizumab (15 mg/kg). After completing the initial treatment, 874 patients (57%) were eligible for the randomization to bevacizumab, pemetrexed, or both with OS as the primary end point. The median OS was 14.4 months for bevacizumab, 15.9 months for pemetrexed, and 16.4 months for the combination, with no significant differences. The combination arm was associated with increased grade ≥3 toxicity (51%) compared to pemetrexed (37%) and bevacizumab (29%). Therefore, with similar OS and higher toxicity for the combination, the results supported the use of either pemetrexed or bevacizumab in the maintenance therapy.

Immunotherapy

Immunotherapy with PD-1 and PD-L1 inhibitors have completely transformed the treatment landscape for advanced NSCLC. Multiple RCTs have demonstrated improvement in patient outcomes with immunotherapy as a single agent or in combination with chemotherapy when compared to chemotherapy alone (Table 30.13).

KEYNOTE-024 was the first randomized trial comparing pembrolizumab (200 mg every 3 weeks) to platinum-based chemotherapy in previously untreated patients with advanced NSCLC with PD-L1 tumor expression ≥50% and no *EGFR* sensitizing mutation or *ALK* translocation.[353] The study met its primary end point of PFS (10.3 months with pembrolizumab vs 6 months with chemotherapy; $P < 0.001$). In the final update of the study with a median follow-up time from randomization to cutoff of 59.9 months, the median OS and 5-year OS were 26.3 months and 31.9%, respectively, for pembrolizumab and 13.4 months and 16.3%, respectively, for chemotherapy.[345] The response rates were 46.1% for pembrolizumab and 31.1% for chemotherapy. The estimated 5-year PFS for patients randomized to pembrolizumab was an unprecedented 12.8%.

In KEYNOTE-042, 1,274 patients with a PD-L1 score ≥1% and wild type for *EGFR* and *ALK* were randomized to pembrolizumab or chemotherapy.[346] The primary end points were OS in patients with PD-L1 ≥50%, ≥20%, and ≥1%, and the study did not allow crossover from chemotherapy to pembrolizumab. Among the study population (PD-L1 ≥51%), the pembrolizumab arm was associated with a significant improvement in the median OS (17.7 vs 13 months). Nevertheless, the benefit was driven by the 47% of patients with PD-L1 ≥50% (20 vs 12.2 months), and the numerically superior OS in the prespecified exploratory subgroup analysis of patients with PD-L1 from 1% to 49% did not reach statistical significance (13.4 months vs 12.1 months). Since then, two other randomized trials showed improved survival for ICIs compared to chemotherapy.

In the IMpower110 trial, atezolizumab was associated with a significant improvement in the median OS compared to chemotherapy in patients with PD-L1 ≥50% (20.2 vs 13.1 months) but not in those with PD-L1 from 1% to 49%.[354]

Similarly, EMPOWER-Lung-1 showed that cemiplimab was associated with improved median OS compared to chemotherapy in patents with stage IIIB to IV NSCLC who had PD-L1 ≥50% and no *EGFR* mutations, *ALK* translocations, or *ROS1* fusions (not reached vs 14.2 months).[347] In an exploratory analysis, the PD-L1 expression subtype was correlated with the depth of changes in tumor measurement, with incremental improvements in ORR, PFS, and OS. Among patients treated with cemiplimab, the median PFS was 15.3 months for patients with PD-L1 ≥90%, 6.2 months for those with PD-L1 >60% to <90%, and 4.3 months for those with PD-L1 from 50% to 60%.

Several RCTs have shown that ICIs can be safely combined with platinum-based chemotherapy doublets with chemoimmunotherapy improving outcomes when compared to chemotherapy alone. These trials did not select patients by PD-L1 status, but most excluded those with sensitizing *EGFR* mutations or *ALK* rearrangements.

In the III KEYNOTE-189 study, 616 patients with metastatic nonsquamous NSCLC were randomized to four cycles of platinum

TABLE 30.13

Selected Randomized Phase III Trials with Immunotherapy in Advanced Stage NSCLC

Trial	Treatment Arm	Histology	PD-L1	Median OS
KEYNOTE-024[345]	Pembro vs chemo	Any NSCLC	≥50%	26.3 vs 13.4 mo (HR, 0.62; 95% CI, 0.48–0.81)
KEYNOTE-042[346]	Pembro vs chemo	Any NSCLC	≥1%	16.7 vs 12.1 mo (HR, 0.81; $P = 0.0018$)
IMpower 110[251]	Atezo vs chemo	Any NSCLC	TC/IC 3	20.2 vs 13.1 mo (HR, 0.59; 95% CI, 0.4–0.89)
EMPOWER-Lung 1[347]	Cemi vs chemo	Any NSCLC	≥50%	NR vs 14.2 mo (HR, 0.57; 95% CI, 0.42–0.77)
KEYNOTE-189[348]	Chemo +/– pembro	Non-squamous NSCLC	Any	22.0 vs 10.6 mo (HR, 0.56; 95% CI, 0.46–0.69)
KEYNOTE-407[349]	Chemo +/– pembro	Squamous NSCLC	Any	17.1 vs 11.6 mo (HR, 0.71; 95% CI, 0.68–0.88)
IMpower 150[350]	Chemo plus atezo, bev, or both	Non-squamous NSCLC	Any	18.6 vs 13.9 mo (HR, 0.79; 95% CI, 0.64–0.98)
CheckMate-227[351]	Nivo plus ipi vs chemo	Any NSCLC	≥1%	17.1 vs 14.9 mo (HR, 0.79; 95% CI, 0.67–0.94)
			<1%	17.2 vs 12.2 mo (HR, 0.64; 95% CI, 0.51–0.81)
CheckMate 9LA[352]	Chemo +/– nivo plus ipi	Any NSCLC	Any	15.6 vs 10.9 mo (HR, 0.66; 95% CI, 0.55–0.80)

NSCLC, non–small-cell lung cancer; PD-L1, programmed cell death ligand 1; OS, overall survival; Pembro, pembrolizumab; Chemo, chemotherapy; HR, hazard ratio; CI, confidence interval; Atezo, atezolizumab; TC, Tumor cells; IC, Immune cells; Cemi, cemiplimab; NR, Not reached; bev, bevacizumab; Nivo, nivolumab; ipi, ipilimumab.

plus pemetrexed with or without pembrolizumab followed by pemetrexed plus pembrolizumab or placebo, respectively, for a maximum of 35 cycles.[348] The chemoimmunotherapy arm was associated with increased median PFS (8.8 months vs 4.9 months) and estimated 12-month OS (69.2% vs 49.9 months). In the most recently published update, the chemoimmunotherapy arm was associated with a significant improvement in the median PFS (9 months vs 4.9 months) and OS (22 vs 10.6 months), both occurring for all subgroups of patients including PD-L1 <1%.[355]

In the KEYNOTE-407 trial, 559 patients with squamous NSCLC were randomized to four cycles of carboplatin plus a taxane (paclitaxel or nab-paclitaxel) with or without pembrolizumab for four cycles followed by maintenance pembrolizumab or placebo, respectively.[349] The chemoimmunotherapy arm was associated with increased median PFS (6.4 months vs 4.8 months) and median OS (15.9 months vs 11.3 months) with the benefit observed in both PD-L1 <1% and ≥1%.

In the IMpower130 trial, 724 patients with metastatic nonsquamous NSCLC were randomized 2:1 to four to six cycles of carboplatin plus nab-paclitaxel with or without atezolizumab followed by maintenance atezolizumab in the experimental arm, with patients in the control arm allowed to receive maintenance pemetrexed according to the treating physician.[356] Patients with *EGFR* mutation or *ALK* rearrangements were allowed after progression on targeted therapy, although the primary end points were PFS and OS in the *EGFR* and *ALK* wild-type population. The chemoimmunotherapy arm was associated with a significant improvement in the median PFS (7 months vs 5.5 months) and OS (18.6 months vs 13.9 months). In the IMpower150 trial, 1,202 patients with metastatic nonsquamous NSCLC were randomized to carboplatin plus paclitaxel plus bevacizumab, atezolizumab, or atezolizumab plus bevacizumab.[350] Similarly to IMpower130, patients with *EGFR* or *ALK* alterations were eligible, but the primary end points of PFS and OS included only wild-type patients. When compared to bevacizumab, atezolizumab plus bevacizumab was associated with a significant improvement in both median PFS (8.3 months vs 6.8 months) and OS (19.2 months vs 14.7 months) in the wild-type population. The median PFS for the 108 patients with *EGFR* or *ALK* alterations included in either arm was significantly increased in the atezolizumab plus bevacizumab arm (9.7 months vs 6.1 months; HR, 0.59; 95% CI, 0.37 to 0.94).

The combination of nivolumab plus ipilimumab was evaluated in the CheckMate-227 trial, where 1,739 patients were randomized according to the PD-L1 expression to nivolumab alone, nivolumab plus ipilimumab, or chemotherapy in case of PD-L1 ≥1% or to nivolumab plus ipilimumab, nivolumab plus chemotherapy, or chemotherapy alone in case of PD-L1 <1%.[351] Nivolumab was dosed at 3 mg/kg every 2 weeks, ipilimumab at 1 mg/kg every 6 weeks, and platinum doublets were administered according to tumor histology. When compared to chemotherapy alone, nivolumab plus ipilimumab was associated with a significant improvement in median OS for all patients (17.1 months vs 13.4 months) in a benefit that was observed for both PD-L1 ≥1% (17.1 months vs 14.9 months) and PD-L1 <1% (17.2 vs 12.2 months).

In an attempt to improve early tumor control, CheckMate-9LA randomized 1,150 patients with advanced-stage NSCLC to nivolumab (360 mg every 3 weeks) plus ipilimumab (1 mg/kg every 6 weeks) and two cycles of chemotherapy or chemotherapy alone.[352] The median OS was significantly increased in the chemoimmuntherapy arm (15.6 months vs 10.9 months), with the benefit observed for both squamous and nonsquamous histologies, as well as for all PD-L1 expression subgroups.

With the exception of patients with tumors harboring targetable gene alterations or autoimmune contraindication, virtually all patients with metastatic NSCLC receive an ICI in the first-line setting.[357] Patients with PD-L1 expression <50% are usually treated with chemoimmunotherapy, and those with expression ≥50% may also be treated with ICI monotherapy. In the latter case, monotherapy may be a better choice for patients with lower tumor burden, whereas combination therapy would be more appropriate for those with higher burden and likely for never-smokers for which ICI monotherapy tends to be less effective.

Second-line Therapy and Beyond

For patients who receive first-line immunotherapy, the second-line therapy typically consists of a platinum-based doublet regimen. For those who received first-line ICI plus platinum-based chemotherapy, the main treatment options include docetaxel alone or in combination with the VEGFR2 antibody ramucirumab based on the REVEL study, which showed improvements in median PFS (4.5 vs 3 months) and OS (10.5 months vs 9.1 months), which were statistically significant but clinically modest.[358]

Another option could be weekly nab-paclitaxel. In the J-AXEL trial, 503 patients with previously treated NSCLC were randomized to weekly nab-paclitaxel 100 mg/m^2 or docetaxel 60 mg/m^2 every 3 weeks.[359] Nab-paclitaxel was associated with a significantly improvement in the ORR (29.9% vs 15.4%) and median PFS (4.2 vs 3.4 months). The median OS was numerically higher (16.2 vs 13.6 months) but not statistically significant (HR, 0.85; 95% CI, 0.68–1.07).

Prior to the use of immunotherapy in the first-line setting, several randomized phase III trials demonstrated a survival benefit from single-agent therapy with nivolumab,[360,361] pembrolizumab,[362] and atezolizumab[363] compared to docetaxel. The approval of nivolumab and atezolizumab extends to all patients regardless of PD-L1 expression in the second-line setting, whereas pembrolizumab is approved in tumors with PD-L1 ≥1% based on the design of the respective clinical trials.

These trials have been recently updated with the inclusion of 5-year PFS and OS. In an updated analysis of CheckMate trials 017 and 057, which compared nivolumab to docetaxel in patients with squamous and nonsquamous histologies, respectively, the 5-year PFS was 8% and the 5-year OS was 13.4%, whereas the 5-year PFS and OS were 0% and 2.6%, respectively, for docetaxel.[364] In the update from the KEYNOTE-010 trial, which randomized 1,034 previously treated NSCLC patients with PD-L1 ≥1% to pembrolizumab or docetaxel, the 5-year OS was 15.6% for pembrolizumab and 6.5% for docetaxel.[365] For patients with PD-L1 ≥50%, the 5-year OS was 25% for pembrolizumab and 8.2% for docetaxel. The 5-year PFS was also remarkably high for patients treated with pembrolizumab, reaching 9.4% for all patients and 18.2% in those with PD-L1 ≥50%. In contrast, the 5-year PFS was 0.7% patients with treated with docetaxel.

In current practice, these agents are rarely used in the second-line setting because most patients with advanced NSCLC receive an immunotherapeutic agent in the first-line setting. Nevertheless, they remain a valid option for patients who have not received prior treatment with an ICI.

New treatment options under development for patients with previously treated NSCLC without actionable gene alterations include ADCs and immunotherapy combinations.

Among the several ADCs currently being evaluated in patients with previously treated NSCLC, those targeting Trop2 have shown promising initial results in phase I trials with response rates and disease control of 23% and 79%, respectively, for datopotamab deruxtecan.[366] The response rates and disease control were 19% and 43%, respectively, for sacituzumab govitecan.[367] Other ADCs currently being tested for previously treated NSCLC include enfortumab vedotin, which targets nectin4; cofetuzumab-pelidotin, which targets PTK-7; SAR408701, which targets CEA-CAM5; lifastuzumab vedotin, which targets NaPi2b; enapotmab vedotin, which targets AXL; and telisotuzumab vedotin, which targets MET.[368,369]

Several ongoing clinical trials are evaluating the role other co-inhibitory receptors in patients with both untreated and previously untreated NSCLC including antibodies against TIGIT, LAG-3, and TIM-3.[370,371] There are also ongoing studies evaluating the role of targeting co-stimulatory receptors including 4-1BB, ICOS, OX40, and GITR. Another area of current research in NSCLC is the targeting of the adenosine pathway through antagonists of the adenosine receptor A2; antibodies against CD39, which convert ATP to adenosine diphosphate and adenosine monophosphate; or against CD73, which converts adenosine monophosphate to adenosine.[372,373]

OLIGOMETASTASES

Although most patients with stage IV NSCLC are treated with palliative systemic therapy, local therapy may be appropriate in selected patients with oligometastatic disease, which has been defined by the presence of one to three metastatic sites by the American Radium Society Appropriateness Use Criteria,[374] or one to five sites by the European Society for Radiotherapy and Oncology (ESTRO) and the American Society of Radiation Oncology (ASTRO).[375]

Oligometastatic disease may be classified as synchronous, when present at the time of the diagnosis of the primary tumor, or metachronous, when the new metastases developed at least 3 months after the initial diagnosis and treatment of the primary tumor.[376] The terms *metachronous* and *oligorecurrence* have been used interchangeably. Oligoprogression is defined as the growth of one of few metastatic sites during systemic therapy, whereas most of the other lesions are stabilized. The ESTRO and the European Organization for Research and Treatment of Cancer (EORTC) consensus recommendation classifies oligometastatic into three major groups including de novo, repeat, and induced (Fig. 30.5).[377] It has been speculated that oligometastatic tumors manifest a different biologic behavior compared to widely metastatic tumors characterized by limited metastatic potential, more indolent disease, and different host response.

Among patients with oligometastases, the best candidates for local therapy, either surgery or SBRT, include those with good performance status, one metastatic site, and metachronous timing, and with available systemic therapy options.[376] Although the sequence for systemic and local therapy is still undefined in oligometastatic disease, the former may be indicated as the initial treatment because it allows an evaluation of the cancer behavior and stratification of the probability of benefit, as those with early disease progression on systemic therapy are less likely to benefit from local ablative therapy.

Isolated Brain Metastases

There is evidence to suggest a benefit from curative treatment of isolated brain metastases in selected patients after a completed evaluated to rule out additional sites of disease including PET-CT and invasive mediastinal staging, as mediastinal lymph node involvement would be a contraindication for brain metastasectomy. The treatment with curative intent of brain metastases may include surgery or stereotactic radiosurgery. Multiple studies have shown that in selected patients with brain metastases treated with curative intent, the 5-year OS is approximately 15% in both synchronous and metachronous presentations.[378]

Extracranial Oligometastatic Non–small-cell Lung Cancer

The staging approach and treatment strategy for patients with isolated adrenal metastases are similar to those for brain metastases. The adrenal metastasis may be resected laparoscopically. The 5-year OS for selected patients with isolated adrenal metastases, with either synchronous or metachronous presentation, treated with curative intent is approximately 27%.[378] In contrast, there is limited data to support treatment of other metastatic extrathoracic sites with curative intent.

Local Therapy for Oligometastatic Disease in the Palliative Setting

Definitive local treatment of oligometastatic NSCLC may lead to an improved PFS. In a multicenter, randomized phase II study, 74

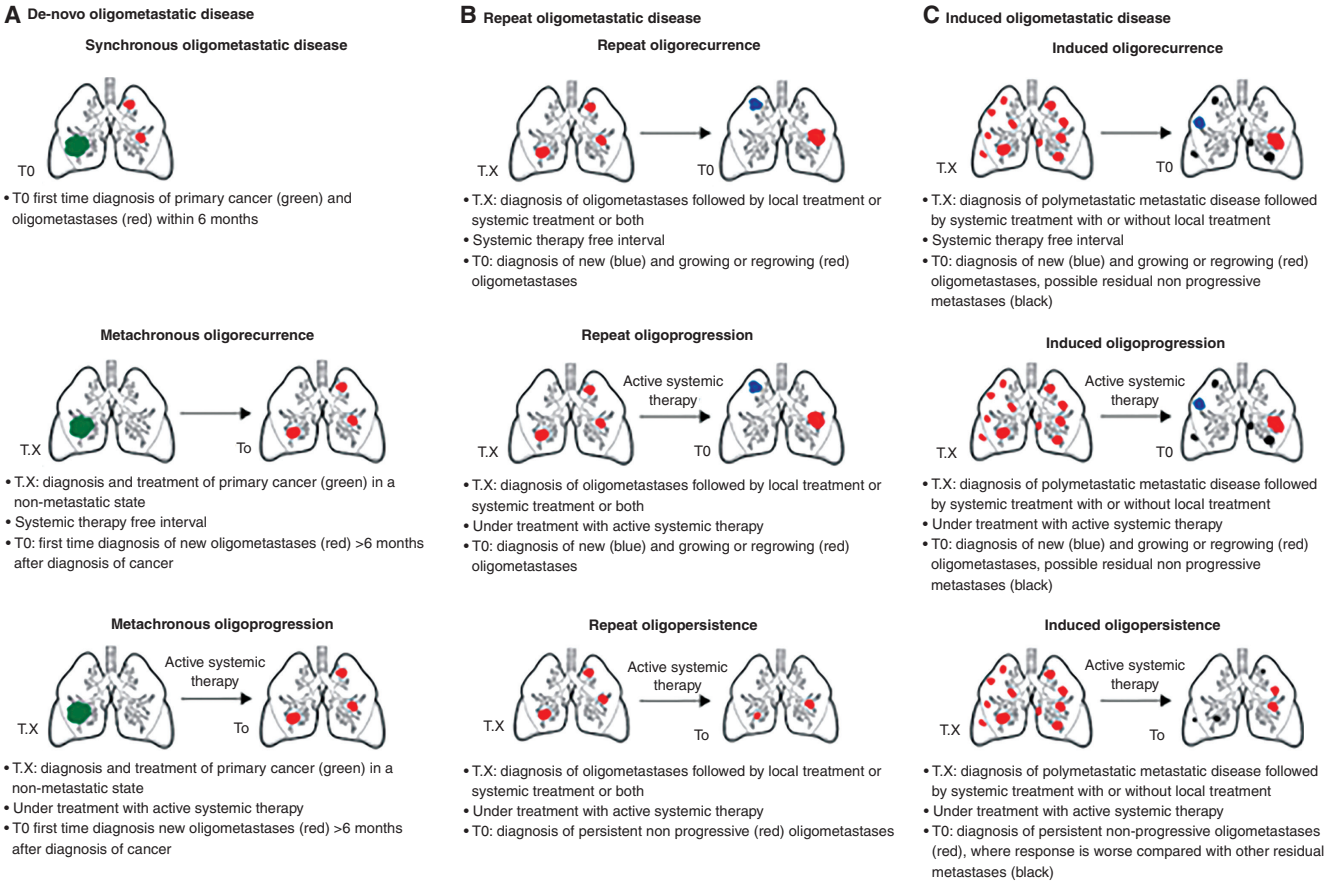


Figure 30.5 Classification of oligometastatic disease. Oligometastatic disease may be classified as de-novo **(A)**, repeat **(B)** or induced **(C)**. T0: Current time; Tx: Any previous point in time. (Reproduced with permission from Guckenberger M, LievensY, Bouma AB, et al. Characterisation and classification of oligometastatic disease: a European Society for Radiotherapy and Oncology and European Organisation for Research and Treatment of Cancer consensus recommendation. *Lancet Oncol*. 2020;21(1):e18–e28.)

patients with stage metastatic NSCLC who had not tumor progression after at least four cycles of platinum doublet chemotherapy or 3 months of targeted therapy for *EGFR* mutations or *ALK* rearrangements were randomized to local consolidation therapy with radiotherapy or surgery for all of the lesions, or maintenance treatment alone that included the option of observation.[379] The study was terminated early after randomization of 49 patients during the annual data safety monitoring committee, with the median PFS significantly increased in the local consolidation arm (11.9 months vs 3.9 months; HR, 0.35; 95% CI, 0.18 to 0.66; $P = 0.005$). In the updated results of the trial, there was a significant improvement in OS with local consolidation therapy (41.2 vs 18.9 months; $P = 0.01$).[380] Similar findings were observed in a smaller study involving 24 patients who were randomized to maintenance chemotherapy alone or SBRT.[381] The median PFS was significantly improved in the SBRT arm (9.7 vs 3.5 months; $P = 0.01$).

MALIGNANT PLEURAL EFFUSION

Malignant pleural effusion (MPE), defined by the presence of malignant cells in the pleural space, occurs in approximately 16% of patients with NSCLC at presentation.[382] Patients with lung cancer and MPE are classified as having metastatic disease (M1a). The goal of treatment for MPE is palliation of symptoms, which include dyspnea, cough, and pain. In a patient with an established MPE, the decision to perform a pleural intervention is determined by the severity of symptoms. Asymptomatic effusions do not necessarily require drainage, whereas symptomatic effusions are usually approached first with therapeutic thoracentesis.[383] The thoracentesis may define whether patients experience symptomatic improvement after drainage, if the underlying lung expands to fill the thoracic cavity or if it entrapped, and how quickly the pleural fluid reaccumulates. If symptoms do not improve after drainage, competing causes of dyspnea such as pulmonary embolism, pericardial effusion, or underlying lung disease should be considered and addressed. If the ipsilateral lung is entrapped and does not expand after thoracentesis, pleurodesis may be contraindicated. Rapid reaccumulation of fluid would argue that a definitive pleural intervention is warranted.

For patients with rapidly recurring, symptomatic MPE who have demonstrated that the underlying lung can expand with apposition of the visceral and parietal pleura, a definitive pleural intervention should be considered. The choices for intervention are pleurodesis or placement of an indwelling pleural catheter (IPC). The two approaches appear to have equivalent benefit in management of dyspnea and no significant difference in survival.[383] IPCs may have an advantage in shorter length of hospital stay and less treatment failure, defined as a need for additional pleural interventions, but are associated with more frequent infectious complications. An IPC can be placed in the ambulatory setting and allows intermittent drainage by the patient or an attendant as needed. For patients with symptomatic MPE and an entrapped lung, failed pleurodesis, or loculated effusion, an IPC may offer some symptom relief. Pleurodesis via surgical or medical thoracoscopy can be done by pleural abrasion alone but more commonly includes instillation of a pleurodesing agent, with talc being the most effective option.[384] Equivalent results are achieved by delivery of talc as a slurry through a chest tube or IPC or by talc poudrage with insufflation of dry powder talc into the pleural cavity.[385] Pleurodesis with an IPC may be achieved spontaneously over the course of weeks without instillation of any agent in approximately 40% to 68% of patients.[386,387]

CONCLUSION

There have been remarkable improvements in the understanding the biology of NSCLC with the development of new treatment modalities. Targeted therapies and immunotherapy, which have been initially used in metastatic disease, are now incorporated into the treatment of patients with earlier stages. There are now approved combinations of chemotherapy and immunotherapy in both the neoadjuvant and adjuvant settings with promising results. Furthermore, the standard treatment for unresectable NSCLC involves the use of chemoradiotherapy followed by consolidation durvalumab with a remarkable improvement in survival. For patients with metastatic disease, there has been an exponential increase in the number of new drugs and targetable gene alterations. For those who are not candidates for targeted therapy, treatment with ICIs has been associated with unprecedented survival, particularly for those with high PD-L1 expression. Nevertheless, virtually all patients receiving targeted therapy develop resistance, and the results from immunotherapy, with or without chemotherapy, are still suboptimal. Therefore, novel targets and immunotherapy drugs are needed to continue the improvement in outcomes, particularly for patients with metastatic disease.

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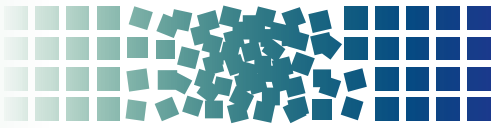
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31 Small Cell and Neuroendocrine Tumors of the Lung



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INTRODUCTION

Neuroendocrine neoplasms (NENs) of the lung account for approximately 20% of all lung cancers and include small cell lung cancer (SCLC), large cell neuroendocrine carcinoma (LCNEC), atypical carcinoid, and typical carcinoid. Despite shared neuroendocrine gene expression and neuroendocrine ultrastructural properties by electron microscopy, SCLC and LCNEC represent an entirely distinct family of tumors from carcinoids as supported by their discrete molecular profiles and differences in histologic, clinical, and epidemiologic characteristics.

SMALL CELL LUNG CANCER

Incidence and Etiology

The incidence of SCLC is slowly declining in the United States, but it remains a major public health problem. Among the estimated 235,760 newly diagnosed lung cancer cases in the United States in 2021, approximately 30,000 were SCLC.[1] Worldwide, there were 2.2 million new diagnoses of lung in 2020 and SCLC is estimated to account for 300,000 of these cases.[2]

Tobacco exposure causes SCLC in over 97% of cases, and SCLC incidence rates in the United States mirror smoking patterns.[3] Peak cigarette consumption occurred in the 1960s but declined following the Surgeon General's report linking smoking to cancer.[4] From 1965 to 2018, the percentage of men who smoke decreased from 51.9% to 15.4%, which was a much greater proportional reduction than in women, whose smoking rate went from 33.9% to 12.0% during the same time period.[5] Correspondingly, the incidence of SCLC in men peaked in 1984 and since has been trending steadily down; in women, the incidence peaked in the mid-2000s and has declined slightly.[6] The gender gap has narrowed, and currently, about half of the patients diagnosed with SCLC are women.

Anatomy and Pathology

SCLC is usually located centrally, adjacent to major airways, but ~5% of cases are peripheral. The histopathologic features of SCLC include dense sheets of small cells with scant cytoplasm, finely granular nuclear chromatin, inconspicuous or absent nucleoli, and frequent mitoses (Fig. 31.1). Necrosis is universally present and frequently extensive. The tumor cells typically measure less than three small resting lymphocytes in diameter. They are round to fusiform in shape and have scant cytoplasm. Nuclear molding is characteristic. Proliferation rate is typically among the highest of all tumor types: mitotic counts average 60 to 80 per 2 mm^2, and Ki67 proliferation rate is consistently >50%, and typically 80% to 100%. Crush artifact is a frequent finding in small biopsy specimens and can make pathologic interpretation difficult.[7–9]

Combined SCLCs account for ~20% of SCLC overall and are defined as tumors containing both SCLC and non–small cell lung cancer (NSCLC) components. The most common NSCLC histology in combined SCLC is large cell neuroendocrine carcinoma (LCNEC), although both adenocarcinoma and squamous cell histology are seen (1% to 9%).[10,11] Given that admixed larger cells are commonly present in SCLC, thoracic World Health Organization (WHO) classification requires LCNEC component to comprise at least 10% of the overall tumor for the diagnosis of combined SCLC/LCNEC, whereas any amount of adenocarcinoma or squamous histology qualifies for a combined SCLC diagnosis.[7]

Although the diagnosis of SCLC can be made based on histopathologic features using the hematoxylin and eosin-stained sections, immunohistochemistry (IHC) is increasingly utilized in pathology practice to support the SCLC diagnosis and to exclude alternative diagnoses. In recent studies, IHC was shown to increase the accuracy of SCLC diagnosis.[10] The IHC markers commonly used to support the diagnosis of SCLC include a panel of three neuroendocrine markers—chromogranin A, synaptophysin, and CD56 (NCAM). Of these, CD56 is most sensitive but is least specific, given that it is expressed in a variety of non-neuroendocrine neoplasms, including various hematolymphoid malignancies. However, 20% of SCLC may be negative for the more specific neuroendocrine markers, synaptophysin and chromogranin A, and up to 10% may be negative for all three markers. The more recent neuroendocrine marker insulinoma-associated 1 (INSM1) has similar sensitivity to CD56 but is more specific, and it is becoming commonly utilized for the diagnosis of SCLC.[11] Recent studies also suggest potential utility of the tuft cell marker POU2F3 in the diagnosis of SCLC lacking the expression of the standard neuroendocrine markers.[11] Thyroid transcription factor 1 (TTF1) is positive in ~80% of SCLC. Notably, TTF1 can be positive in extrapulmonary small-cell carcinomas, so it is not useful in determining the primary site of small-cell carcinomas.[12]

Screening

Screening computed tomography (CT) scans detect NSCLCs at an earlier stage compared to chest x-ray (CXR) and can decrease lung cancer–specific mortality in current or former heavy smokers, as demonstrated in the National Lung Screening Trial (NLST).[13] However, screening CT did not detect SCLC at earlier stages in several lung cancer screening studies, including the NLST, and thus did not impact survival.[14–16] There is currently no recommended screening strategy for early detection of SCLC.

Presentation and Diagnosis

Presenting symptoms in patients with SCLC can be constitutional, pulmonary, the result of extrathoracic spread, or due to paraneoplastic disorders.[17] In one series, fatigue was the most common symptom, with decreased physical activity, cough, dyspnea, decreased appetite, weight loss, and pain all occurring sometime

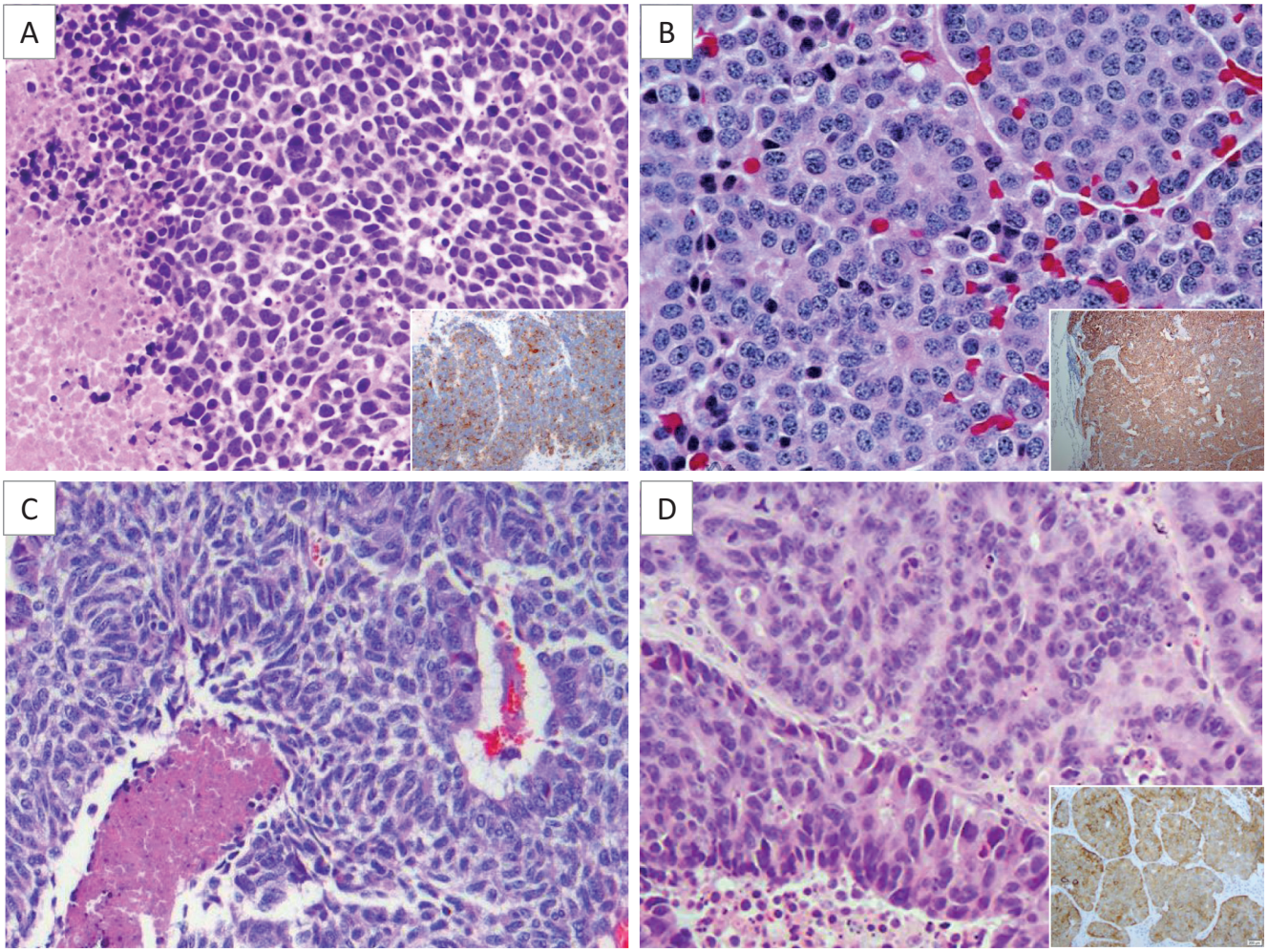


Figure 31.1 Pathologic features of lung neuroendocrine neoplasms. (**A**) small cell carcinoma; (**B**) typical carcinoid; (**C**) atypical carcinoid; and (**D**) large cell neuroendocrine carcinoma. Insets show immunohistochemistry for synaptophysin.

in the course of the illness in the majority of patients.[18] As the primary tumor often presents as a large central mass invading or compressing the mediastinum (Fig. 31.2), superior vena cava obstruction has been reported at diagnosis in 10% of patients with SCLC.[19] Chest imaging typically shows hilar and mediastinal adenopathy. Most patients with SCLC have metastases at diagnosis. Hepatic and adrenal lesions are typically asymptomatic. Brain metastases are reported in up to 18% of patients at diagnosis and are often asymptomatic.[20] Bone involvement is usually characterized by asymptomatic osteolytic lesions, often without elevation of serum alkaline phosphatase.[21]

Paraneoplastic syndromes (PNS) are common in SCLC and differ from those observed with NSCLC. SCLC accounts for approximately 75% of the tumors associated with the syndrome of inappropriate antidiuretic hormone (SIADH). Although serum concentrations of antidiuretic hormone are elevated in most SCLC cases, only approximately 10% of patients fulfill the criteria of SIADH, and symptoms are present in no more than 5%. In some cases, ectopic production of atrial natriuretic factor contributes to the disorder in sodium homeostasis. The primary treatment for hyponatremia in SCLC patients is chemotherapy aimed at treating the disease. Additional management strategies include fluid restriction in mild cases, intravenous (IV) hypertonic saline in severe, symptomatic cases or oral vasopressin V2-receptor antagonists.[22]

Increased serum levels of adrenocorticotropic hormone (ACTH) can be detected in up to 50% of patients with SCLC. Cushing syndrome develops in only 5% of SCLC patients and is associated with very limited survival.[23] The primary treatment of Cushing syndrome in SCLC patients is chemotherapy. Agents such as steroidogenesis inhibitors (ketoconazole, metyrapone, mitotane), centrally acting agents (pergoline), and glucocorticoid receptor antagonists can be used.[24] Hypercalcemia is rare in SCLC.

Neurologic PNS seen in patients with SCLC include sensory, sensorimotor, and autoimmune neuropathies and encephalomyelitis and can be clinically disabling.[25] These syndromes are thought to occur through autoimmune mechanisms, and antinuclear antibodies that bind to SCLC and to neuronal tissues have been identified. Subacute peripheral sensory neuropathy is associated with the anti-Hu antibody. Lambert-Eaton syndrome, characterized by proximal muscle weakness, hyporeflexia, and dysautonomia, is due to autoantibody impairment of voltage-gated calcium channels. Rarer neurologic disorders seen in patients with SCLC include cerebellar degeneration or retinopathy. Symptoms may precede the diagnosis by many months and are often the presenting complaint. In contrast to endocrinologic syndromes that often remit with successful treatment, neurologic symptoms often do not resolve with antineoplastic therapy. Interventions such as plasma exchange and immunosuppressive therapy with corticosteroids, cyclophosphamide, or tacrolimus are standard but typically offer limited benefit.

Clinical and Serologic Predictive and Prognostic Factors

Multivariate analyses suggest that performance status (PS) is a strong and reproducible prognostic factor.[26–28] Poorer PS is

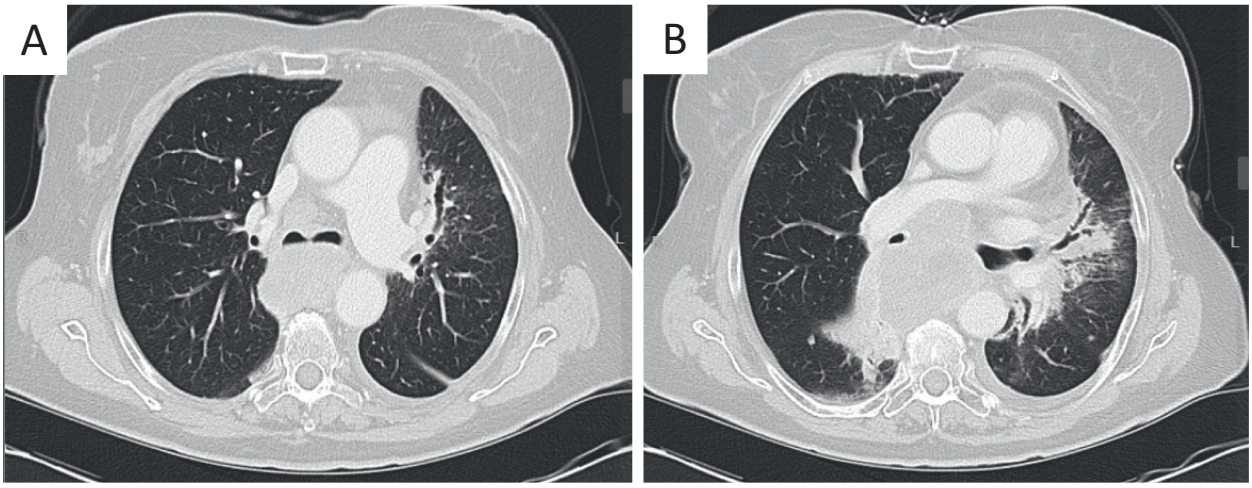


Figure 31.2 Radiographic appearance of small cell lung cancer. Small cell lung cancer often presents as a central mass with regional lymph node involvement. Axial computed tomography images of bilateral mainstem compression due to massive mediastinal lymphadenopathy in a newly diagnosed patient (**A**) and a patient with disease relapse (**B**) within the radiation field after prior chemoradiotherapy.

associated with higher risk for treatment-related complications. Female gender has been associated with improved response and survival.[26–28] Metastases to certain sites, including liver, brain, bone marrow, and bone, as well as the total number of metastatic sites involved have been reported as prognostic factors.[26,29–32] Paraneoplastic Cushing syndrome has been correlated with a poor response to therapy and short survival.[23] Continued use of tobacco during combined modality therapy was identified as an adverse prognostic factor in a group of 186 patients with limited disease.[33] Elevated serum LDH has been reported in 33% to 57% of all SCLC patients and in up to 85% of those with ES disease and is a negative prognostic factor.[27,31,34,35] Circulating tumor cell (CTC) enumeration and characterization have been explored as predictive and prognostic markers in SCLC.[36–38]

Staging

A two-stage system, introduced by the Veterans Administration Lung Study Group (VALSG), has historically been used in the management of SCLC.[39] In the VALSG system, limited stage (LS) is defined as disease confined to one hemithorax that can be "encompassed" in a "tolerable" radiation field. These patients are treated with a combined modality approach. All other patients are considered to have extensive-stage (ES) disease. At presentation, approximately two-thirds of patients with SCLC have ES and one-third LS disease.[40]

The American Joint Committee on Cancer (AJCC) eighth edition is currently the recommended staging system for SCLC, although the VALSG system is still typically used to define clinical trial eligibility and to guide clinical decision making. In both the seventh and eighth AJCC editions, SCLCs are staged using the same definitions as used for NSCLC.[41,42] VALSG LS disease is broadly synonymous with stage I–III (T1–4 N1–3 M0), and ES disease includes patients with M1 disease or disease too extensive to treat with definitive radiation, such as T3 to 4 due to multiple nodules that cannot be encompassed in a tolerable radiation port. The use of the TNM system is most clinically useful for cases of early-stage disease, particularly for patients with stage I SCLC in whom surgical resection may be considered.

Staging studies include a contrast-enhanced CT scan of the chest and abdomen and a brain MRI (or contrast-enhanced head CT). For patients who appear to have LS SCLC, additional tests may include ^{18}F-FDG PET scan and pathological mediastinal LN staging. Pleural effusions should be sampled, to assess whether the effusion is malignant. Effusions too small to permit sampling should not be considered in staging. Osseous abnormalities seen on PET or bone scan should be considered for confirmation with magnetic resonance imaging (MRI) or biopsy if they represent the only site of potential metastasis. SCLC patients who are active smokers should be counseled on tobacco cessation.[33,43]

Patients with an ECOG PS of 0 to 2, and those with ECOG PS 3 or 4 attributable to SCLC, are recommended to receive treatment for SCLC. There is no routine recommendation for treatment of patients who have ECOG PS 3 to 4 not attributable to SCLC. For many patients in this poor PS group, supportive care only and referral to hospice is an appropriate option.

Initial Management of Small Cell Lung Cancer

Due to its strong predilection for metastasis, SCLC should always be presumed systemic, even if imaging supports localized disease. Thus, all patients are recommended to receive systemic therapy, even those with very early-stage disease. Patients with LS SCLC are recommended to receive four cycles of etoposide plus cisplatin concurrently with thoracic radiation, or after surgical resection for stage I disease. Patients with ES SCLC are recommended to receive a platinum agent (cisplatin or carboplatin), etoposide plus a PD-L1 inhibitor as their sole initial therapy. The role of PCI in patients with LS SCLC after completion of initial therapy (surgery and adjuvant chemotherapy or concurrent chemoradiation) is under re-evaluation. The next sections detail key studies of radiotherapy, surgery, and chemotherapy, which have led to the current recommendations in the management of SCLC (Fig. 31.3). A summary of the current SOC approach is included at the conclusion of this section.

Radiotherapy in Limited-Stage Small Cell Lung Cancer

SCLC is exquisitely chemo- and radio-responsive; however, neither modality alone controls all aspects of disease. Concurrent combined-modality therapy is the standard treatment for LS SCLC patients with adequate PS per practice guidelines in Europe and the US. Single-modality or sequential therapy may be appropriate for those who are debilitated or have serious comorbidities. Although gains in radiotherapy delivery and integration with PET imaging have been made, disease failure in local, distant, and sanctuary sites continue to be substantial clinical challenges.

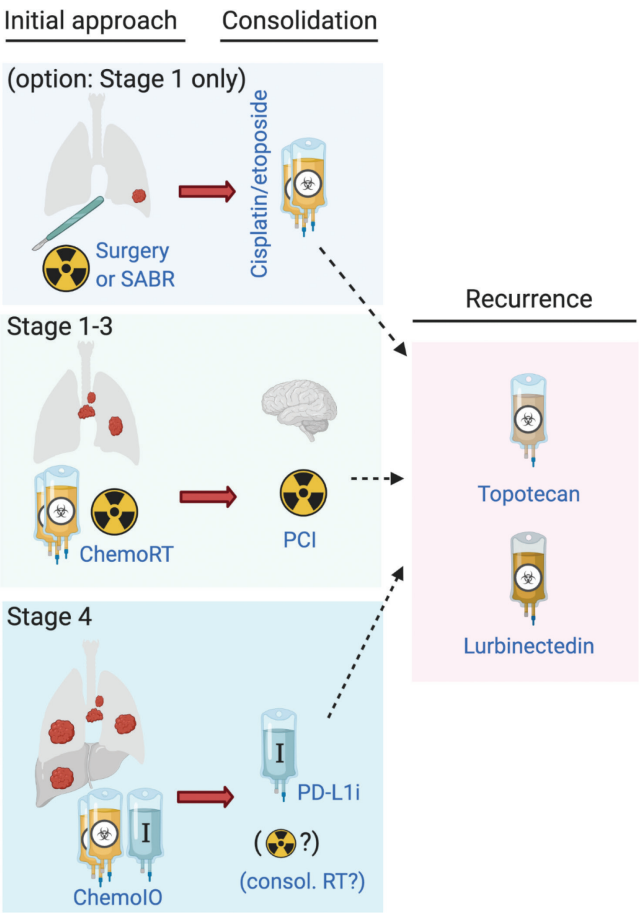


Figure 31.3 Schematic approach to small cell lung cancer treatment by stage. See text for details.

Sequencing of Thoracic Radiotherapy with Chemotherapy

A 1992 meta-analysis evaluated trials in which more than 2,100 patients with LS SCLC were randomized to receive chemotherapy alone or with chest irradiation.[44] Patients given combined-modality therapy had a 14% reduction in death rate and an absolute 5.4% improvement in 3-year survival compared with those who received chemotherapy alone. Both differences were highly significant. A second and independent meta-analysis reached similar conclusions.[45]

Thoracic Radiation Dose and Fractionation

Due to the radio responsiveness of SCLC, modest doses of radiation from 45 to 50 Gy have historically been used. However, with modest-dose radiation therapy, there is a high rate of local failure.[46] For example, in the Intergroup trial, the control arm of 45 Gy given once daily had a 52% rate of intrathoracic relapse.[47] Therefore, higher radiation doses or intensification (i.e., acceleration) have been investigated to improve local control. With conventional (non-conformal and large-field) radiation techniques, the maximum tolerated dose (MTD) of concurrent thoracic chemoradiation seems to be 45 to 51 Gy with a twice-daily approach and 70 Gy daily.[48] However, the advent of highly conformal radiotherapy techniques, such as intensity-modulated radiation therapy (IMRT), and the shift away from targeting clinically uninvolved nodal stations may lead to tolerance of higher radiation doses, and this has prompted clinical trials investigating higher radiation doses using both once-daily and twice-daily regimens (Table 31.1).

SCLC appears to be an ideal neoplasm for twice-daily treatment in that it has a high growth fraction, short cell cycle time, and small-to-absent shoulder on the in vitro cell survival curve, which may allow for a reduction in long-term pulmonary toxicity while maintaining antitumor efficacy.[52] A landmark Intergroup study randomized 417 patients with LS SCLC to receive 45 Gy in either daily fractions of 1.8 Gy or twice-daily fractions of 1.5 Gy, both given concurrently with the first of four cycles of EP.[47] Although fractionation is the obvious variable, the overall duration of radiotherapy varied as well (3 weeks vs 5 weeks), and longer treatment times may exert selective pressure on the emergence of resistant clones. The target volume included the primary tumor, bilateral mediastinal nodes, and the ipsilateral hilum, excluding uninvolved supraclavicular nodes. Local failure was reduced from 52% to 36% with the twice-daily schedule ($P = .06$). Of interest, only 6% of patients with twice-daily radiation failed in both local and distant sites, compared to 23% with daily treatment ($P = .01$). More importantly, although statistically significant differences in survival were not seen at 2 years, by 5 years, the survival was 16% with once-daily treatment, compared to 26% with a twice-daily schedule ($P = .04$).[47] All patients who achieved less than complete response were scored as local failures, but some of those treated with the accelerated scheme who achieved only a partial response survived, as well as those with a complete response, implying that the local failure rate was overestimated by imaging on that arm. There was a higher frequency of grade 3 esophagitis with twice-daily treatment, but overall long-term morbidities were not significantly different between the two arms.

Cancer and Leukemia Group B (CALGB) 39808 determined that 70 Gy delivered with daily fractions of 2 Gy with concurrent chemotherapy was associated with a median survival of 22.4 months and acceptable rates of esophagitis and pneumonitis.[53]

TABLE 31.1

Selected Randomized Radiotherapy Trials in LS SCLC

Trial	Phase	N	Arms	Median OS (mos)	HR	Landmark OS	
						5 year	2 year
Intergroup 0096[47]	3	417	45 Gy BID, in 1.5 Gy fractions 45 Gy qD, in 1.8 Gy fractions	23 19	1.2 *P = 0.04*	26% 16%	
CONVERT[49]	3	547	45 Gy BID, in 1.5 Gy fractions 66 Gy qD, in 2 Gy fractions	30 25	1.18 *P = 0.14*	34% 31%	
CALGB 30610[50]	3	638	45 Gy BID, in 1.5 Gy fractions 70 Gy qD, in 2 Gy fractions	28.5 30.5	0.94 P = 0.59	29% 34%	58% 56%
High-dose vs Standard-dose BID RT in LS SCLC[51]	2	170	45 Gy BID, in 1.5 Gy fractions 60 Gy BID, in 1.5 Gy fractions				48% 74% *(P = 0.0005)*

Other CALGB trials also have used 70 Gy of thoracic irradiation with concurrent chemotherapy.[54] Once-daily fractionation is more familiar for radiation oncologists than twice daily and can be more convenient for patients and radiation therapy departments. Indeed, fewer than a quarter of LS SCLC patients receive twice-daily treatment in the United States.[55] However, some patients prefer the decreased overall treatment duration with a twice-daily schedule.

A phase I trial conducted by the Radiation Therapy Oncology Group (RTOG) and CALGB evaluated the use of 70 Gy delivered in daily fractionation or 45 Gy given in twice-daily fractionation.[56] Although it was intended to establish the MTDs of irinotecan given with cisplatin and thoracic radiation, it also was the first trial to evaluate 70 Gy delivered at cycle 1. Of the 15 patients treated to 70 Gy, 2 had dose-limiting toxicities (DLTs) (diarrhea, esophagitis, and cardiovascular complications).

RTOG 9712 was a phase I trial to establish the MTD of radiation therapy with delayed accelerated hyperfractionation.[57] With this technique, a large field encompassing the gross tumor and mediastinum was treated to 45 Gy. A smaller field that encompassed only the gross tumor was treated as a second daily 1.8 Gy treatment for the last 3 to 11 days. The MTD was found to be 61.2 Gy with this regimen, which subsequently was evaluated in RTOG 0239, a phase II trial.[58] It reported a 2-year overall survival (OS) of 37%, short of the predicted survival of 60%. However, the regimen demonstrated an excellent local control rate of 73%.

These RTOG/CALBG efforts to increase radiation dose using daily fractionation or delayed accelerated hyperfractionation led to a three-arm randomized phase III trial comparing 45 Gy in 1.5-Gy twice-daily fractions (standard; Arm A), 70 Gy in 2-Gy fractions (Arm B), and 61.2 Gy with delayed accelerated hyperfractionation (Arm C). In the first part of this trial, toxicity was to be assessed and the less toxic of Arms B and C was to be compared head to head with the standard arm. Although no clear difference in toxicity was observed between Arms B and C, the delayed accelerated hyperfractionation arm was dropped. Initial results comparing Arms A and B have been presented in abstract form.[50] The study was designed to demonstrate superiority of the higher-dose arms compared to standard twice-daily radiation, but no significant difference was found in overall or progression-free survival.

The CONVERT study was a phase III randomized trial evaluating once- versus twice-daily radiotherapy with concurrent chemoradiotherapy compared 66 Gy in daily 2-Gy fractions against the standard arm of 45 Gy in twice-daily 1.5-Gy fractions.[49] It was also powered to detect superior survival with once-daily treatment; however, no significant survival difference between the two arms was found. Because CONVERT was not designed as a noninferiority study, this does not establish that 66 Gy in daily fractions is equivalent to 45 Gy twice daily. However, in situations where twice-daily radiation is not feasible, both CONVERT and the RTOG/CALGB Intergroup trial support that once-daily radiation to 66–70 Gy achieves comparable survival outcomes.

Given the theoretical advantages of accelerated fractionation in SCLC, and the failure of randomized trials to show improvement with high-dose once-daily fractionation, it remains possible that outcomes could be improved by escalating the dose within the framework of hyperfractionated treatment. A Norwegian randomized trial evaluated the potential for a 60 Gy in 40 fractions (twice daily) to improve survival compared to the standard 45 Gy.[51] This 170-patient study had a primary endpoint of 2-year survival, which was significantly higher with 60 Gy than 45 Gy (74.2% vs 48.1%, $P = 0.0005$). Toxicity outcomes were comparable in the two arms. Though this is a highly promising result, further validation is needed before adopting higher dose hyperfractionation as a standard of care. This was a relatively small phase II, rather than phase III, study and despite the overall survival difference, did not show a significant improvement in progression-free survival or local control between arms.

Radiotherapy for Early-Stage (I or II) SCLC

Though limited-stage SCLC typically presents as bulky disease with nodal involvement, SCLC can also be diagnosed at earlier stages, including as a pulmonary nodule without apparent nodal involvement. Surgical resection rather than definitive radiotherapy is sometimes used as local therapy for this presentation. Another potential local therapy option for Stage I or II, node-negative SCLC is stereotactic body radiation therapy (SBRT), which has demonstrated favorable outcomes in early-stage NSCLC and become a standard treatment for nonoperable patients with that presentation.

Limited prospective and retrospective data suggest that favorable outcomes can be achieved with SBRT in appropriately selected early-stage SCLC patients.[59–62] SBRT has gained use in this context given the appealing convenience and generally low toxicity associated with SBRT compared to conventional thoracic chemoradiotherapy.[63] However, there is a dearth of prospective validation for SBRT in SCLC and concurrent, twice-daily chemoradiation therapy, or surgery followed by adjuvant chemotherapy for stage I, are still considered standards of care. PET-CT staging and mediastinal nodal sampling are strongly advisable if SBRT is being considered, to rule out the possibility of occult nodal involvement.

Early versus Late Thoracic Radiotherapy

Single randomized trials have not defined an optimal time for concurrent irradiation during initial chemotherapy, but most meta-analyses have suggested a benefit to radiation in earlier rather than later cycles of chemotherapy. One such meta-analysis reviewed seven randomized trials with a total of 1,524 patients.[64] Early radiation therapy was defined as beginning before 9 weeks after the initiation of chemotherapy and before the third cycle of chemotherapy. Late radiation therapy began 9 weeks or more after the initiation of chemotherapy or after the beginning of the third cycle. They reported a small but statistically significant improvement in 2-year survival for patients receiving early radiation therapy. A greater benefit was observed in patients receiving hyperfractionated radiation. Another meta-analysis evaluated four randomized trials consisting of 1,056 patients to determine whether the time from the start of chemotherapy until the end of radiotherapy (SER) was a predictor of survival.[65] A significantly higher 5-year survival rate was noted in the treatment arms with a shorter SER. Notably, a low SER was also associated with a higher incidence of severe esophagitis. Together these analyses suggest that an important factor in the treatment of SCLC involves counteracting accelerated repopulation that occurs after treatment with chemotherapy. However, definitive evidence that early integration of radiotherapy leads to improved survival remains elusive. A third meta-analysis of 12 trials of radiotherapy timing (shorter vs longer courses and earlier vs later) shows no overall difference in survival between "earlier or shorter" radiation versus "later or longer" radiation.[66] However, the investigators detected a significant improvement in survival with "earlier or shorter" radiation in patients receiving their planned doses of chemotherapy, whereas "earlier or shorter" radiation was associated with worse survival in patients who did not receive planned chemotherapy. Again a higher incidence of acute toxicity, particularly esophagitis, occurred with "earlier or shorter" radiation schedules.[66]

In clinical practice, it is often prudent to initiate chemotherapy immediately in patients who are highly symptomatic rather than delay treatment until radiation treatment planning is complete. Starting radiotherapy with the second cycle of chemotherapy is more readily achievable than concurrent initiation and should preserve the potential gains in efficacy obtained with early integration of radiotherapy with chemotherapy.

Radiation Therapy Treatment Volumes and Technique

Several analyses have addressed the question of optimizing radiation treatment volumes, aiming to minimize both local treatment

failure and toxicity to normal structures. The Intergroup trial included gross disease, the bilateral mediastinum, and the ipsilateral hilum in the treatment field.[47] The uninvolved supraclavicular area was excluded. RTOG 9712 allowed for the treatment of the uninvolved supraclavicular area in the setting of apical tumors.[57] A small phase II trial suggested that there may be increased elective nodal failure in the supraclavicular area if it is excluded from the treatment field.[67] In contrast a Dutch phase II trial of concurrent chemoradiation limited radiotherapy fields to the primary tumor and clinically involved lymph nodes and reported a relatively low local failure rate of 16%.[68] Favoring a more restricted approach, a number of subsequent prospective and retrospective analyses suggest that involved-field radiation therapy is not associated with an excessive rate of local–regional failure, particularly in an era of routine PET staging.

Utilizing PET to guide the design of limited radiotherapy fields appears to reduce the probability of elective nodal failure due to the improved sensitivity of PET to detect nodal disease compared to CT alone. A follow-up trial from the same group that found a worrisome rate of elective nodal failure when utilizing involved-field radiotherapy in CT-staged patients showed that the rate of isolated nodal failure was only 3% with PET staging, compared to 11% without.[69] A retrospective report similarly showed a low rate of isolated elective nodal failure in a PET-staged cohort of patients treated with IMRT, a technique that gives lower incidental radiation doses to untargeted lymph node regions and, therefore, would theoretically increase the risk of elective nodal failure.[70] Overall, these and other recent publications suggest that with modern staging and treatment techniques, the omission of elective nodal irradiation does not lead to an excessive rate of isolated nodal failure.[71–73] Neither of the contemporary trials of daily versus twice-daily TRT employ elective nodal irradiation (with the exception of hilar lymph nodes in the CALGB trial), indicating that the omission of elective nodal irradiation is increasingly considered an acceptable standard of care for PET-staged patients. The Norwegian randomized study of higher-dose twice-daily radiation mandated PET-CT staging and included only FDG-avid nodes in the radiation treatment field. The lack of apparent excess toxicity in the high-dose arm with this approach, and the promising survival outcomes observed, further support the notion that radiation fields for SCLC need only target clinically suspicious nodes.[51]

In addition to reducing the size of the target volume, another avenue to reduce radiation toxicity is to utilize more highly conformal delivery methods, such as IMRT or proton therapy. These techniques allow for significantly greater control of the location of radiation dose, particularly of incidental radiation dose outside of the target volume. Although this would seem to be an unequivocal advantage, theoretical concerns have persisted regarding the sensitivity of these techniques to respiratory organ motion, and the possibility that decreasing the incidental dose outside of the target volume might increase rates of regional or nodal failure. A retrospective analysis of patients receiving standard three-dimensional conformal radiation therapy and IMRT found no significant difference in OS or disease-free survival (DFS).[74] Proton therapy, which reduces the dose outside of the target volume more fully than is possible with any form of photon radiotherapy, has not yet been widely used in SCLC. A single-institution prospective registry of proton therapy in SCLC suggests proton-beam therapy is feasible and safe for LS disease and can achieve lower mean doses to heart and lung compared with IMRT.[60] Reducing heart dose may prove to be an important goal in SCLC, as this has been shown to be independently predictive of survival after concurrent chemoradiation in NSCLC.[75]

Toxicity of Thoracic Radiotherapy

Chest irradiation can lead to myelosuppressive, pulmonary, and esophageal complications. Esophagitis is a difficult toxicity to compare across studies as trials have used distinct grading systems. However, most trials with concurrent chemoradiation with a cisplatin-based regimen report a 10% to 25% rate of severe esophagitis.[76] A retrospective review suggested that radiation dosimetric parameters including mean esophageal dose and volume of esophagus receiving 15 Gy could be associated with risk of grade 3 or worse esophagitis in patients receiving twice-daily radiation.[77] More recently, a large individual patient data meta-analysis of esophagitis after concurrent thoracic chemoradiotherapy indicated that the best predictor of esophagitis was the volume of esophagus receiving ≥60 Gy, with the highest risk seen when this included 17% or more of the esophagus.[78]

Most recent trials report a radiation pneumonitis (RP) rate of approximately 10% or lower.[49,53,79] Contributors to this improvement relative to trials reported in the 1980s and 1990s may include improved image-guided radiation techniques and better distinction between pneumonitis and other etiologies of respiratory distress, such as infection or tumor recurrence. The volume of lung receiving 20 Gy (V_{20}) is a standard parameter used to predict lung toxicity in patients with NSCLC. A collaborative effort to define consensus dose limits for RP recommended that the V_{20} be kept below 30% to 35% in order to keep the risk of pneumonitis <20%.[80] It has been shown to have value in patients receiving concurrent accelerated hyperfractionated treatment for SCLC as well, with V_{20} values <25% to 30% associated with a lower rate of RP.[81,82]

Prophylactic Cranial Irradiation

Brain metastases are detectable in at least 18% of SCLC patients at diagnosis[20] and eventually are diagnosed in another 20% to 25%, with an increasing likelihood seen with lengthening survival.[83,84] Actuarial analysis reveals a probability of brain metastases ranging from 50% to 80% in patients who survive 2 years. At postmortem examination, brain metastases are found in up to 65% of SCLC patients.[85] Because these metastases are sometimes the sole site of clinical relapse and are frequently clinically disabling, PCI has been recommended by many,[86] but not all,[87] authorities.

Several early prospective randomized trials assessed the benefit of PCI given at or within a few months of diagnosis.[88–92] When these trials were considered together, PCI reduced the frequency of clinically detectable brain metastases from 24% to 6%. However, no significant impact of PCI on survival was seen in any of those studies. Retrospective analyses suggested that virtually all benefit with PCI was confined to patients who achieved a complete remission from their initial treatment.[93]

A meta-analysis of almost 1,000 patients in seven trials between 1977 and 1995 assessed outcomes of patients with and without PCI after initially obtaining a complete response.[94] PCI doses ranged from 24 to 40 Gy in most patients. The meta-analysis suggested an improvement in 3-year survival from 15.3% to 20.7%, with decreased probability of brain metastases and increased DFS. Higher doses appeared to have no impact on survival, although seeming to reduce brain metastases. A trend also was seen toward a decreased risk of brain metastases when PCI was administered earlier. The meta-analysis was not able to assess the impact of PCI on cognitive function, as most of the studies did not include thorough neurocognitive assessments.

An Intergroup trial evaluated standard-dose versus higher dose PCI after complete response for LS disease.[95] A total of 720 patients were randomized to either 25 Gy in 10 fractions or 36 Gy, delivered in 18 daily fractions of 2 Gy, or 24 twice-daily fractions of 1.5 Gy. There was no significant difference in incidence of brain metastases between the standard-dose group and the high-dose group. OS was significantly worse in the higher dose group, although this was due to extracranial disease progression rather than toxicity. The reason for this counterintuitive result remains unclear. This study further established 25 Gy in 10 fractions as a standard regimen of PCI for LS patients.

Due to the now-widespread use of brain MRI screening and surveillance, there is increasing interest in using PCI more selectively

in LS-SCLC. In particular, patients with very early-stage SCLC may have a lower baseline risk of developing brain metastases, and therefore a correspondingly lower potential benefit from PCI. Limited retrospective data in Stage I or II SCLC suggest that omitting PCI may be a reasonable option for these patients, though prospective validation is still needed.[96,97] Phase III clinical trials PCI versus MRI surveillance including limited-stage SCLC are currently underway (NCT04829708; NCT04155034).

Prophylactic Cranial Irradiation for Extensive Stage SCLC

A European Organisation for Research and Treatment of Cancer (EORTC) randomized trial reported a survival benefit with PCI in patients with ES SCLC who had had a response to chemotherapy.[98] A common criticism of this trial is its lack of mandatory pretreatment brain imaging to rule out clinically occult brain metastases, raising the question of whether the apparent survival benefit was due to treatment of patients who already had detectable brain metastases. A Japanese multicenter phase III trial assessed the benefit of PCI in patients undergoing both baseline and regular follow-up brain MRI, who had achieved any response to platinum-based doublet chemotherapy. Though PCI significantly reduced the incidence of brain metastases (33% with PCI vs 59% without at 1 year), there was no survival benefit, and the study was terminated early as a result.[99] These data suggest that the use of brain MRI to rule out occult brain metastasis, followed by regular MRI surveillance, may be a preferable alternative over PCI in most patients with ES SCLC.

Toxicity of Prophylactic Cranial Irradiation

A major concern with PCI is the significant risk of toxicity. Some long-term survivors have neurologic and intellectual impairment as well as abnormalities on CT scan that may be related to PCI.[100,101] In one study, CT scan and CNS abnormalities were significantly more frequent in patients who had received PCI or therapeutic brain irradiation than in those who had not.[102] These findings were especially disturbing because complete responders are at greater risk for complications. Many deficits on neuropsychological testing are unsuspected on casual examination, but some patients have obvious major impairments. CT scan abnormalities can continue to worsen for several years after treatment has ended, although the abnormalities may eventually stabilize.[103] Neurologic abnormalities were most prominent in one series of patients who were given PCI concurrently with high-dose chemotherapy or in individual radiation fractions of 4 Gy.[100]

Complicating assessment of neuropsychological and imaging abnormalities after PCI are other factors, including chemotherapy, micro-metastases, and chronic cigarette and alcohol abuse, that may be important confounders. In one study that evaluated cognitive function in patients before and after chemoradiation but before PCI, deficits were discovered in verbal memory, frontal lobe function, and motor coordination within both groups of patients.[104] A similar, more recent study indicated that 47% of SCLC patients had impaired cognitive function before PCI and that PCI was not associated with persistent declines in cognitive function.[105] The Japanese trial of PCI in ES disease referenced previously, which utilized the now-standard dose of 25 Gy in 10 fractions, reported no differences in cognitive function as assessed with the mini-mental status exam at baseline, 12 months, or 24 months.[99]

A decision analysis suggests that for patients who have had a complete response to initial therapy, PCI offers a better quality-adjusted life expectancy, even if a mild to moderate neurotoxicity rate is assumed.[106]

There has been increasing interest in delivering PCI while sparing the hippocampi, to limit the potential for adverse neurocognitive effects. Hippocampal-sparing brain radiation has been tested in multiple randomized trials of PCI for SCLC, two of which were recently published with apparently conflicting results. A randomized trial of 168 patients in the Netherlands and Belgium compared standard PCI versus hippocampal-avoiding PCI (HA-PCI) in both LS- and ES-SCLC patients after chemotherapy.[107] A negative brain MRI was required prior to enrollment. The primary end point was neurocognitive function, specifically the decline in total recall as assessed by the Hopkins Verbal Learning Test-Revised 4 months after PCI.[107] This Phase III trial demonstrated no significant difference in total recall with HA-PCI vs standard PCI ($P = 1.000$), nor in other cognitive tests. In contrast, a similar randomized trial conducted in Spain did show a significant difference in cognitive function between HA-PCI and standard PCI groups. Neurocognitive function at 3 months after PCI assessed using the Free and Cued Selective Reminding Test demonstrated a significant difference in decline in function with HA-PCI versus standard PCI (5.8% vs 23.5%, $P = 0.003$). The reason for these contradictory results is unclear but could be attributed to the different neurocognitive tests employed.

An additional phase III randomized trial of HA-PCI is being conducted by NRG Oncology (NCT02635009).[108] Though HA-PCI has a sound rationale and preliminary data, there is not yet sufficient support to recommend HA-PCI as a routine standard of care. Brain metastases can occur in the spared hippocampal region, and on both randomized trials above, the HA-PCI group had numerically inferior outcomes on some neurocognitive tests. As the radiation delivery techniques for HA-PCI necessarily entail increasing the dose to certain portions of the brain outside the hippocampal-avoidance area, this raises the possibility that HA-PCI could have unanticipated neurocognitive disadvantages, not only potential advantages. Therefore, HA-PCI should only be offered as part of a clinical trial until additional data are available.

The authors' guidelines for PCI, after a thorough discussion with the patient of potential risks and benefits, are as follows: PCI is typically recommended for LS patients and may be considered for rare ES patients, after a complete or near-complete response to induction therapy has been confirmed and after acute toxicities have resolved, which is typically 2 to 4 weeks after completion of all chemotherapy. The standard fractionation of 25 Gy in 10 fractions is recommended for all patients in whom PCI is deemed appropriate, with a consideration of 20 Gy in 5 fractions for patients for whom, due to borderline PS or significant logistical impediments, shortening overall treatment time is felt to be important. In patients not treated with PCI, surveillance with MRI imaging is recommended. Concurrent administration of memantine, which has been shown to ameliorate neurocognitive toxicity in the setting of whole brain radiation for brain metastases, may also be considered.[109]

Surgery in Early-Stage Small Cell Lung Cancer

Studies evaluating the role of surgery in SCLC are few in number and nonrandomized. Surgery was largely abandoned in the management of SCLC after the British Medical Research Council reported equivalent outcomes between surgery and radiotherapy as a sole modality in resectable SCLC in 1966.[110,111] However, multiple retrospective studies have reported favorable survival in SCLC with stage I disease after surgical resection and adjuvant chemotherapy. AJCC TNM staging should be used when assessing the benefit of surgery as part of multimodality therapy. Nodal (N) and primary tumor (T) status have significant effects on the survival of patients who have undergone resection. The International Association for the Study of Lung Cancer (IASLC) Lung Cancer Staging Project reported on >8,000 SCLC cases, of which 349 (4%) underwent resection and surgical staging. The 5-year survival for stage I, II, and III SCLC were 48%, 39%, and 15%, respectively.[41] Five-year survival decreases with increasing T category in surgically resected patients without nodal metastases.[112]

There are no randomized trials comparing surgery versus surgery followed by chemotherapy; however, observations from

several trials support benefit from adjuvant therapy. The largest experience examining the role of surgery followed by adjuvant therapy in SCLC was a cooperative group trial conducted by the International Society of Chemotherapy Lung Cancer Study Group.[113] Four-year survival rates for completely resected, pathologically staged SCLC patients (n = 183) with N0, N1, and N2 disease who received postoperative therapy were 60%, 36%, and 33%, respectively.

A stage-specific analysis of patients in the National Cancer Database (NCDB) with stage I-IIIA SCLC who received either surgery or chemotherapy-based nonsurgical management has been reported. Of the total 29,994 identified patients a total 2,089 patients were propensity matched; of the patients with N0 disease, surgery was associated with longer survival than nonsurgical management (mOS 38.6 vs 22.9 months, $P < 0.0001$) and the subgroup of stage I/II patients who received a lobectomy and adjuvant chemotherapy had a longer survival than those who received chemoradiation (mOS 48.6 vs 28.7 months, $P = 0.007$).[114] Another NCDB analysis of over 1,500 patients with stage I SCLC who underwent surgical resection reported a significant improvement in survival among patients who received adjuvant chemotherapy alone or chemotherapy and cranial irradiation receiving adjuvant chemotherapy alone (hazard ratio [HR], 0.78; 95% confidence interval [CI], 0.63 to 0.95) or with chemotherapy and cranial irradiation (HR, 0.52; 95% CI, 0.36 to 0.75).[115] Current guidelines support surgery followed by chemotherapy as part of the initial management for patients with stage I disease.[116]

Systemic Chemotherapy

At diagnosis SCLC is typically highly responsive to cytotoxic chemotherapy. The recommended first-line chemotherapy regimen for patients with LS SCLC is cisplatin plus etoposide administered concurrently with thoracic radiation or adjuvantly after surgical resection of stage I disease. Carboplatin can be substituted for cisplatin in elderly patients or those with hearing loss or renal insufficiency but is associated with more hematologic suppression. For patients with ES SCLC, a platinum agent (cisplatin or carboplatin) combined with etoposide and an anti-PD-L1 inhibitor (atezolizumab or durvalumab) for four cycles followed by anti-PD-L1 therapy as maintenance is the recommended first-line regimen. The key studies and observations that led to the establishment of EP as the SOC chemotherapy doublet and the benefit of adding anti-PD-L1 therapy for ES SCLC are summarized in the following sections.

Evolution of Chemotherapy Regimens

The sensitivity of SCLC to chemotherapy agents was recognized over 50 years ago, and systemic treatment continues to have a primary role in the treatment of SCLC. Alkylating agents, anthracyclines, vinca alkaloids, and antifolates have all shown single-agent efficacy in this disease. In the late 1970s CAV became the standard of care regimen based on a study of 358 patients who received CAV followed sequentially by thoracic and brain radiation.[117] Using response assessment parameters of the time, for patients with ES disease, the ORR was 57%, and median survival 26 weeks; for LS disease, 75%, and 52 weeks, respectively.

The combination of etoposide plus cisplatin was first explored clinically in the late 1970s.[118] Studies reported response rates of 55% in patients previously treated with CAV and 86% in newly diagnosed patients.[119] Less myelosuppression occurred with EP compared with CAV, and when given with radiation, EP was better tolerated with less esophagitis and interstitial pneumonitis.[120–122] In the largest trial comparing the regimens, EP produced a better median (15 vs 10 months) and 5-year (10% vs 3%) survival for patients with LS disease.[122] Retrospective analyses and meta-analyses also support the superiority of cisplatin- or carboplatin-containing chemotherapy for SCLC.[123–125] As a result, EP became the standard first-line chemotherapy regimen for SCLC by the 1990s.

Carboplatin has been substituted for cisplatin in SCLC regimens in an effort to decrease nonhematologic toxicities. Randomized trials comparing cisplatin and carboplatin suggest that they have similar efficacy in SCLC. The Hellenic Cooperative Oncology Group randomized 147 patients with either LS or ES disease to receive cisplatin or carboplatin in combination with etoposide.[126] Concurrent radiotherapy was administered for LS patients starting with the third cycle. Response and survival were similar in the two arms, although the sample size of this study was inadequate to confirm equivalency. A meta-analysis that evaluated individual subject data from four randomized trials with a total of 663 SCLC patients found that median OS, median progression-free survival (PFS), and response rates were similar between patients receiving cisplatin or carboplatin.[127] Hematologic toxicities were higher with carboplatin while nonhematologic toxicities were increased with cisplatin.[127] Based on these data, etoposide and carboplatin can be considered an appropriate first-line regimen in patients with ES disease and those who cannot tolerate cisplatin.

In the late 1990s, platinum combinations with topoisomerase-1 inhibitors emerged as potential regimens for initial therapy. The Japan Clinical Oncology Group (JCOG) compared cisplatin and irinotecan (IP) to EP as initial treatment in ES SCLC.[128] The study was terminated after 154 of the planned 230 patients were enrolled because mOS and 2-year survival rates were significantly better in the group treated with IP compared with EP (mOS: 12.8 vs 9.4 months and 2 year survival rates of 19.5% vs 5.2%). Myelosuppression was the most common toxicity in both groups but more frequent with EP, whereas significant diarrhea occurred only in the irinotecan group. Two confirmatory studies were subsequently launched in the United States; one utilized a modified dosing schedule to reduce toxicities while the other used dosing schedule identical to the JCOG study. Neither study confirmed the Japanese result in a US population: both reported equivalent outcomes for EP and IP with respect to ORR and mOS.[129,130] Population-based polymorphisms in uridine 5′-diphospho-glucuronosyltransferase (UDP-UGT1A1), the enzyme responsible for detoxifying the active metabolite of irinotecan, may account for differences observed in the Japanese and American studies.[131] Topotecan plus cisplatin has also been assessed in the first-line setting. In a phase III study, 784 patients were randomized to receive oral topotecan plus cisplatin or EP. The response rates, mOS, and 1-year survival were identical between the arms. Severe neutropenia occurred more often with EP, while treatment with oral topotecan plus cisplatin resulted in more anemia and thrombocytopenia.

Alternating Combinations or Triplet Chemotherapy

The recognition of clonal heterogeneity within a tumor and the intolerability of treatment regimens that included more than four drugs led to trials of alternating chemotherapy combinations.[120,121,132–139] Most of these studies focused on alternating EP and CAV on various schedules, but with some using up to eight cytotoxic drugs. Summarizing the data from these studies, alternating regimens did not offer substantial benefits over initial treatment with EP alone. Similarly, the addition of a third drug (ifosfamide or paclitaxel, among others) to standard EP resulted in more toxicity with little or no improvement in survival.[140–145]

Maintenance Chemotherapy

Several studies have evaluated the use of maintenance chemotherapy after initial induction chemotherapy. These studies have reported mixed results but in general have not demonstrated substantial efficacy when current first-line regimens are employed. ECOG 7593 enrolled 402 eligible patients with ES disease to four cycles of EP and those with at least stable disease (SD) were randomized to four cycles of topotecan or observation.[146] In this study, maintenance therapy with topotecan increased the time to disease progression but without a survival benefit. Two meta-analyses

including 14 or 21 randomized trials evaluated maintenance chemotherapy or biologics and reported small improvements in survival but with increased toxicity.[147,148] Taking these data together, maintenance therapy was not recommended for patients with SCLC.

Dose Intensification

Several studies have explored high-dose chemotherapy in patients with SCLC. Several randomized trials compared standard versus high-dose CAV or EP found no difference in ORR or median survival.[149,150] Hematopoietic growth factors were not used in these trials, and myelosuppression and infections were significantly more severe in the high-dose arms. A randomized trial utilizing granulocyte-macrophage colony-stimulating factor (GM-CSF) with dose escalation found that excess toxicity resulted in lower drug delivery and poorer response and survival rates in the dose-intense arm.[151] In a study of 105 LS patients administration of higher drug doses resulted in improvement in PFS at 2 years (28% vs 8%) and OS (43% vs 26%).[152] Overall dose intensification has not been recommended, as most studies have shown excessive toxicity without substantial benefit.

Dose-dense regimens have also been investigated, including a multicenter study that randomized 300 patients, the majority of whom had LS disease, to six cycles of ifosfamide, carboplatin, etoposide, and vincristine (ICE-V) delivered every 4 versus every 3 weeks (resulting in a 26% increase in dose intensity).[153] Median survival (443 vs 351 days) and 2-year survival (33% vs 18%) were better in the intensified arm ($P = .0014$). However, four other randomized trials have concluded that intensive multidrug regimens are significantly more toxic than standard regimens without improving outcomes.[154–157]

Finally, dose intensification with autologous stem cell rescue has also been assessed in SCLC. ICE given every 4 weeks was compared to ICE given every 2 weeks with support of G-CSF, and autologous blood collected before the cycle was reinfused 24 hours after the chemotherapy.[158] With this approach, the median delivered dose was increased by 82% without a significant increase in toxicity; however, no survival benefit was identified. Several other smaller studies of dose intensification with autologous stem cell rescue have consistently shown that survival is not improved relative to conventional treatment. High-dose chemotherapy with stem cell support does not appear to have a role in SCLC.

Chemotherapy Plus Immunotherapy

Evasion of immune surveillance is a recognized attribute of cancer.[159] Therapies designed to enhance antitumor immune responses, in particular, antibodies targeting the immune-checkpoint proteins, CTLA-4, PD-1, or its ligand PD-L1, have been under investigation in every stage of SCLC management. The earliest studies focused on anti-PD-l and anti-CTLA-4 antibodies in relapsed SCLC and after promising initial results, additional studies expanded to evaluated ICIs in all stages of SCLC, alone and in combination with SOC agents (Table 31.2).

Early Studies of Immune Checkpoint Inhibitors in Relapsed SCLC

KEYNOTE-028 was an open-label, phase Ib multicohort study of pembrolizumab in PD-L1(+) solid tumors and included an SCLC cohort. A total of 24 patients with PD-L1–positive SCLC received treatment with pembrolizumab and had an ORR of 33%.[168] CheckMate 032 was a phase I/II basket study of nivolumab with or without the anti-CTLA-4 antibody, ipilimumab, which also included a cohort of patients with recurrent SCLC. Data from the initial SCLC cohort (n = 216) demonstrated ORRs of 10% for nivolumab and 19% to 23% for two dosing regimens of nivolumab plus ipilimumab.[170] Encouraging 1-year survival rates of 33% and 35% to 43%, respectively, were also reported. The toxicity profiles were similar to those observed in prior studies of these agents. A randomized expansion of the CheckMate 032 SCLC cohort including 242 patients confirmed the efficacy results (ORR of 12% and 21% for nivolumab and nivolumab plus ipilimumab, respectively).[170] Based on CheckMate-032, KEYNOTE-028, and KEYNOTE-158 data, nivolumab and pembrolizumab received accelerated approval by the US FDA in 2018 and 2019, respectively. For both nivolumab and pembrolizumab, confirmatory studies, however, did not meet their primary end points and the FDA approvals have been withdrawn.

Studies of Immune Checkpoint Inhibitors in the First-Line Setting for ES SCLC

Ipilimumab was the first ICI studied in the front-line setting for SCLC and showed promising results when combined with either carboplatin plus paclitaxel or carboplatin plus etoposide in two phase II studies.[171,172] However, a subsequent randomized phase III study of EP plus ipilimumab vs. EP plus placebo in the first-line setting did not demonstrate an improvement in OS (median 11.0 vs 10.9 months, respectively) (Table 31.2).[160] Ipilimumab was associated with increased frequency of immune-related toxicities including diarrhea, rash, and colitis.

Combination EP Plus a PD-L1/PD-L1 Inhibitors

In contrast to results observed with CTLA-4 inhibition, multiple phase III trials now demonstrate the benefit of adding anti-PD-1/PD-L1 checkpoint inhibitors to first-line EP for ES SCLC patients. These studies have established the role of chemo-immunotherapy with PD-L1 inhibitors in the front-line treatment of extensive-stage SCLC (Table 31.2).

The IMpower133 trial randomly assigned 403 patients with extensive-stage SCLC and no prior systemic therapy to the anti–PD-L1 monoclonal antibody atezolizumab or placebo given with carboplatin and etoposide chemotherapy for four cycles.[161] Atezolizumab or placebo was continued as maintenance until disease progression, unacceptable toxicity, or loss of clinical benefit. The primary end point of median OS was 12.3 months (95% CI, 10.8 to 15.8) in the atezolizumab arm versus 10.3 months (95% CI, 9.3 to 11.3) in the placebo arm, and the HR for death was reduced by 24% (HR, 0.76; 95% CI, 0.60 to 0.95).[162] The OS rate at 12 months was 52% in the chemo-immunotherapy arm versus 39% in the placebo arm, a difference that maintained at 18 months (34% vs 21%), thus a subset of patients experienced a durable response from the PD-L1 inhibitor. Grade 3 or 4 immune-related AEs (irAEs) occurred in 8.1% of those in the atezolizumab arm and in 2.6% of those in the placebo arm.

Similarly, the randomized open-label phase III CASPIAN trial compared the addition of the anti–PD-L1 antibody, durvalumab, or both durvalumab and the anti–CTLA-4 antibody, tremelimumab, to an EP backbone as first-line treatment in 805 patients with ES SCLC.[163] Subjects on the combination immunotherapy + EP arms received a maximum of four cycles of chemo-immunotherapy followed by durvalumab maintenance without PCI. In contrast to the IMpower133 study, the CASPIAN study allowed up to six cycles of EP chemotherapy (either carboplatin or cisplatin) and PCI in the control arm. The two primary end points were OS for each investigational arm versus control arm (EP) in the intention-to-treat population. The durvalumab + EP arm demonstrated a significant improvement in mOS (12.9 months vs 10.5 months for chemotherapy alone; HR 0.75, 95% CI, 0.62 to 0.91).[164] While the study was not powered for PFS, a higher PFS rate at 12 months (18% vs 5%) was noted and suggested sustained clinical benefit with the addition of durvalumab. Grade 3 or 4 immune-mediated AEs were reported in 5% of patients who received durvalumab. The Impower133 and CASPIAN trials led to FDA approval for atezolizumab or durvalumab in combination with platinum-based chemotherapy in the first-line treatment of ES SCLC.

TABLE 31.2

Randomized First-Line Chemotherapy Plus Immunotherapy Trials in ES SCLC

Trial (Phase)	N	Arms	ORR (%)	mPFS (mos)	HR	mOS (mos)	HR (P)	Landmark OS		
								12 m	18 m	24 m
EP plus Ipi vs placebo (R-Ph3)[160]	478	EP + Ipi followed by Ipi maintenance	62	4.6	0.85 *P = .0161*	11.0	0.94 *P = .3775*	40		
	476	EP + placebo followed by placebo maintenance	62	4.4		10.9		40		
IMpower133 (R-Ph3)[161,162]	201	EP[a] + Atezo × 4 followed by Atezo maintenance	64	5.2	0.77 *P = 0.02*	12.3	0.76 *P = .0154*	52%	34%	
	202	EP × 4 + placebo followed by placebo maintenance	60	4.6		10.3		39%	21%	
CASPIAN (R-Ph3)[163,164]	268	EP + Durva followed by Durva maintenance	68	5.1	0.78[b] *P = .045*	12.9	0.75[b] *P = .0032*		32%	
	268	EP + Durva plus Treme × 4 followed by Durva maintenance	58	4.9	0.84[b] *(P NR)*	10.4	0.82[b] *P = .045*		31%	
	269	EP + placebo × 4–6	58	5.4		10.5			25%	
KEYNOTE-604 (R-Ph3)[165]	228	EP + Pembro followed by Pembro maintenance	71	4.5	0.75 *P = .0023*	10.8	0.80 *P = .0164*	45%		23%
	225	EP + Placebo followed by Placebo	62	4.3		9.7		40%		11%
EA5161 (R-Ph2)[c166]	80	EP + Nivo × 4 followed by Nivo maintenance	52	5.5	0.65 *P = .012*	11.3	0.67 *P = .038*			
	80	EP × 4 followed by observation	47	4.6		8.5				
ICI Maintenance after Induction EP										
Trial (Phase)	**N**	**Arms**	**ORR (%)**	**mPFS (mos)**	**HR[b]**	**mOS (mos)**	**HR[b]**	**Landmark OS (12 mo)**		
CheckMate-451 (R-Ph3)[167]	261	Nivo for up to 2 years	11	1.9	0.67	10.4	0.84	45%		
	265	Nivo plus ipi × 4 followed by nivolumab up to 2 years (total)	9	1.7	0.72	9.6	0.92 *(P = .37)*	41%		
	263	Placebo for up to 2 years	4	1.4		9.2		40%		

[a]Impower 133 allowed carboplatin only.
[b]HR compared with control.
[c]ITT analysis.
ORR, objective response rate; mPFS, median progression-free survival; H, hazard radio; mOS, median overall survival; R-Ph3, randomized Phase 3; R-Ph2, randomized Phase 2; EP, etoposide plus platinum (cisplatin or carboplatin); Nivo, nivolumab; Ipi, ipilimumab; Atezo, atezolizumab; Durva, durvalumab; Treme, tremelimumab; Pembro, pembrolizumab.

The anti–PD-1 antibodies, pembrolizumab and nivolumab, have also been evaluated in the first-line setting with chemotherapy with similar results. KEYNOTE-604 was a phase III study that randomized 453 patients to pembrolizumab plus EP or placebo plus EP followed by maintenance pembrolizumab or placebo.[165] While the trial did not meet its prespecified end point for OS, the HR for OS favored the addition of pembrolizumab (HR, 0.80; 95% CI, 0.64 to 0.98; median OS, 10.8 vs 9.7 months). The addition of pembrolizumab to EP resulted in a PFS benefit (HR, 0.75; 95% CI, 0.61 to 0.91) as well as improvement in OS at 24 months (22.5% vs 11.2% with chemotherapy alone). The randomized phase II ECOG-ACRIN trial EA5161 investigated the addition of nivolumab to chemotherapy. This study randomized 160 patients to nivolumab plus EP x 4 cycles followed by nivolumab maintenance vs EP x 4 followed by observation. This study met its primary end point of improved PFS (HR, 0.65; 95% CI, 0.46 to 0.91).[166] Results from this smaller phase II trial align with those of the previously mentioned phase III trials, with median PFS of 5.5 vs 4.6 months and median OS of 11.3 vs 8.5 months (HR for OS 0.67; 95% CI, 0.46 to 0.98).

Notably, in an exploratory analysis of 35 patients in IMpower 133 who had treated brain metastases at diagnosis, there was no clear benefit of adding atezolizumab to EP (HR of 0.96; 95% CI, 0.46 to 2.01).[162] Similarly, in subgroup analysis from the CASPIAN study, patients with asymptomatic brain metastases at diagnosis (10%) did not show a clear benefit from durvalumab (HR, 0.79; 95% CI, 0.44 to 1.41).[164] In the absence of larger, confirmatory analyses, patients with brain metastases, treated or not, are still recommended to receive immunotherapy in the first-line setting.

In summary, these four randomized studies support a class effect for PD-1 and PD-L1 inhibition in combination with chemotherapy. While the 2-month improvement in median survival is modest, the maintained separation of survival curves with increased survival observed at 2 years compared with control highlights that a subset of patients experiences a durable benefit from immunotherapy. In these studies, PD-L1 expression or tumor mutation burden did not correlate with benefit. Without a clear predictive biomarker, all newly diagnosed ES SCLC patients without contraindications are recommended to receive chemotherapy plus immunotherapy as a first-line therapy.

Studies of Dual ICI in First-Line Treatment of ES SCLC

Dual ICI therapy has been assessed initially as maintenance after induction EP in CheckMate-451 and concurrently with EP as part

of the CAPSIAN study. In the CheckMate-451 study, 834 patients with stable disease or response after four cycles of EP chemotherapy were randomly assigned to receive nivolumab, nivolumab plus ipilimumab, or placebo. The primary end point was OS comparing nivolumab plus ipilimumab versus placebo. The HR for PFS suggested benefit for combined ICI (HR 0.72 for nivolumab plus ipilimumab (95% CI, 0.60 to 0.87); however, OS was not improved (HR 0.92; 95% CI, 0.75 to 1.12; $P = .37$).[167] The rates of grade 3–4 TRAEs were 52.2% with the nivolumab plus ipilimumab arm versus 8.4% in the placebo arm.

In the CASPIAN study, the addition of durvalumab and tremelimumab to EP did not significantly improve OS over chemotherapy alone (mOS 10.4 with combined ICI vs 10.5 months in the placebo arm; HR, 0.82; 95% CI, 0.68 to 1.00). Higher rates of irAEs were observed with durvalumab/tremelimumab/EP relative to durvalumab/EP. At this time there is no role for adding anti-CTLA4 therapy to anti-PD-L1 in the front-line management of ES SCLC.

Role of Consolidative Irradiation in ES SCLC

The potential role of consolidative chest radiotherapy in patients with ES SCLC has been controversial. Early reviews of nonrandomized outcome data suggested that the addition of chest irradiation plus chemotherapy for patients who have ES SCLC might improve local control in the thorax but without change in survival.[173,174] Several clinical trials have randomized patients with ES disease to chemotherapy alone or in combination with irradiation to the chest as well as to some or all sites of overt distant metastases.[175–178] Only one of these trials reported a survival benefit for consolidative chest irradiation.[176]

A more recent phase III trial randomizing ES SCLC patients who had a response to four to six cycles of chemotherapy to TRT or no TRT was negative for its primary end point of 1-year survival, but a post hoc analysis suggested a survival benefit with TRT at 2 years.[179] Subsequent additional post hoc analyses from this trial further suggested that patients with fewer than three metastases and patients with residual intrathoracic disease after chemotherapy might be most likely to derive benefit from TRT.[180,181]

A related question is whether radiation to all sites of metastasis, not just the primary thoracic disease, may be beneficial as a consolidation strategy in ES SCLC patients with few sites of evident disease. The RTOG performed reported a randomized trial in which patients with ES disease and ≤4 metastatic sites (excluding the CNS) received PCI and consolidative RT to the thorax and all metastatic sites or PCI alone.[182] This study was closed early due to futility, as it was recognized that the experimental arm would not meet the end point of improving survival compared to the control arm. Overall, consolidative radiotherapy to the thorax can be considered in select ES SCLC patients with good systemic response to initial chemotherapy and evident residual disease confined to the chest, but radiotherapy to additional metastatic sites (other than for palliation) is not supported by evidence at this time.

Summary and Recommendations for Initial Management of Small Cell Lung Cancer

Limited Stage SCLC (Stage I–III, that can be Treated with a Tolerable Radiation Port)

The SOC for most LS SCLC patients is concurrent chemoradiotherapy. When possible, TRT should be hyperfractionated (twice daily) and initiated within the first two cycles of chemotherapy. The recommended chemotherapy regimen is cisplatin plus etoposide for four cycles. Carboplatin is an acceptable alternative for patients who are not cisplatin candidates. Good PS patients with T1-2 tumors and node-negative disease confirmed by invasive mediastinal staging may be recommended for surgical resection. Resection should follow guidelines for NSCLC. Adjuvant therapy with four cycles of cisplatin plus etoposide is recommended.

After completion of primary therapy (chemoradiotherapy or resection followed by adjuvant chemotherapy), patients are recommended to have restaging CT scans and brain MRI. Patients who have responded to therapy without evidence of intracranial spread should be considered for PCI. There is no recommended maintenance therapy for LS SCLC. Patients should have surveillance body imaging, blood work, and physical exams every 3 months for the first 2 years, every 6 months during year 3, and then annually.

Extensive Stage SCLC (Stage IV, or T3-4 Disease that Cannot Be Treated with a Tolerable Radiation Port)

The SOC for ES SCLC is four cycles of etoposide plus platinum agent (cisplatin or carboplatin) and an anti-PD-L1 antibody, atezolizumab or durvalumab, followed by maintenance with the anti-PD-L1 antibody. PCI or TRT after first-line therapy might benefit select patients as outlined above. After completion of initial therapy, patients should be monitored closely with surveillance body imaging, blood work, and physical exams approximately every 2 months for the first year, every 3 to 4 months in years 2 and 3, every 6 months in years 4 and 5, and then annually. Good-PS patients should be offered further systemic therapy at the time of progression. Both practice guidelines in Europe and the US support this approach.[183]

Treatment of Relapsed Small Cell Lung Cancer

Most patients diagnosed with SCLC, will relapse, including nearly all those who presented with ES disease and >75% of those who receive definitive treatment for LS disease. Patients with relapsed SCLC are generally categorized as having chemotherapy-sensitive or chemotherapy-refractory disease based on the chemotherapy-free interval (CTFI) they experience after completion of EP, regardless of concomitant administration of PD-L1 inhibitor. Patients who experience a CTFI of >90 days are considered to have chemotherapy (or platinum)-sensitive disease, those with a CTFI of <90 days, chemotherapy-resistant disease.[184] Survival for patients with sensitive relapse averages 6 months. Patients with refractory disease exhibit progression either during initial therapy or within 3 months after its completion and have a shorter median survival than those with sensitive relapse. As in first-line treatment of ES disease, the goals of care for patients with relapsed SCLC are primarily palliative and include symptom management and optimization of quality of life. For most of these patients, coordination with palliative care services should be offered if available and not already part of their management plan. Studies of treatment practices in the United States, Europe, and Japan reported that about 20% of patients with relapsed SCLC receive second-line therapy,[185] a finding reflective of the poor PS and limited therapeutic options for this patient population.

As summarized in the following section, an optimal therapeutic strategy for patients with relapsed SCLC has not been established. Patients whose cancer relapses ≥6 months after initial therapy are often retreated with a first-line chemotherapy doublet. The only FDA-approved therapeutic agents for relapsed SCLC are topotecan and lurbinectedin, the latter of which currently has accelerated approval status. Trilaciclib, a CDK4/6 inhibitor, has received FDA approval to decrease the incidence of chemotherapy-related myelosuppression (CIM) in SCLC patients receiving chemotherapy.

The remainder of agents recommended for relapsed SCLC in national guidelines are based on activity observed in single-arm phase II studies and are considered category 2A recommendations.

FDA-Approved Therapies for the Treatment of Relapsed SCLC

Topotecan: Initial promising phase II studies of topotecan in relapsed SCLC[186] were followed up by randomized phase III studies. In one study, patients who relapsed at least 2 months after initial therapy were randomized to receive topotecan IV or CAV.[187] The response rates for topotecan and CAV were similar (24% vs 18%, respectively) as was the mOS (6 months in both arms); however, symptom improvement was better with topotecan. Based on these results, IV topotecan received FDA approval for sensitive relapsed SCLC. Oral topotecan has also received regulatory approval for second-line therapy of SCLC. In 2006, the Medical Research Council reported that oral topotecan improved survival in relapsed sensitive and refractory SCLC over best supportive care (median 26 vs 14 weeks, respectively) with a slower deterioration of quality of life.[188] In a phase III noninferiority study, treatment with oral topotecan resulted in comparable ORR and mOS compared to IV topotecan. Oral topotecan was associated with a lower incidence of grade 4 neutropenia but a higher incidence of diarrhea. Topotecan crosses the blood–brain barrier and has activity against brain metastases. Fatigue and hematologic toxicities are substantial with topotecan, particularly at the recommended schedule.

Lurbinectedin: Lurbinectedin is an alkaloid analog with pleotropic effects including inhibition of transcription, induction of single- and double-strand DNA breaks, and induction of apoptosis. In a single-arm phase II study, 105 patients with relapsed SCLC after one prior chemotherapy received lurbinectedin 3.2 mg/m^2 IV every 3 weeks. The ORR was 35% with a median DOR of 5.2 months.[189] Higher ORR were observed in those with CTFI of ≥90 days compared with those who had a CTFI of <90 days (45% vs 22%, respectively). Common side effects included low-grade fatigue and nausea, and 23% of patients experienced grade 3–4 neutropenia. Based on these phase II data, in June 2020, lurbinectedin received accelerated approval for patients with metastatic SCLC with disease progression on or after platinum-based chemotherapy.

In September 2021, the results of a phase III study comparing lurbinectedin plus doxorubicin to either topotecan or CAV (ATLANTIS) were reported. This study enrolled 631 patients with relapsed SCLC after one prior platinum-based regimen and a CTFI of ≥30 days. Patients were randomized to receive lurbinectedin 2 mg/m^2 plus doxorubicin 40 mg/m^2 IV every 3 weeks or clinician's choice of either CAV or topotecan.[190] The study did not meet its primary end point of improved OS (HR 0.97). The median PFS was 4 months in each arm and ORRs were similar at approximately 30%. Currently lurbinectedin is recommended as monotherapy only for relapsed SCLC, and a full FDA approval will be contingent on outcomes of additional randomized studies. Ongoing studies of lurbinectedin in SCLC include combinations with immune checkpoint inhibitors (NCT 04610658; NCT 04610658), other chemotherapeutics, or the ATR inhibitor, berzosertib (NCT04802174).

Other Agents for Relapsed SCLC

Other treatment options for relapsed SCLC include several agents that have regulatory approval in other indications that are used for SCLC off label. Most have demonstrated only modest activity, primarily in single-arm phase II studies in recurrent SCLC. Examples include irinotecan, paclitaxel, temozolomide, and gemcitabine. Temozolomide is an oral alkylating agent that crosses the blood–brain barrier. A phase II clinical trial of temozolomide in 64 patients with relapsed sensitive or refractory SCLC reported an ORR of 20% (including response rates of 23% and 13% in sensitive and refractory relapse, respectively). Responses were observed in third-line patients and in those with brain metastases.[191]

Amrubicin, a third-generation synthetic 9-amino-anthracycline, has been studied extensively and has approval in Japan for SCLC treatment. A randomized phase II trial conducted in Japan reported higher response rates and PFS compared with topotecan.[192] Two North American phase II trials of amrubicin showed encouraging results,[193,194] particularly in chemotherapy-refractory patients, and led to a phase III study that compared amrubicin to topotecan.[195] This trial demonstrated no statistically significant difference in OS between the arms. Amrubicin is not available in the United States.

Role of Therapy for the Neuroaxis

For overt metastatic lesions within the CNS, radiation doses of 3 Gy daily to a total of 30 Gy are typically used. Overt intracranial metastases appear to be more difficult to eradicate than intrathoracic disease.[196] Due to the propensity for SCLC patients to develop multifocal brain metastases, stereotactic radiosurgery (SRS) for brain metastases generally has not been recommended due to concern for brain relapse beyond the treated lesion. In the context of prior whole-brain radiotherapy (WBRT) or PCI, SRS may be considered as a salvage option.[197] Emerging, though primarily retrospective, data also suggest that stereotactic radiosurgery (SRS) may be a reasonable treatment option for SCLC patients who do not receive PCI but develop brain metastases.[198] If further validated, this approach (deferring PCI with plan for SRS in the event of intracranial progression) could also support a move away from universally recommending PCI.

Chemotherapy is another option for brain metastases. Small series of patients in whom brain metastases were present at diagnosis and who were treated with standard chemotherapy regimens without radiation confirm that such patients can have clinical and radiographic improvement.[199] Chemotherapy also has been used at the time of relapse, and response rates of 33% to 43% have been reported.[199–201] In previously treated patients, the response to chemotherapy in the brain appears to be comparable to the response rates in other organs. In a phase II study of temozolomide in patients with relapsed or refractory SCLC, intracranial responses in patients with progressive brain metastases that had received prior cranial irradiation were observed.[191] Thus, although brain irradiation remains the standard for patients who have not been previously irradiated, chemotherapy is a reasonable option for those in whom recurrent disease develops after prior brain radiation, particularly if active systemic disease is also present.

Adjunctive Therapy to Mitigate Chemotherapy-Induced Myelosuppression

The cyclin-dependent kinases, CDK4 and CDK6, phosphorylate and inactivate Rb to drive cells from G1 to S phase. Trilaciclib, a reversible CDK4/6 inhibitor, arrests cells in G1 of the cell cycle. When given prior to chemotherapy, trilaciclib can transiently arrest hematopoietic stem cells in G1 and protect them from the cytotoxic effects of chemotherapy. Rb1 loss is a hallmark of SCLC, as such SCLC cells are able proliferate independently of CDK4/6. Trilaciclib has been evaluated in three randomized studies in patients with SCLC, including two first-line studies in combination with either EP or EP/atezolizumab, and with topotecan in relapsed disease. In a pooled analysis of the three studies, trilaciclib was associated with a decreased incidence of chemotherapy-induced myelosuppression (CIM) and did not reduce the efficacy of chemotherapy.[202] Whether this agent is superior to leukocyte growth factors in the supportive management of SCLC patients has not been assessed. Trilaciclib has received FDA approval to decrease the incidence of CIM when administered with an EP regimen or topotecan in SCLC.

Research Efforts in Small Cell Lung Cancer

The management of SCLC described in this chapter is tailored to stage, comorbidities, performance status, and number of prior therapies, but in contrast to NSCLC, the disease itself is in general treated as a single biologic entity. Belying this monolithic approach, since the initial efforts to culture SCLC tumors in the laboratory it has been clear that there are subclasses with distinct molecular features, growth properties, and drug responses. Over the past decade, advances in genomic profiling and the generation of genetically engineered mouse models have revealed a spectrum of oncogenic drivers beyond loss of *RB1* and *TP53* and amplification of *MYC*. In parallel, large scale transcriptional, epigenetic, and proteomic profiling efforts have delineated subtypes of SCLC with distinct biologic properties and drug sensitivities. Re-biopsy programs and advances in circulating tumor cell isolation have resulted in biobanks of patient-derived xenografts and insights into mechanisms of chemoresistance. These advances comprise a "second golden age" of SCLC research envisioned by the late Dr. Adi Gazdar as informing molecular targeted therapies that might ultimately improve patient survival and quality of life.[203] At the same time, advances in single-cell molecular profiling have revealed a new looming challenge, intratumoral heterogeneity, and underlying lineage plasticity that could render SCLC rapidly adaptable and resistant to new therapies. Addressing these challenges will require in-depth understanding of how SCLC undermines the barriers that insulate cell fate decisions and how to restore those barriers.

Molecular Characterization of Small Cell Lung Cancer

Oncogenes and Tumor Suppressors

The first fully sequenced SCLC genome was reported in 2010 from a cell line established at the National Cancer Institute Navy Medical Oncology Branch in 1979, NCI-H209.[204,205] This cell line demonstrated three genomic hallmarks of SCLC: (1) a high tumor mutational burden with a predominant C>A transversion signature due to tobacco carcinogens; (2) biallelic loss of Retinoblastoma 1 (*RB1*), the first gene ever identified as a tumor suppressor; and (3) biallelic loss of tumor protein 53 (*TP53*), the most commonly mutated gene in human cancer.[206,207] In contrast to *TP53*, the frequency of *RB1* mutation across most cancers is relatively low with three notable exceptions: retinoblastoma, osteosarcoma, and SCLC, of which SCLC is by far the most common.[208] Recent genomic profiles of extrapulmonary small cell carcinomas (EPSCCs) arising from the bladder and esophagus, as well as aggressive neuroendocrine prostate cancers (NEPCs), have also revealed loss of *RB1* and *TP53* in most cases.[209–211] This shared genotype underscores the broad molecular, histopathological, and clinical similarities among small cell neuroendocrine carcinomas arising within and outside the lung.

SCLC is rarely resected or re-biopsied, resulting in a scarcity of tumor samples that are suitable for genome sequencing. Over the past decade this challenge has been overcome through dedicated biopsy efforts and multi-institutional collaborations.[212–215] The resulting genomic profiles of SCLC confirmed the loss of *RB1* and *TP53* in nearly all cases, as well as a second feature that was previously discovered in patient-derived cell lines: recurrent amplifications of MYC family transcription factors *MYC*, *MYCN*, or *MYCL*.[216–218] Genetically engineered mouse models have shown that MYC family amplifications can accelerate SCLC tumorigenesis. Deletion of *Rb1* and *Trp53* alone can initiate murine SCLC (mSCLC) tumors after a long latency (12 to 14 months) and when they arise, such tumors often harbor spontaneous amplifications of *Mycl1*.[219–222] The latency of mSCLC emergence can be reduced to 6 months or less by adding alleles that overexpress MYC family genes.[223–226] In-frame fusions between *RLF* and *MYCL* were identified in approximately 5% of human SCLC, and when modeled in mice could also accelerate mSCLC tumorigenesis.[227] *RLF-MYCL* is the first oncogenic gene fusion to be described in SCLC.

During their long latency, *Rb1/Trp53* knockout-driven mSCLC can acquire spontaneous amplifications in nuclear factor 1 B-type (*Nfib*).[221,222] NFIB is a transcription factor involved in neuronal differentiation and cell migration. While *NFIB* is rarely amplified in human SCLC, it is often highly expressed in metastatic lesions and accelerates mSCLC tumorigenesis in combination with *MYCL*.[228,229] In NFIB-negative tumors the drivers of metastasis are not currently known.

Genomic profiles of SCLC have revealed additional recurrent alterations that promote mSCLC tumorigenesis when modeled in the *Rb1/Trp53* knockout background. In small numbers of human SCLC cases, tumor suppressor mutations were found in retinoblastoma-like proteins 1 or 2 (*RBL1* or *RBL2*), or phosphatase and tensin homolog (*PTEN*).[222,230–232] Inactivating mutations were found in genes encoding chromatin remodeling enzymes *CREBBP*, *EP300*, and *KMT2D*, in the NOTCH receptor family (*NOTCH1-4*), and, paradoxically, in myc-associated factor X (*MAX*), which normally heterodimerizes with MYC to activate gene expression.[215,233–236] Recurrent amplifications were found in the stem cell transcription factor *SOX2* and the stimulatory G-protein alpha subunit *GNAS*, but only two alterations have been identified in enzymes with clinically available inhibitors: activating mutations in the alpha-subunit of phosphoinositide 3-kinase (*PIK3CA*), and amplifications in fibroblast growth factor receptor 1 (*FGFR1*).[237–239] Targeting these will be of interest but such studies have not advanced beyond early-phase trials.

These complementary approaches of human SCLC genomics and mouse modeling have revealed a spectrum of oncogenes and tumor suppressors in SCLC and deepened our understanding of SCLC tumorigenesis. One translational obstacle is that beyond *RB1* and *TP53*, no alterations are recurrent in more than 15% of cases, and the paucity of samples and high mutation burden make it a challenge to detect low-frequency events. There are ongoing efforts to target *RB1* loss and *MYC* amplification by synthetic lethality, and promising targets include Aurora kinases, the lysine demethylase KDM5A, and the ubiquitin ligase SKP2.[240–243]

The Aurora kinase A inhibitor, alisertib, has been evaluated in early phase clinical trials. In a multicohort phase II study, patients with relapsed SCLC had a 21% ORR to alisertib.[244] The most frequent drug-related grade 3 to 4 toxicities were neutropenia (43%) and leukopenia (21%). A randomized phase II study for second-line treatment of SCLC combined alisertib with paclitaxel. A total of 178 patients were randomized to paclitaxel plus alisertib or paclitaxel plus placebo. Median PFS was 3.32 months on the investigational arm versus 2.17 on control (HR 0.77, 95% CI 0.557 to 1.067). While this study did not meet its primary end point of PFS in the intention-to-treat population, patients with mutation in cell cycle regulators (CDK6, RBL1, RBL2, RB1) or higher expression of c-Myc had significantly improved PFS when alisertib was added to paclitaxel.[245] The potential predictive biomarkers may warrant prospective study to move forward with this agent; however, grade 3 or higher drug-related AEs were substantially higher with the combination (67% vs 22% in the control arm) as were treatment-related deaths (4 vs 0, combination vs paclitaxel control).[245]

Gene Expression Profiles and Molecular Subtypes

As noted above, the current WHO pathologic classification of SCLC includes only designations for (pure) SCLC and combined SCLC. Although there are no other recognized clinical subtypes, early efforts at cell line derivation in the 1970s and 1980s revealed

dramatic molecular and morphologic differences between different tumors in culture.[246–248] These properties roughly subdivide SCLC into a "classic" category with strong expression of neuroendocrine (NE) markers and frequent *MYCL* amplifications, and a "variant" category more frequently derived from patients following relapse with lower NE marker expression and frequent *MYC* amplifications.[249–251] Clinical samples reflect this division, as expression patterns of individual NE markers such as synaptophysin and chromogranin A are heterogeneous both within and between tumors, and all may be negative in up to 10% of SCLC cases that nevertheless meet diagnostic criteria.[252]

Recent studies have ascribed inter-tumoral heterogeneity in SCLC to differential expression of a small number of transcription factors that contribute to cell lineage determinations during development. Two basic helix-loop-helix (bHLH) transcription factors with roles in neuronal development, achaetescute homologue 1 (*ASCL1*) and neurogenic differentiation factor 1 (*NEUROD1*), are differentially expressed in SCLC cell lines and correlate with "classic" (ASCL1) and "variant" (NEUROD1) morphologies.[59,253–259] Emerging research suggests that a third cohort of SCLC tumors without expression of ASCL1, NEUROD1, or most NE markers is driven by POU class 2 homeobox 3 (*POU2F3*), a transcription factor that normally regulates chemosensory tuft cell development.[260] A small number of cases are negative for all three "ANP" transcription factors and their classification remains in flux. Many ANP-negative SCLC lines were noted to have high expression of yes-associated protein 1 (*YAP1*) and this led to an initial "ANPY" classification schema.[261,262] However, in mSCLC deletion of *Yap* or its paralog *Taz* accelerates tumorigenesis, suggesting tumor suppressor function, and in patient samples or patient-derived xenograft (PDX) models, tumors with exclusive expression of YAP1 were rare or absent.[11,263–265] A second candidate factor discovered in ANP-negative PDX models is the bHLH transcription factor atonal homolog 1 (ATOH1), with assessment in clinical samples awaiting suitable IHC antibodies.[266]

Sub-classification of SCLC by transcription factor expression is advancing rapidly and may inform biomarker-directed therapies and clinical trials.[267] The first clinical application may emerge from discovery of an inflammatory gene signature in ANP-negative cell lines and tumors ("SCLC-I") that correlates with durable responses to immune checkpoint blockade.[268–270]

Additional subtype-specific targets associated in SCLC may include BCL2 and DLL3, both of which are associated with ASCL1 expression. BCL-2 is highly expressed in most SCLC tumors.[271–273] Clinical trials of BCL2–directed therapy either with antisense oligonucleotides or small-molecule inhibitors were disappointing in unselected populations of SCLC, but predated recognition of transcriptionally defined subtypes.[274–279] Several preclinical studies have shown combinatorial efficacy in targeting mTOR and BCL-2 in SCLC, which may prompt further investigation.[280,281] Initial studies of the DLL3-targeted antibody drug conjugate (ADC) rovalpituzumab teserine showed promising activity,[282] but later phase trials in the relapsed, maintenance, and first-line settings revealed unacceptable toxicity.[283,287] DLL3 remains a target of substantial clinical interest, and is overexpressed in ASCL1+ SCLC; agents in active clinical trials include the anti-DLL3 T-cell engagers AMG-757 and HPN328.

Notably these lineage-defining factors are not as strongly mutually exclusive as the oncogenic driver mutations in lung adenocarcinoma. Although most established cell lines can be assigned to one of these subtypes, this is less true for PDX models or primary tumor samples, where up to 30% show heterogeneous expression of two or even three ANP factors.[11,261,265,266,288,289] Underlying this heterogeneity may be epigenetic plasticity that permits subtype switching, as has been observed during *Myc*-driven mSCLC tumorigenesis.[290] Epigenetic plasticity is emerging as a hallmark feature of SCLC implicated in early metastasis and therapy resistance.

Intra-tumoral Heterogeneity, Lineage Plasticity, Cell of Origin, and Transdifferentiation

Further enumeration of SCLC subpopulations has been aided by single-cell RNA sequencing. Comparison of primary lung cancer specimens demonstrated that SCLC tumors have greater transcriptional intra-tumoral heterogeneity than lung adenocarcinomas, and comparison of SCLC PDX models suggested that this feature may contribute to treatment resistance, as heterogeneity is further increased after relapse.[291,292] Tumor microenvironments also influence subpopulations, as both tumor-stroma and tumor-immune interactions increase SCLC heterogeneity over isolated growth in culture.[293]

Some of this intratumoral heterogeneity is evident when SCLC specimens are placed in culture. Some disaggregated tumors grow as mixtures of floating clusters and adherent monolayers, and this mixed phenotype can persist.[250] Adherent and suspension subpopulations carry the same mutations; typically adherent cells downregulate expression of NE markers.[294,295] SCLC PDX and mSCLC models also produce mixed cultures when disaggregated, enabling investigation of the roles of each subpopulation in tumorigenesis, metastasis, and resistance to therapy. The NE-high population is enriched for "tumor propagating cells" that can reconstitute the entire tumor.[296] In contrast, adherent NE-low cells proliferate more slowly but have been implicated in tumor characteristics including chemoresistance, secretion of paracrine factors that promote metastasis, and formation of tumor-derived blood vessels in a process termed "vasculogenic mimicry."[297–300] Subpopulations may be spatially distinct, as shown in mSCLC driven by NFIB.[301] Single-cell RNA sequencing of primary SCLC tumors revealed a recurrent stem-like subpopulation marked by the phospholipase C enzyme *PLCG2* that expressed genes involved in metastasis and portended worse overall survival.[291] Strategies to selectively target these subpopulations are beginning to emerge. A recent analysis of cell death pathways in SCLC found that non-NE cells are dependent on glutathione peroxidase 4 (GPX4) to prevent ferroptosis whereas NE cells are dependent on the thioredoxin (TRX) antioxidant pathway.[302] Combined GPX4/TRX inhibition prevented escape of one cell type and represents the first preclinical example of subpopulation-targeted therapy.

Lineage plasticity, an erosion of barriers between the pathways that define cell fate, may contribute to the diversity of cell types in SCLC tumors. Selective targeting of distinct cell types in the mouse pulmonary epithelium demonstrates that multiple cell types can give rise to mSCLC, though with markedly different efficiencies.[303,304] SCLC is diagnosed in combination with adenocarcinoma or squamous cell carcinoma in up to 10% of cases, and recent sequencing efforts suggest that these combined tumors share common precursors.[12,305–307] Lineage plasticity is more likely driven by epigenetic factors than by subclonal genomic alterations, as estimates of intratumoral genomic heterogeneity have been surprisingly low.[215,288] In 3% to 10% of cases, *EGFR*-driven lung adenocarcinomas acquire resistance to targeted therapy through transdifferentiation to SCLC, losing dependence and often expression of the *EGFR*-mutant oncoprotein.[308,312] Ongoing clinical trials are attempting to address lineage plasticity in the context of EGFR-mutant lung cancers transforming into SCLC (NCT03567642; NCT04538378).

Response and Resistance to Therapy

Determinants and Correlates of Platinum Sensitivity

Treatment-naïve SCLC is remarkably sensitive to DNA damaging chemotherapy, with clinical response rates that nearly double those of NSCLC.[127,313] However, these responses are usually transient,

and the emergence of cross-resistance renders second-line agents far less effective.[184] As noted above, chemotherapy-free interval (CTFI) after first-line chemotherapy is a strong predictor of benefit from second-line therapy, with a CTFI of more than 90 days delineating platinum-sensitive from platinum-resistant disease.[184] The factors determining both initial sensitivity and acquired resistance are not sufficiently understood and are major areas of ongoing research.[314,315]

The determinants of differential initial chemosensitivity in SCLC have not been defined, although some correlative associations have been made. Surprisingly, 40% to 44% of SCLC patients were found to have germline pathogenic variants in genes that predispose to cancer, and tumors in these patients were significantly more platinum sensitive than other SCLC.[317] The most common germline variants, present in 5% of patients, were found in the base excision repair gene mutY DNA glycosylase (*MUTYH*). These patients had extensive family histories of cancer including multiple relatives with SCLC.

Targeted proteomic profiling may offer insight into determinants of chemosensitivity. A broad comparison of cell lines revealed that DNA damage response genes are highly expressed in SCLC versus NSCLC or normal tissue, including the checkpoint kinases *ATM*, *ATR*, *DNA PKcs*, *CHK1*, and *WEE1*.[271] Inhibition of these key checkpoints has the potential to enhance platinum sensitivity. Selective inhibitors against several of these kinases have been developed and are now under active clinical investigation in SCLC. The WEE1 inhibitor ZN-c3 is being evaluated in a Phase I/II study that includes SCLC (NCT04158336). The CHK1 inhibitor prexasertib has been evaluated in a phase II parallel cohort study of relapsed SCLC including a platinum-sensitive and platinum-resistance cohort (NCT02735980).[317] Prexasertib demonstrated little activity as a single agent, and future trials will require patient-selection and/or combinatorial strategies. Clinical studies of the ATR inhibitor, berzosertib, are discussed later in this section.

A notable distinguishing feature between SCLC and NSCLC in proteomic analysis was high expression of poly(ADP-ribose) polymerase 1 (*PARP1*), and SCLC cell lines are highly sensitive to PARP inhibitors despite little evidence of defects in intrinsic DNA damage repair.[215,318] PARP inhibition has been assessed in clinical trials in the first-line, maintenance, and relapsed settings. In relapsed SCLC, addition of the PARP inhibitor veliparib to temozolomide (TMZ) substantially increased response rate (39% veliparib + TMZ vs 14% placebo + TMZ),[319] and a similarly high response rate was observed with olaparib + TMZ (42%) in a single-arm study.[320] Veliparib has also been studied in the first-line setting in two randomized phase II studies; both studies were initiated prior to the use of immunotherapy first line. In ECOG2511, the addition of veliparib to cisplatin plus etoposide was associated with an improved PFS.[321] Similarly, in combination with carboplatin plus etoposide, veliparib significantly improved PFS in first-line treatment of ES SCLC but without a corresponding improvement in OS.[322]

SCLC mouse models, cell lines and PDXs each have advantages in studying chemosensitivity and resistance. An advantage of PDX models is that they can largely recapitulate the drug responses of their donor patients, tethering experimental results to clinical outcomes.[288,323–328] Until recently these models were derived almost exclusively before chemotherapy due to scarcity of post-relapse biopsies. This limitation was overcome by the discovery that live circulating tumor cells (CTCs) could be isolated from blood and directly injected in mice to generate PDXs.[325] CTCs have been used to generate large biobanks of relapsed SCLC models as well as serial pre-treatment/post-relapse models from the same patients.[266,288,292] Comparison of PDXs from chemo-naïve and post-relapse patients revealed increased expression of MYC transcriptional targets with resistance, and in a separate cohort, upregulation of genes involved in epithelial-to-mesenchymal transition (EMT).[288,292] Both features echo previous findings in cell lines and mSCLC models.[226,329,330] Post-relapse PDXs demonstrate greater intra-tumoral heterogeneity of gene expression, and direct analysis of CTCs suggests that heterogeneity of gene copy number also increases.[292,299] In serial PDX models, acquired upregulation of soluble guanylate cyclase (sGC) contributed to chemoresistance, nominating sGC as a new target in relapsed SCLC.[331]

Established SCLC cell lines have been interrogated extensively for drivers of chemoresistance. The first candidates were multidrug resistance (MDR) efflux pumps, which are overexpressed or amplified as cell lines adapt to escalating drug concentrations in the laboratory.[332] Recurrent MDR gene amplifications have also been found in relapsed patient samples, but MDR pumps have proven difficult to target without excessive toxicity.[333,334] More recently, cell line screens revealed that Schlafen 11 (*SLFN11*) expression and promoter methylation strongly correlated with sensitivity to DNA damage.[323,335–338] Replication stress causes SLFN11 to block replication forks irreversibly, triggering cell death.[339,340] This effect compounds the toxicity of chemotherapy-induced replication stress, and suppression of SLFN11 confers drug resistance.[340–342] In PDX models serially drugged to platinum-resistance, *SLFN11* expression was recurrently suppressed but could be restored by inhibition of the histone methyltransferase EZH2.[343] EZH2 inhibitors are under active clinical investigation as a strategy to overcome chemoresistance: the EZH1/2 inhibitor, DS3201b, is being studied in a phase I/II study in combination with irinotecan in patients with relapsed SCLC (NCT03879798). Among the resistant PDX models with unchanged *SLFN11* expression, recurrent upregulation occurred in Twist-related protein 1 (*TWIST1*), a key driver of EMT. This corroborated the EMT signature identified in post-relapse PDX models and suggests that lineage plasticity also contributes to chemoresistance.[292]

SCLC patients are contributing directly to the investigation of acquired chemoresistance by undergoing elective biopsies after relapse.[333] These samples have confirmed recurrent amplifications in MDR genes and found deletions in DNA mismatch repair genes *MSH2* and *MSH6*. A central finding from this growing repository was consistent upregulation of WNT signaling, through both genomic alterations and gene expression changes. To validate this feature as a resistance mechanism, the *APC* gene was inactivated in SCLC cell lines and conferred resistance to etoposide. *ASCL1* expression was also diminished significantly, suggesting decreased expression of NE markers. This finding was echoed in pre-treatment biopsies for a phase II trial of topotecan plus the ATR inhibitor berzosertib in patients with relapsed SCLC.[344] This trial was heavily enriched for patients with platinum-resistant disease, yet the combination yielded a promising ORR of 36%. Differential gene expression analysis revealed that responsive tumors had upregulation of DNA repair genes, consistent with the trial hypothesis that ATR inhibition would target replication stress. In contrast, nonresponsive tumors had decreased expression of NE markers, echoing the *ASCL1*-low cohort and suggesting transformation to a NE-low lineage.

Immune Evasion and Resistance to Immune Checkpoint Blockade

The addition of ICIs to platinum-doublet chemotherapy introduced for the first time the possibility of long-term survival for ES-SCLC patients and represented a long-awaited advance in care.[161,163] However, the fraction of patients that benefit from ICIs is much smaller than for NSCLC. This was surprising, as many of the features that make NSCLCs susceptible to ICI are even more pronounced in SCLC, including heavy tobacco exposure, high tumor mutational burden (TMB), and absence of the recurrent oncogenic driver mutations that correlate with resistance.[3,345–348]

Factors implicated in immune evasion in SCLC include suppression of antigen presentation, interferon response deficits, and mechanisms of immune effector exclusion. Early comparisons with NSCLC cell lines demonstrated that SCLC lines were less

immunogenic when injected into immunocompetent mice and expressed very low levels of MHC class I.[349–351] Tumor pathology series prior to the advent of ICIs have consistently demonstrated that lymphocytic infiltrates are minimal in most tumors but if present correlate strongly with early stage and improved prognosis.[352–355] More recently, SCLC tumors with inflammatory gene expression signatures, including antigen presentation genes, and low neuroendocrine expression were found to be enriched for durable responses to ICI.[268–270,355] Absence of MHC class I expression also necessitates evasion of NK cell surveillance. Recent evidence in mSCLC and patient samples suggests this is accomplished through repression of surface ligands for the NK cell activating receptor *NKG2D*, and when re-expressed these ligands delay tumor growth and prevent metastasis.[356,357]

Suppression of both MHC class I and NK cell ligand expression in SCLC appears to be primarily epigenetic. Exogenous interferon can restore MHC class I expression,[358–360] and although interferon treatment did not improve survival in SCLC,[361–363] these trials were conducted before the advent of ICIs and could have been confounded by interferon-stimulated expression of PD-L1.[364] Recent studies demonstrated that inhibitors of polycomb repressor complex 2 (PRC2) components EZH2 or EED could restore MHC-I expression in some SCLC cell lines and tumors models and could work in synergy with interferon.[269,365] As noted above, stimulation of NOTCH signaling in SCLC cells can drive a Non-NE expression pattern, and when this occurs it also de-represses MHC-I expression.[300,366] Finally, induction of RAS signaling in SCLC can drive loss of NE expression, endogenous production of type I interferon, and expression of both MHC class I and PD-L1.[269,367,368] Collectively, these investigations suggest that immune evasion in SCLC depends on a NE gene expression program that represses inflammation and silences antigen presentation via PRC2, requires low NOTCH and requires RAS signaling for lineage stability. A clear mechanistic understanding of this regulatory network may lead to improved immune therapy for SCLC.

Combination Immunotherapy and Strategies to Enhance Immune Response

Combinatorial immunotherapy strategies are under clinical evaluation in every stage of SCLC. In LS SCLC, the phase II ETOP/IFCT 4–12 STIMULI trial randomized patients after chemoradiation and PCI to best supportive care or to nivolumab plus ipilimumab for four cycles followed by nivolumab up to 12 months (Table 31.3).[369] The study closed early due to slow accrual. A total of 222 patients were enrolled, 153 of whom were randomized to ICI versus observation. No difference in PFS was found (median PFS 10.7 months ICI arm vs 14.5 months in the observation arm, HR 1.02, $P = 0.93$). At a median follow-up of 35 months, the OS was not significantly different (median OS ICI arm NR vs 32 months in observation arm, HR 0.95, $P = 0.37$).[369] The median time to discontinuation of ICI therapy was 1.7 mos and toxicities were substantial. Grade ≥3 AEs were reported in 62% of the ICI arm and in 25% in the observation arm. Notably, patients who received twice daily radiation as part of their concurrent chemoradiation appeared to have more benefit from ICI maintenance than those who received once-daily radiation. This observation underscores the need to define the best radiation schedule. Longer term data are pending, but the significant rate of toxicities may outweigh potential benefit.

Ongoing studies evaluating ICI monotherapy or dual checkpoint blockade after completion of or concurrently with chemoradiation include ADRIATIC, ACHILES, and NRG LU005 (Table 31.3). ADRIATIC is a phase III study that randomized 728 patients with inoperable LS-SCLC and stable disease or better after four cycles of concurrent chemoradiation to up to

TABLE 31.3

Randomized Studies of ICI in LS SCLC

COMPLETED						
ICI after Concurrent Chemoradiation						
Trial (Phase)	**N[a]**	**Arms**	**mPFS (mos)**	**HR**	**mOS[b] (mos)**	**HR (P)**
STIMULI (RPh2)[369]	78	Nivo plus Ipi × 4 followed by nivo × 12 months	10.7	1.02 *(P = 0.93)*	NR	0.95 *(P = 0.82)*
	75	**Observation**	14.5		32.1	
ONGOING						
Concurrent Chemoradiation Plus ICI						
Trial (Phase; NCT)	**N**	**Arms**	**Endpoints**			
NRG-LU005 (R-Ph2/3; (NCT03811002)	506	1) EP + Atezo with concurrent radiation followed by Atezo maintenance × 1 year 2) Chemoradiation followed by observation	Primary: OS Secondary: PFS, ORR, local control			
ICI after Chemoradiation						
ADRIATIC (R-Ph3; NCT03703297)	728	1) Durva plus placebo × 4 cycles followed by Durva × 2 years 2) Durva plus Treme × 4 cycles followed by Durva × 2 years 3) Placebo × 4 cycles followed by placebo × 2 years	Primary: PFS and OS for Durva vs placebo arms Secondary: OS and PFS for Durva/Treme arm vs placebo ORR, landmark PFS Completed enrollment Dec. 2021			
ACHILES (R-Ph2; NCT03540420)	212	1) Atezo maintenance × 1 year 2) Observation	Primary: 2-year survival Secondary: PFS, best response			

[a]Of 222 enrolled, 153 were randomized.
[b]Median follow up at the time of publication, 35 mos.

2 years of durvalumab, durvalumab plus tremelimumab for four cycles followed by durvalumab up to 2 years, or placebo. PCI was offered at the investigator's discretion. The co-primary end points are PFS and OS on durvalumab monotherapy versus placebo (NCT03703297). This study completed enrollment in 2021 and results are awaited. ACHILES is a randomized phase II study of best supportive care versus atezolizumab for up to 12 months after completion of chemoradiation. This study is currently enrolling and has a planned enrollment of 212 patients. The primary end point for this study is 2-year survival (NCT03540420). NRG LU005 is a phase II/III trial for LS SCLC of atezolizumab given concurrently with chemoradiation and as maintenance for up to 1 year compared to chemoradiation followed by observation. The primary end point of the phase II portion of this study is PFS and of the phase III portion, OS. The study is current enrolling (NCT03811002).

KEYLNK-013 is a randomized placebo-controlled phase III study evaluating pembrolizumab concurrently with chemoradiation followed by pembrolizumab for 12 months and concurrent chemoradiation/pembrolizumab followed by pembrolizumab plus olaparib. The primary end points are PFS and OS from the time of randomization. The study plans to enroll 672 patients and is currently open (NCT04624204).

In the first-line setting for ES SCLC, multiple studies adding to EP/ICI backbone are under way including additional immunotherapy agents, chemotherapy, or targeted agents. SKYSCRAPER 2 is a randomized phase III study of EP/atezolizumab plus the anti-TIGIT antibody, tiragolumab. The primary end points are investigator-assessed PFS and OS. The study has completed enrollment (NCT04256421). IMforte is an open-label phase III study investigating the benefit of adding lurbinectedin to atezolizumab maintenance after four cycles of EP-atezolizumab for patients with ES SCLC, with co-primary end points of PFS and OS. Planned enrollment is 690 and the study is ongoing (NCT05091567). KEYNOTE-B99 is a randomized phase II study of EP/pembrolizumab followed by pembrolizumab plus either an anti-CD27 antibody (MK-5890), lenvatinib, or an anti-ILT4 antibody (MK-4830). The study has a planned enrollment of 120 patients with primary end points of ORR and 6-month PFS (NCT04924101). SWOG 1929 is a randomized phase II study of maintenance atezolizumab versus atezolizumab in combination with the PARP inhibitor, talazoparib in patients with SLFN11 (+) ES SCLC. The primary end point is PFS. The study is ongoing with a planned enrollment of 94 patients (NCT04334941).

T-cell redirecting therapy is also being explored in SCLC. Tarlatamab (AMG-757) is a bispecific T-cell engager that binds both DLL3 and CD3 that is under phase I evaluation in relapsed SCLC. Initial results have been promising with an acceptable safety profile and clinical responses.[370] The phase I study is ongoing (NCT03319940). Additional planned studies of AMG-757 include a phase II study at the RP2D and combination strategies with ICIs.

Finally, CD47, a cell-surface molecule expressed on normal and malignant cells, acts as a ligand for signal regulatory protein α (SIRPα), a regulatory transmembrane protein expressed on macrophages and dendritic cells. CD47 is highly expressed in human SCLC and may contribute to immune escape mechanisms.[371] Disruption of CD47/SIRPα binding by anti-CD47 antibodies or targeted inactivation of CD47 results in inhibition of SCLC tumor growth in preclinical models.[371] A phase I study of Hu5F9-G4 (magrolimab), an anti-CD47 antibody in advanced solid tumors, reported that it was well tolerated and had clinical activity.[372] Multiple additional studies with magrolimab are underway, though none specific to SCLC at this time.

Survivorship

A small minority of SCLC patients attain long-term survival. Patients are at greatest risk of dying from SCLC during the first 24 months after diagnosis. This risk declines between years 2 and 3 and is further reduced beyond the year 3. Excessive mortality in long-term survivors is also impacted by the development of second primary tumors, mainly NSCLC, and other illnesses associated with cigarette smoking.[373,374] Late relapse can occur in approximately 10% of patients at 5 years.[375] Overall, the relative risk of a second primary tumor in patients who survive beyond 2 years is increased 3.5-fold. The second lung cancer risk is increased 13-fold among those who received chest irradiation.[373] The risk of a second primary tumor increases significantly over time and with continued smoking. The risk of a second lung cancer in patients who continue to smoke was approximately fourfold greater than those who stopped before the diagnosis of SCLC and twofold greater in patients who received chest irradiation. The cumulative risk of a second lung cancer was 32% at 12 years and continued to increase beyond that time point. Patients successfully treated for SCLC constitute a high-risk group for second malignancies and require close surveillance. The cigarette smoking status of survivors should be periodically assessed, and smoking cessation efforts should be instituted for any patient who expresses an interest in quitting.

Long-term survivors are also at increased risk for noncancer-related problems, including long-term sequelae of therapy. In a Danish analysis of patients surviving 5 years or more, there was a sixfold increased risk of death from noncancer causes, particularly cardiovascular and pulmonary diseases.[31] In a French study of patients surviving beyond 30 months, treatment-related sequelae included neurologic impairment in 13% of the patients, pulmonary fibrosis in 18%, and cardiac disorders in 10%.[376] Chest irradiation may accelerate coronary artery disease. Neurologic impairment is a potential long-term toxicity of cranial irradiation and, in some cases, a permanent sequela of a neurologic paraneoplastic syndrome. Tobacco-related illnesses, including heart disease, stroke, and chronic obstructive pulmonary disease, may also impact survival in these patients. Return to work was possible in 40% of survivors and was not influenced by the presence of late treatment-related complications.[376] Physicians who care for survivors of SCLC should aggressively implement strategies to reduce cardiovascular disease.

TYPICAL CARCINOID AND ATYPICAL CARCINOID TUMORS

Incidence and Etiology

Carcinoids are rare, well-differentiated NETs that can originate in multiple primary sites including the gastrointestinal (GI) tract (midgut NETs), pancreas, lungs, ovaries, thyroid, pituitary, and adrenal glands. Lung is the second most common site of carcinoids, accounting for 25% of overall cases. Pulmonary carcinoids comprise about 2% of all primary lung tumors.[377] Over the past 30 years, the age-adjusted incidence of carcinoids has increased, possibly related to increased detection; this trend applies to pulmonary carcinoid as well.[378]

Pulmonary carcinoids are further categorized as typical or atypical, the latter of which are less common (10% to 15% of cases), have a higher likelihood of metastatic spread or relapse after surgery, and an overall poorer prognosis.[379–381] While there is a wide distribution in age of incidence, patients with typical carcinoids are on average approximately 10 years younger than those with atypical carcinoids, which peak in incidence in the sixth decade of life.[382,383] Carcinoids are not clearly attributed to tobacco exposure, like SCLC and LCNEC, although some series suggest that a higher percentage of patients with atypical carcinoid smoke than patients with typical carcinoid tumors.[379,382] Pulmonary carcinoids rarely occur in association with multiple endocrine neoplasia type 1 (MEN1) syndrome. Some sporadic pulmonary carcinoid tumors demonstrate inactivation of the *MEN-1* gene.[384]

Anatomy and Pathology

The majority (75%) of pulmonary carcinoids are central (endobronchial or peribronchial), whereas 25% are peripheral. The histologic appearance of typical carcinoids and atypical carcinoids (Fig. 31.1) is similar, characterized by a uniform population of monotonous, bland tumor cells arranged in organoid nests, trabecular and rosettes. The tumor cells have finely granular (so-called "salt and pepper") nuclear chromatin and a moderate amount of cytoplasm. There are a wide variety of histologic patterns in these tumors, including spindle cell, oncocytic, glandular, follicular, clear cell, and melanocytic. Stromal ossification can occur as well.[8,9]

By thoracic WHO criteria, the distinction between typical and atypical carcinoids is based on mitotic counts and the presence of punctate necrosis, with mitotic counts of ≥2 per 2 mm^2 and/or the presence of necrosis distinguishing atypical from typical carcinoids.[7] Notably, the WHO criteria for typical versus atypical carcinoids are applicable only to surgically resected specimens given that accurate mitotic count cannot be performed in small biopsy samples.[7]

Although Ki67 assessment is not part of the WHO 2021 criteria for distinguishing typical and atypical carcinoids, its role as an independent predictor of prognosis in carcinoids is becoming increasingly recognized.[9,385] In WHO 2021 guidelines, reporting of Ki67 has been recommended as "desirable."[7] The usual range of Ki67 in typical carcinoids is <1% to 5% and in atypical carcinoids – 5% to 20% to 30%.[7,9] Ki67 rates of >5% in typical carcinoids and >10% overall are associated with increased risk of postsurgical recurrence.[9,386,387] In addition, Ki67 is useful in distinguish carcinoids from SCLC in bronchoscopic biopsies, where crush artifact is common, and can cause carcinoids to mimic the morphologic appearance of SCLC. In this setting, assessment of mitotic figures can be challenging, and IHC for Ki67 provides critical diagnostic information as it allows a separation of carcinoids (usually <20%) and SCLC (>50%).[388,389]

Recent studies have identified a hyper-proliferative variant of carcinoid, with Ki67 proliferation rates exceeding 20% and/or mitotic counts exceeding 10 per 2 mm^2.[390] Such variants are rare in primary carcinoid tumors but are relatively common at metastatic sites, with proliferative escalation documented in >30% of matched primary and metastatic samples.[391,392] Similar proliferative escalation in metastases has been documented in enteropancreatic neuroendocrine tumors.[393] Pathologic criteria and nomenclature for such tumors are still being defined.[9]

Carcinoid tumorlets are morphologically identical to carcinoids but measure <0.5 cm in size. Isolated tumorlets usually are incidental pathologic findings without clear clinical significance. Diffuse idiopathic pulmonary neuroendocrine cell hyperplasia (DIPNECH) is a relatively common but under-recognized condition defined as lung involvement by multiple carcinoid tumorlets (with or without carcinoids) and widespread proliferation of neuroendocrine cells within the airways.[7,9] This occurs almost exclusively in women. Patients with extensive involvement may have severe respiratory symptoms. Usually, carcinoids that arise in DIPNECH are typical rather than atypical. DIPNECH has a nearly pathognomonic appearance on high-resolution CT scans, manifesting as air trapping and thickened airways in addition to numerous small nodules.[394,395] Notably, radiologically, DIPNECH may mimic the presentation of bilateral miliary metastases, such as in patients on surveillance for known malignancies.[394,395]

Genetics

Large-scale sequencing of pulmonary carcinoid tumors was first reported in 2014 including copy number analysis, genome/exome sequencing, and transcriptome sequencing of 69 samples.[396] The most prominent alterations occurred in chromatin-remodeling genes, including histone modifiers and subunits of the SWI/SNF complex that were mutated in 40% and 22% of cases, respectively. *TP53* and *RB1* mutations were rare, suggesting that pulmonary carcinoids arise through mechanisms distinctly different than SCLC.[396] High-coverage targeted or whole exome sequencing of 148 lung NETs, including 53 typical carcinoid, 35 atypical carcinoid, 27 LCNEC, and 33 SCLC samples, also demonstrated a high prevalence of mutations in chromatin-remodeling gene (45.5% of carcinoids and 55% of LCNEC/SCLC). Carcinoids had approximately 10% rate of mutation in *TP53*, *RB1*, and *ATM* compared with 70% in carcinomas. Mutations in PI3K/AKT/mTOR pathway were rare (2.3%) in carcinoid tumors. *RB1* loss detected by CNA was observed in 24.5% of typical carcinoid, 11.4% of atypical carcinoid, and 70% of SCLC. MEN1 alterations were almost exclusively seen in carcinoid tumors, and CNA prevalence was significantly lower in carcinoids.

More recently, an integrative analysis of transcriptome and methylome data was reported on 257 pulmonary neuroendocrine neoplasms, including pulmonary carcinoids (n = 119), LCNEC (n = 75), and SCLC (n = 66).[397] This study identified three clusters with 10-year OS ranging from 88% (Cluster A1), 83% (Cluster A2), and 60% (Cluster B). Cluster A1 was noted to express ASCL1 and DLL2 and had a high dendritic cell infiltrate. Cluster A2 harbored recurrent mutations of EIF1AX and downregulation of SLIT1 and ROBO1 genes, which could in turn sensitize to CXCR4 inhibition. Cluster B appeared to have a more suppressive immune context and was characterized by mutations in MEN1. A separate supracarcinoid group had carcinoid morphology but other features resembling LCNEC, including high neutrophil infiltration and expression of immune checkpoint receptors and ligands.[397] Application of this classification prospectively on patients undergoing therapy, particularly on trial, better define subsets of patient most likely to response to therapeutics.

Screening

Screening for pulmonary carcinoids is not recommended. They represent a very small percentage of all lung malignancies and have an indolent natural history. In one study, atypical carcinoid was diagnosed in only 2 of 31,567 (0.006%) asymptomatic patients undergoing baseline lung cancer screening CT scans.[398]

Diagnosis, Presentation, and Imaging

Two-thirds of pulmonary carcinoids develop in the major bronchi. As a result, the most common presenting symptoms include obstructive pneumonia, atelectasis, dyspnea, and cough.[379] Hemoptysis may occur in 10% to 20%. Up to 30% of patients with pulmonary carcinoid tumors are asymptomatic at presentation. Carcinoid syndrome, manifested as facial flushing, diarrhea, and wheezing due to the production of vasoactive peptides, is uncommon in pulmonary carcinoid, presenting in approximately 8% of cases.[399] Cushing syndrome, due to ectopic corticotropin production, is rare (1% to 2% of carcinoid patients).[400] Carcinoid tumors are the most common known cause of extrapituitary secretion of GHRH, which can lead to acromegaly.[401]

Initial workup proceeds in a similar fashion as with other lung tumors, but once a diagnosis of carcinoid is confirmed, more specific radiologic and serologic evaluations for these NETs is recommended. Biopsy of central tumors can be easily obtained by bronchoscopy. Biopsy of peripheral lesions by fine-needle aspiration can be performed, but a definitive diagnosis may be difficult to ascertain in cytology samples.

On CT scan, approximately 5% to 20% of typical carcinoid tumors are associated with hilar or mediastinal lymphadenopathy; in many cases, however, this is attributable to local inflammation rather than metastasis. The positive predictive value of CT in assessing hilar and mediastinal lymph nodal involvement in carcinoid has been reported to be as low as 20% to 45%.[402] The majority

of pulmonary carcinoids express somatostatin receptors (SSR) and can be imaged using radiolabeled somatostatin analogs. Early studies demonstrated that indium-111 (^{111}In)-DTPA-D-Phe-1-octreotide (used in octreoscans) localized 86% of carcinoid tumors.[403] This imaging is typically fused with single-photon emission CT (SPECT) to improve anatomic localization of the radiotracer. PET-based SSR imaging using several radionuclides (^{68}Ga-DOTATATE, ^{68}Ga-DOTATOC or ^{64}Cu-DOTATATE) has FDA approval for routine use since 2016.[404] (Fig. 31.4) Comparison studies between ^{68}Ga DOTATATE-PET and octreoscan have demonstrated superiority of ^{68}Ga DOTATATE-PET in detecting NETs, particularly in liver and bone lesions.[405,406] At this time SSR-PET/CT or SSR-PET/MRI are routinely used in the staging of NETs.

Pulmonary carcinoids occasionally secrete bioactive products such as chromogranin A or 5-hydroxytryptophan. Carcinoid syndrome, due to release of vasoactive substances (serotonin and bioactive amines) is uncommon in pulmonary carcinoids compared with midgut (small bowel) NETs. In SEER analysis of 3,000 patients with pulmonary carcinoid tumors, 8% presented with carcinoid syndrome.[399] In the setting of advanced or metastatic disease, chromogranin A levels can follow disease activity, but this has not been a consistent finding. The overall utility of following chromogranin A levels is limited and is not generally recommended as part of diagnostic work-up or post-treatment surveillance per practice guidelines in Europe and the US.

Staging and Prognosis

Pulmonary carcinoids are staged according to the AJCC TNM classification used for other lung cancers. Applying this staging system to cases in the NCI SEER registry and cases submitted to the IASLC database, it was determined that 5-year OS for patients with stage I was 93%, stage II was 74% to 85%, stage III was 67% to 75%, and stage IV was 57%.[409] Survival is significantly better for typical than atypical carcinoid. Five- and 10-year survival rates have been reported at 87% and 87%, respectively, for typical carcinoid and 56% and 35%, respectively, for atypical carcinoid.[407] A recently published SEER analysis of 124,969 patient with lung NENs from 1988 to 2015 included 8,146 with typical carcinoid and 656 with atypical carcinoid.[408] The average 5-year survival rate in this analysis was 84% for patients with typical carcinoid and 64% for those with atypical carcinoid.

Negative predictors of survival include advanced stage, larger tumor size, higher mitotic rate (i.e., atypical subtype), and age older than 60 years.[409] A nomogram to predict 3, 5, and 10 year survival based on age, grade, histologic type, N and M stage, tumor size, and treatment as variables has been proposed.[410]

Management by Stage

Stages I, II, and III

Surgery is the primary treatment modality and the only curative option for patients with pulmonary carcinoids. Most patients undergo lobectomy; however, in cases with proximal tumors pneumonectomy or bilobectomy may be required.[383,411] For patients with favorable prognostic features, such as typical histology and absence of lymph node involvement, more limited resections have been proposed.[412,413] Atypical carcinoids should be resected using the same principles guiding surgery for NSCLC including complete mediastinal lymph node dissection.[414,415] Multiple series report decreased incidence of local recurrence and improved survival when complete mediastinal lymph node dissection is performed.[381,416] Stereotactic body radiation is a rational alternative for patients with small peripheral carcinoid tumors who are not candidates for surgery, although there are no prospective trials that verify benefit.

Adjuvant chemotherapy after surgical resection is not recommended for patients with stage I to III typical carcinoid or stage I to II atypical carcinoid as the risk of recurrence is low and the impact of adjuvant chemotherapy is unknown.[415] In a series of 291 resected typical carcinoids, only 3% of patients developed recurrence.[383,417] Systemic recurrence occurs more frequently in patients with atypical carcinoid with mediastinal lymph node involvement (stage III); however, there is no consensus on the use of adjuvant platinum after margin-negative surgical resection and lymph node dissection. Most guidelines suggest consideration of adjuvant chemotherapy and individualized care. The use of adjuvant radiation therapy for nodal disease is also of undefined utility but is generally recommended for R1 resection of atypical carcinoid.[415]

There are no category 1 recommendations for patients with unresectable locally advanced atypical carcinoid, though definitive chemoradiation is a consideration particularly if the tumor harbor aggressive features, such as a high Ki-67 index. For patients with unresectable typical carcinoid, radiation is a consideration for those with symptomatic masses.

Stage IV Carcinoid

Data regarding the efficacy of systemic therapies specifically in pulmonary carcinoid are generally lacking. Pulmonary carcinoid is rarely studied independently and has been omitted from some carcinoid trials. The overall management approach for patients with metastatic pulmonary carcinoid includes local therapies for

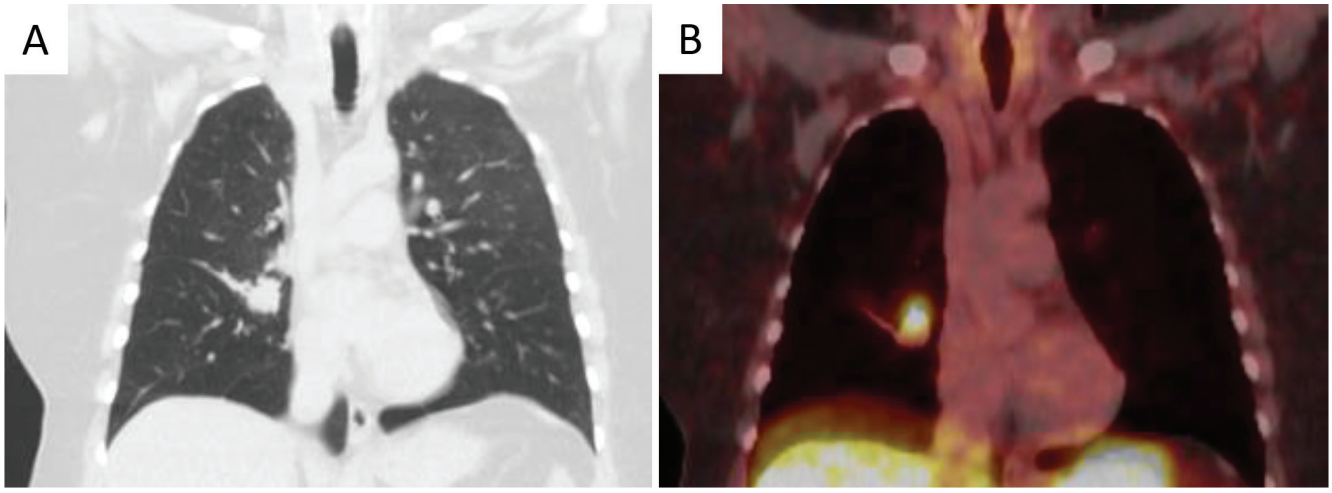


Figure 31.4 Imaging for carcinoid tumors. Carcinoid tumors can be imaged by computed tomography scans (**A**) or somatostatin receptor–based imaging (**B**) such as Gallium-68 (^{68}Ga) DOTATATE positron emission tomography.

oligometastatic disease, including liver-directed therapies, and systemic therapies for widely metastatic disease. Various chemotherapeutic agents have been studied in limited series and are specified below. No therapies have been shown to consistently induce regression. An option for asymptomatic patients with low tumor burden, advanced pulmonary carcinoid is close surveillance with routine imaging and clinical assessment. Systemic therapy can be initiated at the time of symptomatic progression.

Somatostatin Receptor–Targeted Therapies

Carcinoid tumors typically express the somatostatin receptor, and somatostatin analogues (SSAs) have been shown to inhibit growth and prolong time to disease progression. The PROMID study demonstrated that long acting–release octreotide acetate (somatostatin LAR) significantly prolonged time to tumor progression compared with placebo in patients with well-differentiated midgut NETs (14.3 vs 6 months) but did not improve OS.[418] Similarly, the somatostatin analogue, lanreotide autogel/depot, was studied in a randomized placebo controlled trial in patients with metastatic enteropancreatic NETs and was found to significantly prolong progression free survival.[419] Based on these studies, long-acting SSAs are routinely used for the initial management of patients with metastatic typical or atypical carcinoid.

More recently, the SPINET study comparing lanreotide depot versus placebo was initiated for patients with lung carcinoid (NCT02683941). The study was terminated early due to poor recruitment but analyses from the 77 patients randomized (2:1) have been presented. The median PFS was 16.6 versus 13.6 months in the experimental versus placebo arms, respectively (HR0.9 [046 to 1.88]) and ORR was 14 versus 0%.[420] The conclusion of this study was that in SSR(+) patients, lanreotide was an appropriate treatment option. A likely contributor to the poor enrollment of this study was that many practitioners were already treating lung carcinoid patients with long-acting SSA based on prior randomized studies.

Since the early 1990s, early-phase trials showed promising results with peptide receptor radionuclide therapy (PRRT) using radiolabeled somatostatin analogs such as ^{111}In-DTPA-octreotide, yttrium-90 (^{90}Y)-DOTATOC, and lutetium-177 (^{177}Lu)-Dotatate.[421,422] The phase III NETTER-1 trial randomized 229 patients with advanced somatostatin receptor–positive GI NETs to ^{177}Lu-Dotatate plus 30 mg octreotide LAR or high-dose (60 mg) octreotide LAR.[423] The study met its primary end point of PFS benefit. Treatment with ^{177}Lu-Dotatate was associated with statistically significant reduction in the risk of disease progression or death versus high-dose octreotide.[423] In January 2018, ^{177}Lu-Dotatate was approved for patients with somatostatin receptor–positive gastroenteropancreatic NETs. Smaller, nonrandomized studies have reported activity of ^{177}Lu-Dotatate in bronchial NETs.[424] Of 34 lung carcinoid patients treated with ^{177}Lu-Dotatate, the DCR was 80%, CR 6%, PR 27%, and 47% with SD. The median PFS in this group was 20.1 months. While the FDA approval for ^{177}Lu-Dotatate is specific to patients with SSR(+) gastroenteropancreatic NETs,-^{177}Lu-Dotatate appears to have similar activity in patients with bronchial carcinoids.

Mammalian Target of Rapamycin-Directed Therapy

The mTOR pathway, which regulates cell growth, proliferation, and metabolism, has been implicated in the pathogenesis of NETs. A randomized, placebo-controlled phase III study of everolimus plus octreotide LAR (RADIANT-2) included 429 patients with low-grade or intermediate-grade carcinoids, of which 44 were of pulmonary histology.[425] Median PFS was 16.4 months (95% CI, 13.7 to 21.2 months) for the group receiving the combination compared to 11.3 months (95% CI, 8.4 to 14.6 months) in the octreotide LAR–only group (HR, 0.77; 95% CI, 0.59 to 1.0; $P = .026$). For the 44 pulmonary carcinoid patients, median PFS was 13.6 and 5.6 months in the combination versus the octreotide LAR-only group, respectively.

The RADIANT-4 study was an international double-blind placebo-controlled phase III study that randomized 302 patients 2:1 with nonfunctional lung or GI NETs to receive everolimus or placebo.[426] The median PFS was 11.0 months in the everolimus arm and 3.9 months in the placebo arm (HR for progression or death was 0.48 favoring everolimus). Grade 3/4 drug-related toxicities included stomatitis, infections, diarrhea, and anemia, all of which occurred in <10% of patients.

A post hoc analysis of the 90 patients with pulmonary NETs who had enrolled on RADIANT-4 was reported. The median PFS in this cohort was 9.2 versus 3.6 months and tumor shrinkage of 58% versus 13% in patients who received everolimus (n = 63) and placebo (n = 27), respectively. Grade 3/4 stomatitis, hyperglycemia, and infections were more common in patients who received everolimus.[427] Based on this subgroup analysis, everolimus received FDA approval for use in patients with lung NETs in 2016.

Chemotherapy

Typical and atypical carcinoid are less chemosensitive than SCLC; however, for lack of alternatives, regimens typically used for SCLC often are recommended for advanced and symptomatic disease per practice guidelines in Europe and the US. In several small series that included patients with typical and atypical carcinoid, treated with chemotherapy (mostly etoposide and cisplatin), the ORR was approximately 20% to 30%.[428–431]

Newer agents are being studied in bronchial carcinoids. Temozolomide has been evaluated either alone or in combination with other agents including capecitabine. In a retrospective study of 36 patients, of which 13 had bronchial carcinoid, treatment with temozolomide monotherapy was associated with a 31% PR rate and a 67% rate of clinical benefit.[432] Another study that included 31 patients with typical or atypical pulmonary carcinoids treated with temozolomide monotherapy, PR and SD were reported in 3 (14%) and 11 (52%) patients, respectively; median PFS and OS were 5.3 and 23.2 months, respectively.[433]

Temozolomide has been safely combined with capecitabine (CAP-TEM) with high ORRs in carcinoids. In one study, a total of 18 patients with pulmonary carcinoid and pancreatic NETs metastatic to the liver, the CAP-TEM regimen had a reported ORR of 61%, and 3 of 4 pulmonary carcinoid patients attained clinical benefit (complete response, 1; partial response, 1; SD, 1).[434] A meta-analysis of 15 studies included 384 patients with well-differentiated (G1/G2) and poorly differentiated (G3) NENs treated with CAPTEM. Of studies that included only well-differentiated NETs, the DCR with CAP-TEM was 91.07%.[435] Together these results suggest capecitabine plus temozolomide is active and may prolong survival in this malignancy.

Immunotherapy

The PD-1 inhibitor, spartalizumab, was evaluated in a phase II trial that enrolled 30 patients with well-differentiated thoracic carcinoid (28 lung and 2 thymic). The ORR of the 20 thoracic NETs was 20% and SD was achieved in 53%.[436] PD-L1 expression was 19% in thoracic carcinoid tumors. This ORR for spartalizumab was higher than that published for advanced NETS in the multicohort KEYNOTE-028 and KEYNOTE-158 studies of pembrolizumab. While lung carcinoid–specific data were not available, the ORR to pembrolizumab was 12% in KEYNOTE-028 carcinoid cohort in which 9 of the 25 patients enrolled had lung primary.[437] In KEYNOTE-158, 14 of 107 had lung primary NETs, and the ORR was 3.7%.[438]

Patients with pulmonary carcinoid have been included in two basket studies of nivolumab plus ipilimumab with disparate results but very small numbers.[439,440] The DART-SWOG 1609 was a prospective phase II trial evaluating nivolumab plus ipilimumab in patient with rare tumors and included 32 patients with nonpancreatic NETs. Of the three patients with lung carcinoid there were

no responses.[439] CA209-568, a phase II study of nivolumab plus ipilimumab in rare tumors given at the same dose and schedule as the DART study. In contrast with the DART study experience, of nine patients with atypical carcinoid, three (33%) had objective responses (two CRs and one PR) and four had SD. The three responders had duration of response of >20 months at the time of study publication.[440] Consistent with prior reports of combined ICI therapy, grade ≥3 immune-related AEs were reported in 34% of study patients. In a phase II single-institution study of nivolumab plus ipilimumab in advanced well-differentiated NETs, three of six patients with lung carcinoid (50%) had responses, including one CR.[441]

Summary

Patients with resectable stage I–III disease pulmonary carcinoid tumors should be referred for surgical management. Patients who are not surgical candidates should be considered for SABR (especially for early-stage peripheral tumors). No adjuvant therapy has shown benefit after R0 resection; however, patients with N2 involvement and whose tumors exhibit aggressive features may be considered for adjuvant chemotherapy. Patients with R1 resection should be referred to discuss adjuvant radiation with a radiation oncologist. After primary treatment, the recommended interval for surveillance is CT imaging and physical visits every 12 to 24 months.

In the approach to metastatic bronchial carcinoids, SSA are used as initial management for patient with low tumor burden in the setting of a positive somatostatin receptor scan. For patients who progress on octreotide LAR, treatment with ^{177}Lu-Dotatate may be considered, if available; everolimus is also a consideration for these patients. Chemotherapy is recommended for those with more rapidly progressing and symptomatic tumors or those who have progressed on less toxic treatments. For patients with liver-dominant disease, resection or liver-directed therapies including transarterial chemoembolization or radioembolization are often utilized. Additional prospective, randomized studies are needed for traditional cytotoxic, molecularly targeted agents, and immunotherapy in this disease. Palliative radiation therapy can be used for symptomatic lesions.

LARGE CELL NEUROENDOCRINE CARCINOMA

Incidence and Etiology

LCNEC of the lung was first described in 1991 as a form of high-grade NSCLC with neuroendocrine features.[442] LCNEC accounts for about 3% of surgically resected lung cancers.[443] In previously reported series, LCNEC patients have a median age of 62 years (range, 33 to 87 years), and a majority of patients are male current or former cigarette smokers.

Anatomy and Pathology

According to thoracic WHO classification, LCNECs are diagnosed based on the following criteria: (1) neuroendocrine morphology with organoid nesting, palisading, or rosette-like structures (Fig. 31.1); (2) mitotic rate of >10 per 2 mm^2 (average, 60 to 80 mitoses per 2 mm^2); (3) non–small-cell cytologic features, including large cell size, low nuclear or cytoplasmic ratio, nucleoli, or vesicular chromatin; and (4) neuroendocrine differentiation by IHC or electron microscopy, although the latter is rarely used in current practice.[7,444] Expression of at least one of the three standard neuroendocrine markers (synaptophysin, chromogranin A, CD56) is required for the diagnosis of LCNEC; however, there is no minimal requirement for the extent of neuroendocrine marker expression, provided that neuroendocrine morphology is convincing. The role of the more recently identified neuroendocrine markers (INSM1, ASCL1) in the diagnosis of LCNEC has not been standardized. The distinction of LCNEC from SCLC requires consideration of multiple histologic features, such as cell size, nucleoli, chromatin pattern, and nuclear to cytoplasmic ratio, rather than a single criterion. The distinction of LCNEC from SCLC can be difficult in 5% or more of cases, even for expert thoracic pathologists.[445] Similarly, LCNEC may have morphologic features resembling adenocarcinoma and nonkeratinizing squamous cell carcinoma.[444] Combined LCNECs are defined as tumor with both LCNEC components and other NSCLC histologic components, including adenocarcinoma or squamous cell carcinoma. Adenocarcinoma is the histologic type found most often in combined LCNEC.[446] Ki67 proliferation index is high in LCNEC (generally >50%). However, the mean Ki67 rate of LCNEC is generally lower than that of SCLC, and some LCNEC may have Ki67 index below 50%,[9,444] especially those showing NSCLC-type genomic profiles[447] or combined histology.[446] TTF1 is positive in 50% to 60% of LCNECs.[448]

An important consideration in the diagnosis of LCNEC is that some conventional lung adenocarcinomas, squamous cell carcinomas, and large-cell carcinomas can express neuroendocrine markers in the absence of neuroendocrine features at morphologic level (~15% of NSCLC overall). Usually this occurs as labeling for a single marker in focal distribution; however, more extensive expression can occasionally occur.[444,449] The significance of this phenomenon remains unclear, and the diagnosis of LCNEC is reserved for tumors with overt neuroendocrine morphology.[7] Nevertheless, it is likely that there exists a spectrum among NSCLCs in which neuroendocrine differentiation manifests only at gene expression level to those with full neuroendocrine morphology. This concept is supported by the observation that a subset of LCNEC expresses low levels of exocrine marker Napsin A[450] and harbors genomic alterations typical of adenocarcinoma (*STK11*, *KRAS*).[447,451] Conversely, another major subset of LCNEC identified by genomic studies suggests a relationship with SCLC.[447,451]

The diagnosis of LCNEC in small biopsies and cytology samples has historically been difficult because the characteristic neuroendocrine morphologic pattern may be difficult to demonstrate in minute pieces of tissue. However, in recent years, larger volumes of tissue are characteristically obtained in lung biopsies to enable predictive molecular testing, which is making the diagnosis of LCNEC more feasible.[7,445] Specific criteria in biopsies for LCNEC based on the combination of morphologic features and IHC have been proposed.[452]

Molecular Characterization

Genetic studies of LCNEC have provided more insight into the underlying biology of this disease. Targeted NGS of 241 cancer-related genes of 45 match tumor/normal pairs revealed that *TP53* (78%), *RB1*(38%), *STK11*, *KEAP1*, and *KRAS* were commonly mutated.[447] Further, the 45 tumors segregated into three groups: an SCLC-like group characterized by either co-alteration of *TP53*/*RB1* or *MYCL* amplification; an NSCLC-like group that harbored mutations in genes such as *STK11*, *KRAS*, or *KEAP*; and a carcinoid-like group with *MEN1* mutations and low mutation burden.

Targeted capture sequencing of all coding exons of 244 cancer-related genes was performed in 78 LCNEC (including 10 mixed LCNEC/adenocarcinoma) samples.[453] Similar to prior reports, a high prevalence of inactivating mutations in *TP53* (73%) and *RB1* (26%) were observed. Protein loss of Rb was found in 74% of cases. *MYCL1* amplification was observed more commonly in advanced stage cases versus early-stage tumors (23% versus 8%). Alterations in PI3K/AKT/mTOR were also observed in 15% of tumors. Activating mutations in *KRAS*, *FGFR1*, *KIT*, *HER2*, *HRAS*, or *EGFR* occurred in <5% of cases.

Using clinical data from the Netherlands Cancer Registry and Pathology Registry, survival outcomes for LCNEC patients treated with either a NSCLC regimen (gemcitabine/taxane [GEM/TAX]) or an SCLC regimen (EP) were correlated with Rb status. Overall *RB1* mutation or protein loss was found in 47% and 72% of cases. In patients whose tumors maintained wild-type Rb, survival was superior with GEM/TAX relative to EP (OS 9.6 vs 5.8 month). In context of *RB1* mutation or loss, no difference in OS was observed between the treatment regimens.[454]

Integrative analysis of genomic (n = 60) and transcriptomic profiling (n = 69) of 75 LCNEC samples identified two genomic subgroups: Type 1 (37%) that harbor biallelic *TP53* and *STK11/KEAP1* alterations and Type 2 (42%) that harbor biallelic inactivation of *TP53* and *RB1*).[451] The transcriptomic profiles, when compared with data from SCLC, pulmonary carcinoid, and NSCLC histologic subtypes (adenocarcinoma and squamous cell carcinoma) revealed that LCNEC expression clustered with that of SCLC. Type 1 LCNEC were ASCL1 high, DLL3 high, and NOTCH low, like SCLC despite having a genomic profile more like NSCLC. Conversely, the type II subgroup was ASCL1-low, DLL3-low, and NOTCH-high, despite a genomic profile more like SCLC. Taken together, these data further define LCNEC as a distinct entity from both SCLC and other histologies of NSCLC.

Diagnosis and Presentation

LCNECs generally present as peripheral tumors; thus cough, hemoptysis and postobstructive pneumonia are infrequent symptoms at presentation.[455] On CT scan these tumors are often well defined and lobulated, without air bronchograms or calcifications; spiculated margins are uncommonly observed.[456,457] Inhomogeneous enhancement is found in larger diameter (33 mm or larger) tumors secondary to necrosis. LCNEC typically have high FDG uptake on PET scans, which is helpful in locating extrathoracic metastases.[457] A definitive diagnosis may be difficult to make in small specimens or by cytology, as indicated previously; thus, a surgical biopsy is often needed in LCNEC.

Staging

LCNEC is incorporated in the TNM classification used for all NSCLC. Stage for stage, survival for LCNEC is significantly worse than for non-LCNEC NSCLC.[382] In a recent report of 195,148 lung cancer cases identified in the SEER database from 2010 to 2015, 1,681 were LCNEC (0.9%).[458] LCNEC was diagnosed more commonly in men. With respect to stage, 18.7% had stage I disease, 8.9% stage II, 17.7% stage III, and 54.6% stage IV at diagnosis. The most common sites of metastases were liver and brain. The 2 year and mOS were: stage I-II 63% 2-year survival (mOS 42 months), Stage III disease 36% 2-year survival (mOS 14 months), and 8% 2-year survival (mOS 5 months).[458] Several groups have reported that the survival curve of LCNEC can be superimposed on that of SCLC with minimal differences in survival, stage for stage, between the two.[459,460]

Initial Management by Stage

Patients with LCNEC are generally managed similarly to those with other histologies of NSCLC, stage for stage. A primary controversy in LCNEC centers on the choice of chemotherapy, with both typical SCLC and NSCLC regimens being employed. Published reports on chemotherapy regimens for LCNEC are of limited utility as essentially all are small, retrospective series.

Stages I, II, and III

For patients with early-stage LCNEC, resection is recommended. The modalities of choice are either lobectomy or pneumonectomy, with systematic nodal dissection.[461] Given the aggressive nature of LCNEC, surgery alone is not generally considered sufficient for its treatment. A study of surgical outcomes for patients with resected LCNEC identified 6,092 patients with stage IA-IIIA LCNEC in the NCDB from 2004 to 2015.[462] The majority of patients underwent surgical resection, and complete resection was achieved in >85% of cases. The 5-year survival for patients with resected stage I disease was 50%, stage II 45%, stage IIIA 36%, and N2 disease 32%. Outcomes by stage were better when compared with available data using SBRT (stage I patients) or chemoradiation (stage II-IIIA and N2).[462]

Several small retrospective studies have attempted to discern the role of neoadjuvant or adjuvant chemotherapy in LCNEC, using SCLC-based regimens. A retrospective review of 100 patients with Stage I–III resected LCNEC evaluated the use of perioperative chemotherapy reported no significant association between OS and receipt of neoadjuvant or adjuvant platinum-based regimens. When this analysis was restricted to patients with R0 resection of stage IB–IIIA disease, receipt of platinum-based chemotherapy was marginally associated with improved mOS.[463] A retrospective analysis of 144 surgical LCNEC cases reported that perioperative chemotherapy led to a statistically significant survival benefit across all stages compared to surgery alone ($P = .04$).[459]

Although data are limited, SCLC-based chemotherapy regimens are typically recommended in the adjuvant treatment of LCNEC.[464,465] Interpretation of these reports is limited by the lack of randomized prospective data and small sample sizes. Given the absence of randomized controlled studies in this context, adjuvant platinum-based chemotherapy following either a SCLC or NSCLC standard of care is acceptable.

Data regarding use of radiation therapy in LCNEC are sparse. For patients who have unresectable locally advanced disease, guidelines recommend definitive chemoradiation. In retrospective analyses, chemoradiation provides better ORR and outcomes than chemotherapy alone. A recent analysis included 5,797 patients identified in the NCDB; 54% received chemoradiation and 46% received chemotherapy alone.[466] The mOS was 11.9 months for patients who received chemotherapy alone compared with 16.1 months for those who received chemoradiation. In the absence of study data specific to LCNEC, following the current NSCLC paradigm of chemoradiotherapy followed by durvalumab maintenance is a reasonable option.

Stage IV

The optimal chemotherapeutic regimen in relapsed or stage IV LCNEC is similarly not defined, although treatment with platinum-based regimens is advocated. Several studies have suggested that the response rate of LCNEC to cisplatin-based chemotherapy is comparable or lower than that of SCLC. A retrospective series of 20 patients with stage III and IV LCNEC treated with platinum-based therapy showed a response rate of 50% (1 complete response, 9 partial responses).[467] Response rates for chemotherapy-naïve patients was better (64%) than those who were previously treated (17%). In patients with metastatic disease, SCLC-based chemotherapy (platinum doublet) appeared to be more effective than traditional NSCLC regimens (median survival of 51 versus 21 months, respectively; $P < .001$). In a retrospective review of 45 patients with pathologically confirmed LCNEC receiving first-line chemotherapy, response rates and outcomes were compared between subjects who received SCLC (n = 11) and NSCLC (n = 34) regimens.[468] Those treated with SCLC regimens had nonsignificantly higher response rates, median PFS, and OS compared to patients in the NSCLC chemotherapy group (73% vs 50%, $P = .19$; 6.1 vs 4.9 months, respectively, $P = .41$; 16.5 vs 9.2 months, respectively, $P = .10$). These studies preceded the current standard of ICI plus platinum-doublet for patients with ES SCLC, and many recent studies have not provided specific data on outcomes of patients with LCNEC who may have enrolled.

Taken together, the treatment and outcomes of LCNEC have been derived from small studies and suggested that these cancers should be treated similarly to SCLC, although the response rates may be lower, and outcomes are generally inferior. The impact of adding ICI to EP in LCNEC is unknown and the use of chemoimmunotherapy may be practice specific. Larger, randomized prospective studies are needed to determine the optimal treatment regimen for this disease.

Treatment of Relapsed Large Cell Neuroendocrine Carcinoma

There are no specific recommendations for relapsed LCNEC. Practitioners can reasonably choose to follow treatment recommendations for SCLC or NSCLC. SCLC studies, however, generally have not included patients with LCNEC, thus the ability to extrapolate findings in SCLC to LCNEC is unknown. Two of the therapeutic approaches that have shown promise in SCLC, namely targeting immune checkpoints and targeting DLL3, have been studied to a limited extent in LCNEC.

Cases of LCNEC response to PD-1 therapy have been reported. In a retrospective study of 10 patients with advanced LCNEC who received PD-1 inhibitor therapy at relapse, 6 had a partial response, 1 patient had SD, and the median PFS was 57 weeks.[469] These observations need to be confirmed in larger series but were encouraging. Patients with LCNEC were also included in a basket study of combined ICI.[439] The DART-SWOG 1609 was a prospective phase II trial evaluating nivolumab plus ipilimumab in patient with rare tumors and included a cohort of 32 patients with nonpancreatic NETs. Of these, 18 patients had high-grade neuroendocrine disease, including 9 with LCNEC. The combination of nivolumab plus ipilimumab demonstrated a 44% ORR in patients with high-grade neuroendocrine carcinoma including LCNEC.

DLL3 is expressed in approximately 80% of LCNEC cases, but the number of patients included on the phase I study of the DLL3-targeted antibody drug conjugate, Rova-T, was too small for meaningful separate analysis.[282] A basket study of Rova-T in DLL3-expressing solid tumors included a cohort of patients with NET and neuroendocrine carcinoma.[470] For the primary analysis, patients with NET/NEC were pooled and the ORR in DLL3-high (≥50%) tumors was 17.1% (compared with 8.8% in DLL3-low tumors). Of the 12 LCNEC patients 2 had PR, 9 had SD, and 1 had PD as their best change in tumor size. Clinical development of Rova-T has been terminated; however, other DLL3-based therapies may be considered on a research basis given the prevalence of high DLL3 expression in LCNEC.

Summary

Patients with resectable LCNEC disease should be referred for surgical resection and mediastinal lymph node dissection followed by adjuvant chemotherapy. Patients who are not surgical candidates or with unresectable locally advanced disease should receive definitive chemoradiation. Platinum doublet chemotherapy is recommended, and maintenance immunotherapy can be considered.

Patients with stage IV LCNEC are recommended to receive platinum-based chemotherapy or chemoimmunotherapy, using either SCLC or NSCLC chemotherapy regimens. The benefit of ICI with EP in the first-line setting specifically for LCNEC is unknown. There is no standard regimen for relapsed LNCEC. Off-label use of second-line agents for relapsed NSCLC or SCLC are often used. Nivolumab plus ipilimumab is included in some guidelines, based on the high ORR observed in the DART study. Prospective studies are critically needed in this disease, if possible, with correlative studies such as subtyping, to better personalize therapy.

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32 Neoplasms of the Mediastinum

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THYMIC NEOPLASMS

Incidence and Etiology

Thymic epithelial neoplasms (TETs), predominantly thymomas, constitute 30% and 15% of anterior mediastinal masses in adults and children, respectively. Etiologies of mediastinal masses, in general, are listed in Table 32.1 by location. Surveillance, Epidemiology, as well as several European and Asian Cancer Registries suggest that 1.3 to 2.6 TETs occur for every 1,000,000 person-years.[1,2] TETs are considered rare tumors with only approximately 390 cases annually in the United States (U.S.). TETs are more common in American Asians and Pacific Islanders (2.5/1,000,000) as well as African Americans (1.8/1,000,000).[1] The incidence of TETs increases with age from age 18 into the eighth decade of life peaking with an incidence of 4.2 to a peak of 10.6 cases/1,000,000. Over time the overall incidence has fluctuated with changes attributed to changes in reporting, pathology techniques, and classification (Fig. 32.1).[1,2] The incidence of TETs by histopathology subtype are shown in Figure 32.2.[2] No specific etiology or risk factors for TETs are known, but recognized associations between thymomas and myasthenia gravis (MG), pure red cell aplasia, and hypogammaglobulinemia do exist as well as an association primarily between neuroendocrine thymus tumors (NETTs) and multiple endocrine neoplasia, type 1 (MEN1).[3] Thymic carcinoma (TC) accounts for 20% of U.S. TETs with a similar peak incidence in the eighth decade of life (2.5 cases/1,000,000 person-years). Unlike thymoma, however, TC only infrequently is associated with MG or other paraneoplastic syndromes, such as Sjögren syndrome and syndrome of inappropriate antidiuretic hormone (SIADH).[4]

Anatomy

The thymus is an incompletely understood lymphoepithelial organ that functions in T-lymphocyte maturation. It is composed mainly of thymocytes, lymphocytes, and an epithelial stroma. Although lymphomas, carcinoid tumors, and germ cell tumors all may arise within the thymus, only thymomas, thymic carcinomas,

TABLE 32.1

World Health Organization Classification System for Thymic Epithelial Tumors

Thymoma Subtype	Obligatory Criteria (Thymoma)	Optional Criteria (Thymoma)
Type A	Occurrence of bland, spindle-shaped epithelial cells (at least focally); paucity[a] or absence of immature (TdT^+) T cells throughout the tumor	Polygonal epithelial cells $CD20^+$ epithelial cells
Atypical type A variant	Criteria of type A thymoma; in addition: comedo-type tumor necrosis, increased mitotic count (>4/2 mm^2), nuclear crowding	Polygonal epithelial cells $CD20^+$ epithelial cells
Type AB	Occurrence of bland, spindle-shaped epithelial cells (at least focally); abundance[a] of immature (TdT^+) T cells focally or throughout tumor	Polygonal epithelial cells $CD20^+$ epithelial cells
Type B1	Thymus-like architecture and cytology: abundance of immature T cells, areas of medullary differentiation (medullary islands); paucity of polygonal or dendritic epithelia cells without clustering (i.e., fewer than three contiguous epithelial cells)	Hassall corpuscles; perivascular spaces
Type B2	Increased numbers of single or clustered polygonal or dendritic epithelial cells intermingled with abundant immature T cells	Medullary islands; Hassall corpuscles; perivascular spaces
Type B3	Sheets of polygonal slightly to moderately atypical epithelial cells; absent or rare intercellular bridges; paucity or absence of intermingled TdT^+ T cells	Hassall corpuscles; perivascular spaces
Rare others[b]		
Thymic carcinomas	Squamous cell carcinoma; basaloid carcinoma; mucoepidermoid carcinoma; lymphoepithelioma-like carcinoma; clear cell carcinoma; sarcomatoid carcinoma; adenocarcinomas (several); NUT carcinoma; undifferentiated carcinomas; rare others	

Note: Diagnostic criteria of the major thymoma subtypes in the 2015 WHO classification, separating obligatory from optional histological thymoma features. Thymic carcinomas exhibit a histologic spectrum similar to equally labeled carcinomas in other organs.
[a]Paucity versus abundance: any area of crowded immature T cells or moderate numbers of immature T cells in >10% of the investigated tumor are indicative of "abundance."
[b]Micronodular thymoma with lymphoid stroma, metaplastic thymoma, microscopic thymoma, sclerosing thymoma, lipofibroadenoma.
NUT, nuclear protein in testis.
Adapted from Marx A, Chan JK, Coindre JM, et al. The 2015 World Health Organization classification of tumors of the thymus: continuity and changes. *J Thorac Oncol* 2015;10(10):1383–1395, with permission.